
Math60033/70033: Algebraic curves

Module description

This module is meant as a first encounter with algebraic geometry, through the study of affine and projective plane curves over the field of complex numbers. We will also discuss some complex-analytic aspects of the theory (Riemann surfaces). Important results include the definition of local intersection multiplicities and Bézout's theorem, inflection points and the classification of plane cubics, linear systems of curves, and the degree-genus formula.

Books

William Fulton, *Algebraic curves: an introduction to algebraic geometry*, 2008.

Available at <https://dept.math.lsa.umich.edu/~wfulton/CurveBook.pdf>.

Frances Kirwan, *Complex algebraic curves*, 1992.

Available at <https://doi.org/10.1017/CB09780511623929>.

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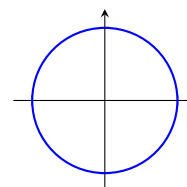
1 Basic definitions and examples

This module is intended as an introduction to algebraic geometry, which in a broad sense uses geometric methods to study the solution sets to systems of polynomial equations. We begin with some definitions and examples.

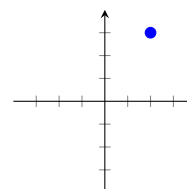
Definition 1.1. Let $\mathbb{K}$ be a field, and let $f_1, \dots, f_m \in \mathbb{K}[x_1, x_2, \dots, x_n]$ be polynomials in n variables with coefficients in $\mathbb{K}$. Then the *affine algebraic set* or *algebraic variety* defined by $f_1, \dots, f_m$ is the set

$$\mathbb{V}_{\mathbb{K}}(f_1, \dots, f_m) = \left\{ (a_1, \dots, a_n) \in \mathbb{K}^n \left| \begin{array}{l} f_1(a_1, \dots, a_n) = 0 \\ f_2(a_1, \dots, a_n) = 0 \\ \vdots \\ f_m(a_1, \dots, a_n) = 0 \end{array} \right. \right\}.$$

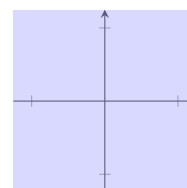
Example 1.2. The set $\mathbb{V}_{\mathbb{R}}(x^2 + y^2 - 1)$ is a unit circle.



Example 1.3. The point $(2, 3) \in \mathbb{R}^2$ is equal to $\mathbb{V}_{\mathbb{R}}(x - 2, y - 3)$.



Example 1.4. The set $\mathbb{V}_{\mathbb{R}}(0)$ is all of affine space $\mathbb{R}^n$.



Example 1.5. Let $S = \{a_1, \dots, a_n\} \in \mathbb{C}$ be a finite set. Then the polynomial

$$p(x) = (x - a_1)(x - a_2) \dots (x - a_n) \in \mathbb{C}[x]$$

is zero at x if and only if $x \in S$, so $S = \mathbb{V}_{\mathbb{C}}(p(x))$. Is $\mathbb{Z} \subset \mathbb{C}$ algebraic?

Our main objects of study will be the simplest interesting case of these, namely curves in the plane.

Definition 1.6. An *affine plane curve* over $\mathbb{K}$ is an affine variety cut out by a single non-constant polynomial in two variables: we write

$$C = \mathbb{V}_{\mathbb{K}}(P) = \{(x, y) \in \mathbb{K}^2 \mid P(x, y) = 0\}.$$

We will look for general techniques to understand the following sorts of questions:

- How can we describe the set of points of a given plane curve?
- What can we say about the intersection of two plane curves?
- When do two polynomials define the same plane curve?

For general fields, this last question is hard. For example, over $\mathbb{R}$ we have

$$\mathbb{V}_{\mathbb{R}}(x^2 + 1) = \mathbb{V}_{\mathbb{R}}(1) = \emptyset$$

even though the two defining polynomials don't seem related. On the other hand, $\mathbb{V}_{\mathbb{C}}(x^2 + 1) = \{i, -i\}$ is different from $\mathbb{V}_{\mathbb{C}}(1) = \emptyset$. We will mostly stick to algebraically closed fields, especially $\mathbb{C}$, to avoid such issues.

Definition 1.7. The *degree* of an affine plane curve $C = \mathbb{V}_{\mathbb{K}}(P)$ is the degree of the corresponding polynomial $P \in \mathbb{K}[x, y]$. If we write

$$P(x, y) = \sum_{i,j \geq 0} a_{ij} x^i y^j$$

then this is the maximal value of $i + j$ over all pairs (i, j) with $a_{ij} \neq 0$.

Let's look at some examples of plane curves of low degree over $\mathbb{R}$.

Example 1.8. If $\deg(C) = 1$, then C must be defined by a linear equation $ax + by = c$, where at least one of a and b is nonzero. Then C is a line in $\mathbb{R}^2$.

Example 1.9. If $\deg(C) = 2$, then C is defined by a polynomial of the form

$$P(x, y) = ax^2 + bxy + cy^2 + dx + ey + f,$$

making it a *conic section*. These include

- circles, such as $P(x, y) = x^2 + y^2 - 1$;
- ellipses, such as $P(x, y) = x^2 + 4y^2 - 1$;
- hyperbolas, such as $P(x, y) = x^2 - y^2 - 1$;
- parabolas, such as $P(x, y) = y - x^2$;

and degenerate examples where P is a product of two degree-1 factors, e.g., a pair of intersecting lines ($xy = 0$) or even a single line ($x^2 = 0$).

Understanding the points of a plane curve can often have interesting applications:

Question 1.10. A *Pythagorean triple* is a triple of nonzero integers (a, b, c) satisfying $a^2 + b^2 = c^2$. How can we describe all Pythagorean triples?

Certainly any Pythagorean triple (a, b, c) has $c \neq 0$, so we can divide through by c to see that $(x, y) = (\frac{a}{c}, \frac{b}{c})$ satisfies $x^2 + y^2 = 1$, or in other words

$$\left(\frac{a}{c}, \frac{b}{c}\right) \in \mathbb{V}_{\mathbb{Q}}(x^2 + y^2 - 1).$$

Thus what we really want is an explicit way to describe all points of the plane curve

$$C = V_{\mathbb{K}}(x^2 + y^2 - 1)$$

in the case $\mathbb{K} = \mathbb{Q}$. Since C is a curve of low degree, this is actually possible!

To start, the point $p_0 = (-1, 0)$ always belongs to C . Any other point of $\mathbb{K}^2$ lies on a unique line through p_0 , say the line

$$L_t = \{(x, y) \mid y = t(x + 1)\} = V_{\mathbb{K}}(y - t(x + 1))$$

of some slope $t \in \mathbb{K}$, so we can try to figure out where (other than p_0) the curve C intersects L_t . The system of equations

$$\begin{aligned} x^2 + y^2 &= 1 \\ y &= t(x + 1) \end{aligned}$$

can be solved by substituting the second equation into the first to get

$$x^2 + t^2(x + 1)^2 = 1 \iff (x + 1)((t^2 + 1)x + (t^2 - 1)) = 0.$$

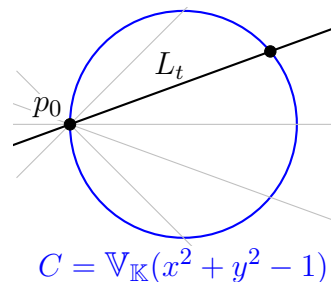
This has roots $x = -1$ and $x = \frac{1-t^2}{1+t^2}$, and plugging these into $y = t(x + 1)$ gives

$$C \cap L_t = \left\{ (-1, 0), \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right) \right\}.$$

In other words, C can be parametrized away from p_0 by a pair of rational functions:

$$V_{\mathbb{K}}(x^2 + y^2 - 1) = \{(-1, 0)\} \cup \left\{ \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right) \mid t \in \mathbb{K} \right\}.$$

Writing $t = \frac{r}{s}$ and clearing denominators, we have $(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}) = (\frac{s^2-r^2}{s^2+r^2}, \frac{2rs}{s^2+r^2})$, and this now includes the point $(-1, 0)$ as the case $(r, s) = (1, 0)$. Thus any Pythagorean triple (a, b, c) must be a multiple of $(s^2 - r^2, 2rs, s^2 + r^2)$ for some integers r and s .



Question 1.11. Why was the fact that $\deg(x^2 + y^2 - 1) = 2$ important in finding this parametrization? What goes wrong for degrees 3 and above?

2 Affine varieties

2.1 Some algebra and topology

We wish to develop some basic facts about the topology of an affine variety, but first we need to recall some algebra.

Definition 2.1. An *ideal* of a commutative ring R is a subset $I \subset R$ that is closed under addition and under multiplication by elements of R . In other words, if $a, b \in I$ and $r \in R$ then $a + b \in I$ and $ra \in I$.

The ideal *generated* by a subset $X \subset R$, denoted $I(X) \subset R$, is the smallest ideal containing all of X . If $X = \{r_1, r_2, \dots\}$ then we may write $I = \langle r_1, r_2, \dots \rangle$.

We will use the following important fact from commutative algebra without proof.

Theorem 2.2: Hilbert basis theorem, special case

For any field $\mathbb{K}$, any ideal $I \subset \mathbb{K}[x_1, x_2, \dots, x_n]$ has a finite generating set.

Generalizing the previous definition of an affine algebraic set, if $I \subset \mathbb{K}[x_1, \dots, x_n]$ is an ideal then we may write

$$\mathbb{V}(I) = \{(a_1, \dots, a_n) \mid f(a_1, \dots, a_n) = 0 \text{ for all } f \in I\}.$$

Theorem 2.2 says that I is finitely generated, so we can write

$$I = \langle f_1, f_2, \dots, f_m \rangle.$$

Then we can check that

$$I = \{g_1 f_1 + g_2 f_2 + \dots + g_m f_m \mid g_1, \dots, g_m \in \mathbb{K}[x_1, \dots, x_n]\},$$

because the set on the right is an ideal, and because if I contains each of $f_1, \dots, f_m$ then by definition it must contain each of the elements of that set, so that set is the smallest ideal containing each f_i .

We should check that $\mathbb{V}(I)$ really does generalize our definition of an affine variety.

Lemma 2.3. If $I = \langle f_1, \dots, f_m \rangle \subset \mathbb{K}[x_1, \dots, x_n]$, then $\mathbb{V}(I) = \mathbb{V}(f_1, \dots, f_m)$.

Proof. We first show that $\mathbb{V}(I) \subset \mathbb{V}(f_1, \dots, f_m)$: if $(a_1, \dots, a_n) \in \mathbb{V}(I)$, then for each $i = 1, \dots, m$ we have $f_i \in I$ and so $f_i(a_1, \dots, a_n) = 0$. Thus $(a_1, \dots, a_n) \in \mathbb{V}(f_1, \dots, f_m)$ by definition.

For the reverse inclusion, we take $(a_1, \dots, a_n) \in \mathbb{V}(f_1, \dots, f_m)$. An arbitrary element $f \in I$ can be written

$$f = g_1 f_1 + \dots + g_m f_m$$

with each $g_i \in \mathbb{K}[x_1, \dots, x_n]$, so we have

$$f(a_1, \dots, a_n) = \sum_{i=1}^m g_i(a_1, \dots, a_n) \cdot \underbrace{f_i(a_1, \dots, a_n)}_{=0} = 0.$$

Thus $(a_1, \dots, a_n) \in \mathbb{V}(I)$ as well. □

Proposition 2.4. *Let V_1 and V_2 be affine algebraic sets in $\mathbb{K}^n$. Then their union $V_1 \cup V_2$ and their intersection $V_1 \cap V_2$ are also affine algebraic sets.*

Proof. Write $V_1 = \mathbb{V}_{\mathbb{K}}(f_1, \dots, f_m)$ and $V_2 = \mathbb{V}_{\mathbb{K}}(g_1, \dots, g_\ell)$. Then we have

$$\begin{aligned} V_1 \cap V_2 &= \{(a_1, \dots, a_n) \in \mathbb{K}^n \mid (a_1, \dots, a_n) \in V_1 \text{ and } (a_1, \dots, a_n) \in V_2\} \\ &= \left\{ (a_1, \dots, a_n) \in \mathbb{K}^n \mid \begin{array}{l} f_i(a_1, \dots, a_n) = 0 \text{ for } i = 1, \dots, m \\ g_j(a_1, \dots, a_n) = 0 \text{ for } j = 1, \dots, \ell \end{array} \right\} \\ &= \mathbb{V}_{\mathbb{K}}(f_1, \dots, f_m, g_1, \dots, g_\ell). \end{aligned}$$

Exercise: prove that $V_1 \cup V_2 = \mathbb{V}_{\mathbb{K}}(f_i \cdot g_j \mid 1 \leq i \leq m, 1 \leq j \leq \ell)$. □

From the proof of Proposition 2.4 we see that the union of two affine plane curves is again an affine plane curve, since

$$\mathbb{V}_{\mathbb{K}}(P) \cup \mathbb{V}_{\mathbb{K}}(Q) = \mathbb{V}_{\mathbb{K}}(PQ).$$

Is the intersection of two affine plane curves also an affine plane curve?

Remark 2.5. The intersection $\bigcap_{\alpha} \mathbb{V}(I_{\alpha})$ of arbitrarily many affine varieties is again an affine variety, because if we write $I_{\alpha} = I(X_{\alpha})$ for all α and let $I = I(\bigcup_{\alpha} X_{\alpha})$, then I is finitely generated (by the Hilbert basis theorem) and $\mathbb{V}(I) = \bigcap_{\alpha} \mathbb{V}(I_{\alpha})$.

This lets us define the *Zariski topology* on $\mathbb{K}^n$, in which a set is open if and only if it is the *complement* of an affine algebraic set. One can check that this is a topology: the intersection of two open sets is open, and the union of arbitrarily many open sets is also open. When $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ this is weaker than the usual topology on $\mathbb{R}^n$ or $\mathbb{C}^n$, because Zariski-open sets are open in the usual sense but not vice versa.

None of the above requires plane curves to be nonempty, but under nice circumstances we can guarantee that they do in fact have points. We are about to insist that $\mathbb{K}$ be algebraically closed, meaning that any polynomial $f(x) \in \mathbb{K}[x]$ has a root in $\mathbb{K}$. For example:

Theorem 2.6: Fundamental theorem of algebra

The field $\mathbb{C}$ of complex numbers is algebraically closed.

Proposition 2.7. *If $\mathbb{K}$ is algebraically closed, then any plane curve $C \subset \mathbb{K}^2$ has infinitely many points. If in fact $\mathbb{K} = \mathbb{C}$ then C is unbounded.*

Proof. We write $C = \mathbb{V}_{\mathbb{K}}(P)$ for some nonconstant polynomial $P \in \mathbb{K}[x, y]$. Since P is not constant, some monomial of P has a positive power of either x or y , say x without loss of generality. If $d \geq 1$ is the highest exponent of x in any monomial of P , then we can write

$$P(x, y) = Q_d(y)x^d + Q_{d-1}(y)x^{d-1} + \cdots + Q_1(y)x + Q_0(y),$$

where each $Q_i(y) \in \mathbb{K}[y]$ is a polynomial only in y , and where $Q_d(y)$ is not zero.

Letting $e = \deg_y(Q_d)$, there are at most e values of y such that $Q_d(y) = 0$. At any other $y_0 \in \mathbb{K}$, the polynomial $P(x, y_0)$ is a degree- d polynomial in x , since its leading coefficient is $Q_d(y_0) \neq 0$. Then since $\mathbb{K}$ is algebraically closed, there is some $x_0 \in \mathbb{K}$ (depending on y_0) such that $P(x_0, y_0) = 0$, or equivalently $(x_0, y_0) \in C$. Now C is infinite since there are infinitely many such y_0 , and if $\mathbb{K} = \mathbb{C}$ then the set of such y_0 is unbounded, hence so is C . $\square$

Proposition 2.7 fails if $\mathbb{K}$ is not algebraically closed, since $\mathbb{V}_{\mathbb{R}}(x^2 + 1)$ is empty.

2.2 Irreducible plane curves

We recall the following definition from algebra.

Definition 2.8. A polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$ is *irreducible* if whenever we write f as a product $f = g \cdot h$, at least one of g and h is a constant.

The polynomial ring $\mathbb{K}[x_1, \dots, x_n]$ is a *unique factorization domain*, meaning that any element $P(x_1, \dots, x_n)$ can be written uniquely as a product

$$P = f_1 \cdot f_2 \cdot \dots \cdot f_k$$

of irreducible polynomials, up to reordering the f_i and multiplication by constants. Then we have

$$\mathbb{V}_{\mathbb{K}}(P) = \mathbb{V}_{\mathbb{K}}(f_1) \cup \mathbb{V}_{\mathbb{K}}(f_2) \cup \cdots \cup \mathbb{V}_{\mathbb{K}}(f_k).$$

Repeated factors don't change anything: if $\alpha \geq 1$ then $\mathbb{V}_{\mathbb{K}}(f^\alpha) = \mathbb{V}_{\mathbb{K}}(f)$, and so

$$\mathbb{V}_{\mathbb{K}}\left(\prod_{i=1}^k f_i^{\alpha_i}\right) = \mathbb{V}_{\mathbb{K}}\left(\prod_{i=1}^k f_i\right) = \bigcup_{i=1}^k \mathbb{V}_{\mathbb{K}}(f_i).$$

So it seems that in some sense, the most basic varieties are those of the form $\mathbb{V}_{\mathbb{K}}(f)$ where f is irreducible. We will make this notion precise in the following way.

Definition 2.9. A plane curve $C \subset \mathbb{K}^2$ is *irreducible* if any time we can write $V = V_1 \cup V_2$ as a union of plane curves, we have either $V = V_1$ or $V = V_2$.

In order to relate irreducibility for polynomials and for plane curves, we will need the following theorem, which we will not prove here. It is a substantial strengthening of Proposition 2.7.

Theorem 2.10: Weak Nullstellensatz

Let $\mathbb{K}$ be an algebraically closed field. Then for any ideal $I \subset \mathbb{K}[x_1, \dots, x_n]$, the variety $\mathbb{V}(I)$ is empty if and only if $1 \in I$.

The weak Nullstellensatz is false if we do not require $\mathbb{K}$ to be algebraically closed: once again, over $\mathbb{R}$ the variety $\mathbb{V}_{\mathbb{R}}(x^2 + 1)$ is empty even though $1 \notin \langle x^2 + 1 \rangle$.

Proposition 2.11. Let $\mathbb{K}$ be algebraically closed, and let $f, g \in \mathbb{K}[x_1, \dots, x_n]$ be polynomials. If f is irreducible and $\mathbb{V}_{\mathbb{K}}(f) \subset \mathbb{V}_{\mathbb{K}}(g)$, then f divides g .

Proof. By hypothesis the subsets of $\mathbb{K}^n$ where $f = 0$ and where $g \neq 0$ are disjoint. We use the “Rabinowitsch trick”: we know that

$$g(x_1, \dots, x_n) \neq 0 \iff t \cdot g(x_1, \dots, x_n) = 1 \text{ for some } t \in \mathbb{K},$$

so we add an extra variable t , view f and g as elements of $\mathbb{K}[x_1, \dots, x_n, t]$, and note that

$$\mathbb{V}_{\mathbb{K}}(f) \cap \mathbb{V}_{\mathbb{K}}(tg - 1) = \mathbb{V}_{\mathbb{K}}(f, tg - 1) \subset \mathbb{K}^{n+1}$$

is empty. Then the weak Nullstellensatz tells us that

$$1 \in \langle f, tg - 1 \rangle,$$

so there must be polynomials $a, b \in \mathbb{K}[x_1, \dots, x_n, t]$ such that

$$a(x_1, \dots, x_n, t)f(x_1, \dots, x_n) + b(x_1, \dots, x_n, t)(tg(x_1, \dots, x_n) - 1) = 1.$$

We make the substitution $t = \frac{1}{g}$, resulting in the rational expression

$$a(x_1, \dots, x_n, g^{-1})f(x_1, \dots, x_n) = 1,$$

and multiply by some large power g^N to clear denominators. The result is a polynomial relation in $\mathbb{K}[x_1, \dots, x_n]$ of the form $\tilde{a} \cdot f = g^N$. Since f is irreducible and $\mathbb{K}[x_1, \dots, x_n]$ has unique factorization, it follows that f must divide g . $\square$

Corollary 2.12. *Let $\mathbb{K}$ be algebraically closed, and let $f, g \in \mathbb{K}[x_1, \dots, x_n]$ be non-constant, irreducible polynomials. Then $\mathbb{V}_{\mathbb{K}}(f) = \mathbb{V}_{\mathbb{K}}(g)$ if and only if $f = \lambda g$ for some nonzero $\lambda \in \mathbb{K}$.*

Theorem 2.13

Let $\mathbb{K}$ be algebraically closed. A plane curve $C \subset \mathbb{K}^2$ is irreducible if and only if $C = \mathbb{V}_{\mathbb{K}}(P)$ for some irreducible, nonconstant polynomial $P \in \mathbb{K}[x, y]$.

Proof. Suppose that $C = \mathbb{V}_{\mathbb{K}}(P)$ is irreducible, and that we can write

$$P = f_1^{\alpha_1} \cdot f_2^{\alpha_2} \cdot \dots \cdot f_k^{\alpha_k}$$

where the polynomials f_i are irreducible and $\alpha_i \geq 1$. Then we know that

$$C = \mathbb{V}_{\mathbb{K}}(f_1) \cup \mathbb{V}_{\mathbb{K}}(f_2) \cup \dots \cup \mathbb{V}_{\mathbb{K}}(f_k),$$

and since C is irreducible it must be equal to one of the $\mathbb{V}_{\mathbb{K}}(f_i)$, as desired.

Conversely, suppose that $C = \mathbb{V}_{\mathbb{K}}(P)$ where P is irreducible, and that we can write C as a union of two plane curves

$$C = \mathbb{V}_{\mathbb{K}}(P_1) \cup \mathbb{V}_{\mathbb{K}}(P_2).$$

Equivalently, we have $\mathbb{V}_{\mathbb{K}}(P) = \mathbb{V}_{\mathbb{K}}(P_1 P_2)$, so Proposition 2.11 tells us that P divides $P_1 P_2$. Since P is irreducible, it must then divide one of the P_i , say P_1 without loss of generality. Similarly, if f is any irreducible factor of P_1 then

$$\mathbb{V}_{\mathbb{K}}(f) \subset \mathbb{V}_{\mathbb{K}}(P_1) \subset \mathbb{V}_{\mathbb{K}}(P_1 P_2) = \mathbb{V}_{\mathbb{K}}(P),$$

so again by Proposition 2.11 we see that f divides P . Since P is irreducible we have $P = \lambda f$ for some nonzero $\lambda \in \mathbb{K}$, and then $\mathbb{V}_{\mathbb{K}}(f) = \mathbb{V}_{\mathbb{K}}(P)$. In the above chain of inclusions we must therefore have $\mathbb{V}_{\mathbb{K}}(P_1) = \mathbb{V}_{\mathbb{K}}(P) = C$, and so C is an irreducible plane curve. $\square$

Theorem 2.13 tells us that if $P \in \mathbb{K}[x, y]$ can be written as a product

$$P = \lambda \cdot P_1^{e_1} \cdot P_2^{e_2} \cdot \dots \cdot P_k^{e_k}$$

of irreducibles, with $\lambda \in \mathbb{K}$ nonzero and each $e_i \geq 1$, then the plane curve $C = \mathbb{V}_{\mathbb{K}}(P)$ can be written as a union of finitely many irreducible plane curves, of the form

$$C = C_1 \cup C_2 \cup \dots \cup C_k$$

where $C_i = \mathbb{V}_{\mathbb{K}}(P_i)$. Moreover, any irreducible plane curve $C' \subset C$ can be described as

$$C' = \mathbb{V}_{\mathbb{K}}(f)$$

where $f \in \mathbb{K}[x, y]$ is irreducible, and then $\mathbb{V}_{\mathbb{K}}(f) \subset \mathbb{V}_{\mathbb{K}}(P)$ implies by Proposition 2.11 that f divides P . But then f is a scalar multiple of one of the P_i , and we have $C' = C_i$. In other words, the decomposition of a plane curve into its irreducible components is unique up to reordering, and in fact we have shown the following.

Theorem 2.14

Let $\mathbb{K}$ be algebraically closed, and let $f, g \in \mathbb{K}[x, y]$ be two polynomials. Then $\mathbb{V}_{\mathbb{K}}(f) = \mathbb{V}_{\mathbb{K}}(g)$ if and only if there are irreducible polynomials

$$P_1, \dots, P_k \in \mathbb{K}[x, y],$$

positive integers $a_1, \dots, a_k$ and $b_1, \dots, b_k$, and nonzero $\lambda, \mu \in \mathbb{K}$ such that

$$\begin{aligned} f &= \lambda \cdot P_1^{a_1} P_2^{a_2} \dots P_k^{a_k}, \\ g &= \mu \cdot P_1^{b_1} P_2^{b_2} \dots P_k^{b_k}. \end{aligned}$$

Then the irreducible components of $\mathbb{V}_{\mathbb{K}}(f)$ are precisely the curves $\mathbb{V}_{\mathbb{K}}(P_i)$.

As a special case of Theorem 2.14, we see that if the irreducible factorizations of f and g each have no repeated factors, then we have

$$\{(x, y) \in \mathbb{K}^2 \mid f(x, y) = 0\} = \{(x, y) \in \mathbb{K}^2 \mid g(x, y) = 0\}$$

if and only if $f = \lambda g$ for some nonzero $\lambda \in \mathbb{K}$.

3 Projective varieties

3.1 Projective space

Consider the pair of affine plane curves

$$\begin{aligned} L_1 &= \{ax + by + c = 0\}, \\ L_2 &= \{dx + ey + f = 0\} \end{aligned}$$

in $\mathbb{K}^2$. These intersect at a single point if and only if the system of linear equations

$$\begin{pmatrix} a & b \\ d & e \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -c \\ -f \end{pmatrix}$$

has one solution, which happens if and only if $ae - bd \neq 0$. If $ae = bd$ then the two lines may never meet, even though if we stand on the plane $\mathbb{K}^2$ they seem to come together at the horizon. We can make sense of this by adding “points at infinity” to compactify the plane. This gives a new setting that we call the *projective plane*.

Definition 3.1. The n -dimensional projective space $\mathbb{P}^n = \mathbb{P}^n(\mathbb{K})$ is the set of 1-dimensional linear subspaces of $\mathbb{K}^{n+1}$.

A line $L \subset \mathbb{K}^n$ through 0 is uniquely determined by any nonzero point $p \in L$, and two nonzero points $p, q \in \mathbb{K}^{n+1}$ determine the same line iff they are proportional, meaning that $p = \lambda q$ for some $\lambda \in \mathbb{K}^\times$. So we can think of $\mathbb{P}^n$ as a quotient

$$\mathbb{P}^n = \frac{\mathbb{K}^{n+1} \setminus \{0\}}{\mathbb{K}^\times},$$

where $\mathbb{K}^\times$ acts on $\mathbb{K}^{n+1} \setminus \{0\}$ by $\lambda \cdot (x_0, \dots, x_n) = (\lambda x_0, \dots, \lambda x_n)$. We will write

$$[x_0, \dots, x_n] \in \mathbb{P}^n$$

for the equivalence class of the point $(x_0, \dots, x_n)$.

Example 3.2. When $n = 0$ there is exactly one 1-dimensional linear subspace of $\mathbb{K}^1$, namely $\mathbb{K}^1$ itself, so $\mathbb{P}^0$ is a point.

Example 3.3. When $n = 1$ each point $[x, y] \in \mathbb{P}^1$ has either

- $y \neq 0$, in which case we can write $[x, y] = [\frac{x}{y}, 1]$;
- or $y = 0$ but $x \neq 0$, in which case $[x, y] = [x, 0] = [1, 0]$.

So $\mathbb{P}^1$ looks like a copy of $\mathbb{K}$, namely $\{[x, 1] \mid x \in \mathbb{K}\}$, together with the extra point $[1, 0]$ “at infinity”. The same analysis lets us decompose $\mathbb{P}^n$ into a union $\mathbb{K}^n \cup \mathbb{P}^{n-1}$, corresponding to $x_n \neq 0$ and $x_n = 0$ respectively.

We can write projective space $\mathbb{P}^n$ as a union of $n+1$ copies of affine space $\mathbb{K}^n$, called *affine charts*: for each $j = 0, 1, \dots, n$ we let

$$U_j = \{[x_0, x_1, \dots, x_n] \in \mathbb{P}^n \mid x_j \neq 0\},$$

and we note that $\mathbb{P}^n = U_0 \cup U_1 \cup \dots \cup U_n$ because every $[x_0, x_1, \dots, x_n] \in \mathbb{P}^n$ has $x_j \neq 0$ for some j . We define a map $i_j : \mathbb{K}^n \rightarrow \mathbb{P}^n$ by

$$i_j(x_1, \dots, x_n) \mapsto [x_1, x_2, \dots, x_j, 1, x_{j+1}, \dots, x_n].$$

Lemma 3.4. *The map i_j gives a bijection of $\mathbb{K}^n$ onto U_j .*

Proof. Certainly the image of i_j lies in U_j , so we define an inverse $\phi_j : U_j \rightarrow \mathbb{K}^n$ by

$$\phi_j([x_0, x_1, \dots, x_n]) = \left(\frac{x_0}{x_j}, \dots, \frac{x_{j-1}}{x_j}, \frac{x_{j+1}}{x_j}, \dots, \frac{x_n}{x_j} \right).$$

We have to check that this is well-defined – it does not change if we replace $[x_0, \dots, x_n]$ with $[\lambda x_0, \dots, \lambda x_n]$ – and that $\phi_j \circ i_j = \text{id}_{\mathbb{K}^n}$ and $i_j \circ \phi_j = \text{id}_{U_j}$. $\square$

Example 3.5. We can represent each point of $\mathbb{P}^n(\mathbb{C})$ by coordinates

$$[x_0, x_1, \dots, x_n] \in \mathbb{C}^{n+1} \text{ such that } |x_0|^2 + |x_1|^2 + \dots + |x_n|^2 = 1.$$

The space of all such coordinates is the unit sphere S^{2n+1} in $\mathbb{C}^{n+1} \cong \mathbb{R}^{2n+2}$. But this choice isn't unique: the unit circle S^1 in $\mathbb{C}$ acts freely on this sphere by

$$\lambda \cdot (x_0, x_1, \dots, x_n) = (\lambda x_0, \lambda x_1, \dots, \lambda x_n),$$

and two points of S^{2n+1} represent the same point of $\mathbb{P}^n(\mathbb{C})$ if and only if they're in the same orbit. So we have a continuous, surjective map

$$\begin{aligned} S^{2n+1} &\rightarrow \mathbb{P}^n(\mathbb{C}) \\ (x_0, x_1, \dots, x_n) &\mapsto [x_0, x_1, \dots, x_n] \end{aligned}$$

where the preimage of any point $[x_0, \dots, x_n]$ is the S^1 -orbit of $(x_0, \dots, x_n) \in S^{2n+1}$. (A topologist would say that we have an S^1 -bundle $S^{2n+1} \rightarrow \mathbb{P}^n(\mathbb{C})$.) We can conclude that $\mathbb{P}^n(\mathbb{C})$ is compact, since S^{2n+1} is. A good exercise if you've studied manifolds is to prove that it is a manifold, and in particular that it's Hausdorff.

It doesn't quite make sense to evaluate polynomials on $\mathbb{P}^n$, because the coordinates of a point $[x_0, x_1, \dots, x_n]$ are only well-defined up to rescaling, but under some circumstances we can still talk about when a polynomial is zero, and thus define plane curves.

Definition 3.6. A polynomial $f \in \mathbb{K}[x_0, \dots, x_n]$ is *homogeneous* of degree d if we have

$$f(\lambda x_0, \lambda x_1, \dots, \lambda x_n) = \lambda^d f(x_0, x_1, \dots, x_n)$$

for all $\lambda \in \mathbb{K}$ and all $(x_0, \dots, x_n) \in \mathbb{K}^n$.

For example, a single monomial $c x_0^{e_0} x_1^{e_1} \dots x_n^{e_n}$ satisfies

$$c(\lambda x_0)^{e_0} (\lambda x_1)^{e_1} \dots (\lambda x_n)^{e_n} = \lambda^{e_0 + \dots + e_n} \cdot c x_0^{e_0} x_1^{e_1} \dots x_n^{e_n}$$

and is therefore homogeneous of degree $e_0 + \dots + e_n$. Then any sum of monomials, each of which has total degree d , is a homogeneous of degree d .

Example 3.7. The polynomials $f(x, y, z) = x^2 + y^2 + z^2$ and $g(x, y, z) = xy + yz + zx$ are homogeneous of degree 2, but $h(x, y, z) = x^2 + y^2 - 1$ is not homogeneous.

Lemma 3.8. If $f \in \mathbb{K}[x_0, \dots, x_n]$ is homogeneous, then the set

$$\{[x_0, x_1, \dots, x_n] \in \mathbb{P}^n \mid f(x_0, x_1, \dots, x_n) = 0\}$$

is well-defined.

Proof. We need to make sure that this does not depend on the representative $(x_0, x_1, \dots, x_n)$ we chose of our point $[x_0, x_1, \dots, x_n] \in \mathbb{P}^n$. Supposing that f is homogeneous of some degree d , and that $f(x_0, x_1, \dots, x_n) = 0$, then for any nonzero $\lambda \in \mathbb{K}$ we have

$$f(\lambda x_0, \lambda x_1, \dots, \lambda x_n) = \lambda^d f(x_0, x_1, \dots, x_n) = \lambda^d \cdot 0 = 0$$

as well. Thus the condition $f = 0$ does not depend on which representative of the equivalence class we chose. $\square$

Lemma 3.8 lets us extend the notion of an algebraic set from $\mathbb{K}^n$ to $\mathbb{P}^n$.

Definition 3.9. A *projective variety* is a set $V \subset \mathbb{P}^n$ defined as the vanishing locus of some homogeneous polynomials $f_1, \dots, f_m \in \mathbb{K}[x_0, \dots, x_n]$, meaning that

$$V = \left\{ [x_0, x_1, \dots, x_n] \in \mathbb{P}^n \mid \begin{array}{l} f_1(x_0, \dots, x_n) = 0 \\ f_2(x_1, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, \dots, x_n) = 0 \end{array} \right\}.$$

Example 3.10. Let $f(x, y) \in \mathbb{K}[x, y]$ be a homogeneous polynomial of degree $d > 0$, and write

$$f(x, y) = c_d x^d + c_{d-1} x^{d-1} y + \cdots + c_1 x y^{d-1} + c_0 y^d.$$

Then the projective variety

$$V = \{[x, y] \in \mathbb{P}^1 \mid f(x, y) = 0\}$$

contains the point $[1, 0]$ at infinity if and only if $f(1, 0) = 0$, which is equivalent to $c_d = 0$ and hence to $y \mid f(x, y)$. It contains a point $[t, 1]$ if and only if $f(t, 1) = 0$, or equivalently if t is a root of the polynomial

$$f(x, 1) = c_d x^d + c_{d-1} x^{d-1} + \cdots + c_1 x + c_0.$$

In fact, when $\mathbb{K}$ is algebraically closed, we have the following.

Proposition 3.11. *Let $f(x, y) \in \mathbb{K}[x, y]$ be a homogeneous polynomial of degree $d > 0$. If $\mathbb{K}$ is algebraically closed, then f can be factored into a product*

$$f(x, y) = \prod_{i=1}^d (a_i x + b_i y)$$

of nonzero linear terms, and the points of the projective variety $V = \{f = 0\} \subset \mathbb{P}^1$ are precisely

$$[b_1, -a_1], [b_2, -a_2], \dots, [b_d, -a_d].$$

Proof. Certainly if $f(x, y)$ factors as claimed then $[x, y] \in V$ if and only if $a_i x + b_i y = 0$ for some i , which is equivalent to $[x, y] = [b_i, -a_i]$. Thus we will spend the rest of the proof factoring f .

We write $f(x, y) = y^{d-e} g(x, y)$, where g is homogeneous of degree $e \geq 0$ and y does not divide g . Then we can write

$$g(x, y) = c_e x^e + c_{e-1} x^{e-1} y + \cdots + c_1 x y^{e-1} + c_0 y^e,$$

with $y \nmid g$ telling us that $c_e \neq 0$. Now we have

$$g(x, y) = y^e g\left(\frac{x}{y}, 1\right) = y^e \left(c_e \left(\frac{x}{y}\right)^e + c_{e-1} \left(\frac{x}{y}\right)^{e-1} + \cdots + c_1 \left(\frac{x}{y}\right) + c_0 \right),$$

and the polynomial in parentheses is a degree- e polynomial in $\frac{x}{y}$. Since $\mathbb{K}$ is algebraically closed this polynomial has e roots, say $t_1, \dots, t_e$ when counted with

multiplicity, and then we can write

$$f(x, y) = y^{d-e} g(x, y) = y^{d-e} \cdot c_e \prod_{i=1}^e \left(\frac{x}{y} - t_i \right) = c_e y^{d-e} \cdot \prod_{i=1}^d (x - t_i y),$$

which has the desired form after we absorb the c_e factor into one of the linear terms. $\square$

As with affine varieties, we will focus on plane curves in particular.

Definition 3.12. A *projective plane curve* of degree $d > 0$ is the set

$$C = \{[x, y, z] \in \mathbb{P}^2 \mid P(x, y, z) = 0\}$$

defined by a nonconstant, homogeneous polynomial $P \in \mathbb{K}[x, y, z]$ of degree d .

Much of what we already know about affine plane curves can be extended straightforwardly to projective plane curves. For example, we can say that a projective plane curve $C \subset \mathbb{P}^2$ is *irreducible* if any time we can write $C = C_1 \cup C_2$ as a union of projective plane curves, then either $C = C_1$ or $C = C_2$. Then:

Proposition 3.13. Let $\mathbb{K}$ be an algebraically closed field, and $C \subset \mathbb{P}^2(\mathbb{K})$ a projective plane curve. Then

- C contains infinitely many points;
- C is irreducible if and only if $C = \{P = 0\}$ for some irreducible homogeneous polynomial $P \in \mathbb{K}[x, y, z]$;
- if $P, Q \in \mathbb{K}[x, y, z]$ are irreducible, nonconstant homogeneous polynomials, then the projective plane curves $\{P = 0\}$ and $\{Q = 0\}$ are equal if and only if $P = \lambda Q$ for some nonzero λ in $\mathbb{K}$;
- if $C = \{P = 0\}$, where $P(x, y, z)$ has irreducible factorization $P = f_1^{\alpha_1} f_2^{\alpha_2} \dots f_k^{\alpha_k}$ with each $\alpha_j \geq 1$, then the irreducible components of C are precisely the projective plane curves $\{f_j = 0\}$.

Implicit in the last assertion of Proposition 3.13 is the following claim, which we leave as an exercise: if a homogeneous polynomial f can be written as a product $f = gh$, then both g and h are homogeneous.

When $\mathbb{K} = \mathbb{C}$ we see one genuinely new feature in the projective setting. Recall from Proposition 2.7 that affine plane curves over $\mathbb{C}$ are unbounded, hence are never compact as subsets of $\mathbb{C}^2$.

Proposition 3.14. *Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a projective plane curve over $\mathbb{C}$. Then C is compact.*

Proof. The complex projective plane $\mathbb{P}^2(\mathbb{C})$ is compact in the quotient topology inherited from $\mathbb{C}^3 \setminus \{0\}$. Then $C \subset \mathbb{P}^2(\mathbb{C})$ is closed in this topology, because we can write $C = \{P = 0\}$ for some homogeneous $P \in \mathbb{C}[x, y, z]$ and the corresponding set $P^{-1}(0) \subset \mathbb{C}^3 \setminus \{0\}$ is closed. In particular C is a closed subset of a compact space, hence compact. $\square$

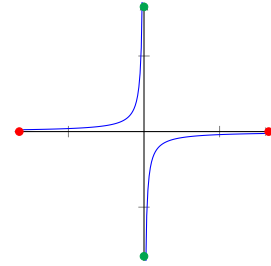
3.2 Projective curves and affine curves

We recall that the projective plane $\mathbb{P}^2$, say with coordinates $[x, y, z]$, is covered by a union of three affine charts

$$\{x \neq 0\} \cup \{y \neq 0\} \cup \{z \neq 0\}.$$

On the chart $\{z \neq 0\}$, every point can be given unique coordinates of the form $[x, y, 1]$ with $x, y \in \mathbb{K}$, and in doing so we can restrict a projective curve $C \subset \mathbb{P}^2$ to this chart by setting $z = 1$ to get a well-defined plane curve.

Example 3.15. The projective plane curve $C = \{xy + z^2 = 0\} \subset \mathbb{P}^2$ restricts to the affine plane curve $\{xy + 1 = 0\} \subset \mathbb{K}^2$. The points of C “at infinity”, meaning outside this chart, have $z = 0$ by definition: these look like $[x, y, 0]$ with $xy + 0^2 = 0$, or equivalently $xy = 0$. There are exactly two such points, namely $[1, 0, 0]$ and $[0, 1, 0]$.



Example 3.16. The projective plane curve $C = \{z = 0\} \subset \mathbb{P}^2$ restricts to the affine plane curve $\{1 = 0\} \subset \mathbb{K}^2$, which is empty. In other words, the curve C is disjoint from this affine chart: it is equal to the “line at infinity” in $\mathbb{P}^2$.

The process of restricting a projective plane curve to an affine plane curve can be reversed: any affine plane curve $\mathbb{V}_{\mathbb{K}}(P) = \{P(x, y) = 0\}$ can be realized as the restriction of a projective plane curve, obtained by *homogenizing* the polynomial P . We will call the corresponding projective curve the *projectivization* of $\mathbb{V}_{\mathbb{K}}(P)$.

To do this, we note that a homogeneous polynomial of degree d should satisfy

$$F(x, y, z) = z^d F\left(\frac{x}{z}, \frac{y}{z}, 1\right),$$

at least when z is nonzero, so if f has degree d then we define its homogenization as

$$F(x, y, z) = z^d \cdot f\left(\frac{x}{z}, \frac{y}{z}, 1\right) \in \mathbb{K}[x, y, z].$$

Example 3.17. The polynomial $f(x, y) = x^3 + xy + 1$ has degree 3, so its homogenization is

$$\begin{aligned} F(x, y, z) &= z^3 \left[\left(\frac{x}{z} \right)^3 + \left(\frac{x}{z} \right) \left(\frac{y}{z} \right) + 1 \right] \\ &= x^3 + xyz + z^3 \end{aligned}$$

and $\{F = 0\} \subset \mathbb{P}^2$ is the projectivization of $\{f = 0\} \subset \mathbb{K}^2$. We can check that $F(x, y, 1) = x^3 + xy + 1 = f(x, y)$.

Lemma 3.18. *Let $f \in \mathbb{K}[x, y]$ be a polynomial of degree d . Then its homogenization $F \in \mathbb{K}[x, y, z]$ is a homogeneous polynomial of degree d satisfying $F(x, y, 1) = f(x, y)$, and F is not a multiple of z .*

Proof. We should first check that $F(x, y, z) = z^d f(\frac{x}{z}, \frac{y}{z}, 1)$ is an honest polynomial. Any nonzero monomial in f has the form $cx^i y^j$, where $i + j \leq d$, and homogenization turns it into

$$z^d \cdot c \left(\frac{x}{z} \right)^i \left(\frac{y}{z} \right)^j = cx^i y^j z^{d-i-j},$$

whose exponents are all nonnegative since $d - i - j \geq 0$. The total degree of this new monomial is exactly d , so we have

$$c(\lambda x)^i (\lambda y)^j (\lambda z)^{d-i-j} = \lambda^d \cdot cx^i y^j z^{d-i-j},$$

and summing over all monomials we get $F(\lambda x, \lambda y, \lambda z) = \lambda^d F(x, y, z)$, proving homogeneity. The relation $F(x, y, 1) = f(x, y)$ comes from setting $z = 1$ above, since

$$cx^i y^j \cdot 1^{d-i-j} = cx^i y^j$$

recovers the original monomial. Finally, to see that F is not a multiple of z , we note that there must be some monomials of f whose total degree is exactly d , and homogenization turns any such $cx^i y^{d-i}$ into a monomial $cx^i y^{d-i} z^0$ that is not a multiple of z . $\square$

Proposition 3.19. *There is a bijection of the form*

$$\phi : \left\{ \begin{array}{l} \text{projective plane curves } \overline{C} \subset \mathbb{P}^2(\mathbb{K}) \\ \text{not containing the line } \{z = 0\} \end{array} \right\} \rightarrow \{ \text{affine plane curves } C \subset \mathbb{K}^2 \}$$

defined by sending the projective curve $\overline{C} = \{P(x, y, z) = 0\}$ to the affine curve $C = \{P(x, y, 1) = 0\}$.

Proof. We define the inverse map ψ by sending the affine curve $\{f(x, y) = 0\}$ to the projective curve $\{F(x, y, z) = 0\}$, where F is the homogenization of f . In order to prove that ϕ is a bijection, we need to check that ϕ and ψ are really inverses.

Let $C = \{f(x, y) = 0\}$ be an affine plane curve, where f has degree $d \geq 1$, and let $F(x, y, z)$ be the homogenization of f . Lemma 3.18 says that z does not divide $F(x, y, z)$, so by Proposition 3.13 we know that $\psi(C) = \{F(x, y, z) = 0\}$ does not contain the line $\{z = 0\}$ at infinity, and then

$$\phi(\psi(C)) = \{F(x, y, 1) = 0\} = \{f(x, y) = 0\} = C,$$

using the relation $F(x, y, 1) = f(x, y)$ of Lemma 3.18.

In the other direction, let $\overline{C} = \{F(x, y, z) = 0\}$ be a projective plane curve, with F homogeneous of degree $d \geq 1$. If $\overline{C}$ does not contain the line $\{z = 0\}$ then this means again that z does not divide F , so F contains some nonzero monomial of the form $cx^i y^{d-i}$. Then the polynomial $f(x, y) = F(x, y, 1)$ contains the same degree- d monomial, so f has degree d as well; in particular it is not constant, so $\phi(\overline{C}) = \{f = 0\}$ is an affine plane curve. Moreover, the homogenization of f by definition

$$z^d f\left(\frac{x}{z}, \frac{y}{z}\right) = z^d F\left(\frac{x}{z}, \frac{y}{z}, 1\right) = F(x, y, z),$$

where the last inequality follows from F being homogeneous of degree d , so now

$$\psi(\phi(\overline{C})) = \psi(\{f = 0\}) = \{F = 0\} = \overline{C}.$$

Thus ϕ and ψ really are inverses, as claimed. $\square$

3.3 Points at infinity

Unlike in the affine case, we'll eventually see that any pair of projective plane curves must intersect somewhere in $\mathbb{P}^2$. For example, consider the affine lines

$$\begin{aligned} L_1 &= \{ax + by + c = 0\}, \\ L_2 &= \{ax + by + d = 0\} \end{aligned}$$

where $c \neq d$. These do not intersect in $\mathbb{K}^2$, but their projectivizations are

$$\begin{aligned} \overline{L}_1 &= \{ax + by + cz = 0\} \\ \overline{L}_2 &= \{ax + by + dz = 0\}, \end{aligned}$$

which do intersect along the line $L_\infty = \{z = 0\}$ at infinity, namely at $[b, -a, 0]$.

We can determine the points at infinity systematically. Let $C = \{f(x, y) = 0\}$ be an affine plane curve. We can break the defining polynomial $f \in \mathbb{K}[x, y]$ into homogeneous parts of each degree up to d , writing

$$f(x, y) = f_d(x, y) + f_{d-1}(x, y) + \cdots + f_1(x, y) + f_0(x, y),$$

where each $f_i \in \mathbb{K}[x, y]$ is the sum of all monomials of f with total degree i ; then f_i is homogeneous of degree i . For example, we might have

$$f(x, y) = x^3 + xy^2 + 3x - 5y + 8 \implies \begin{cases} f_3(x, y) = x^3 + xy^2 \\ f_2(x, y) = 0 \\ f_1(x, y) = 3x - 5y \\ f_0(x, y) = 8 \end{cases} \quad (3.20)$$

We claim that the points at infinity of the projectivization are determined completely by the leading-order terms f_d .

Proposition 3.21. *Let $C = \{f(x, y) = 0\}$ be an affine plane curve of degree $d > 0$, and write*

$$f(x, y) = f_d(x, y) + f_{d-1}(x, y) + \cdots + f_1(x, y) + f_0(x, y),$$

where each f_i is homogeneous of degree i . Let $\overline{C} \subset \mathbb{P}^2$ be the projectivization of C . Then there is a bijection

$$\begin{aligned} \overline{C} \cap L_\infty &\longleftrightarrow \{[x, y] \in \mathbb{P}^1 \mid f_d(x, y) = 0\} \\ [x, y, 0] &\longleftrightarrow [x, y]. \end{aligned}$$

Proof. The homogenization of f is given by

$$F(x, y, z) = f_d(x, y) + f_{d-1}(x, y) \cdot z + f_{d-2}(x, y) \cdot z^2 + \cdots + f_0(x, y) \cdot z^d,$$

and $\overline{C} = \{F = 0\} \subset \mathbb{P}^2$ is the projectivization of C . Along the line $L_\infty = \{z = 0\}$ we therefore have

$$\begin{aligned} \overline{C} \cap L_\infty &= \{[x, y, 0] \mid F(x, y, 0) = 0\} \\ &= \{[x, y, 0] \mid f_d(x, y) = 0\}. \end{aligned} \quad \square$$

The points of $\{f_d = 0\} \subset \mathbb{P}^1$ are described by Example 3.10. If $\mathbb{K}$ is algebraically closed then this is made more explicit in Proposition 3.11, and in this case we can say that $\overline{C} \cap L_\infty$ consists of exactly $d = \deg(C)$ points when counted with multiplicity. For example, in (3.20) we have

$$f_3(x, y) = x^3 + xy^2 = (x + 0y)(x + iy)(x - iy)$$

and so $\overline{C} \cap L_\infty$ consists of the three points $[0, 1, 0]$, $[i, -1, 0]$, and $[-i, -1, 0]$.

4 Singular points and smooth curves

When is a plane curve over $\mathbb{R}$ smooth? In analysis, we might use the following:

Theorem 4.1: Implicit function theorem, special case

Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth function, and fix a point $(a, b) \in \mathbb{R}^2$ with $F(a, b) = 0$. If $\frac{\partial F}{\partial y}(a, b) \neq 0$, then there is neighborhood U of (a, b) and a smooth function

$$f : (a - \epsilon, a + \epsilon) \rightarrow \mathbb{R}$$

such that the zero set of F is defined locally by the graph of f :

$$U \cap \{F = 0\} = \{(x, f(x)) \mid x \in (a - \epsilon, a + \epsilon)\}.$$

In other words, the (non-algebraic!) plane curve $\{F = 0\}$ is smooth at (a, b) if $\frac{\partial F}{\partial y} \neq 0$, because then we can express it as the graph of a smooth function $y = f(x)$. The same is true if $\frac{\partial F}{\partial x} \neq 0$, because then it's the graph of some function $x = f(y)$ instead. The only way it can fail to be smooth is if all partial derivatives of F are zero. Since the derivative of a polynomial is again a polynomial, we can make sense of this condition as a *definition* of smoothness for algebraic curves.

In what follows we will use subscripts to denote partial derivatives, as in $F_x = \frac{\partial F}{\partial x}$.

Definition 4.2. Let $C = \{f(x, y) = 0\} \subset \mathbb{K}^2$ be an affine plane curve. Then C is *smooth* at a point (a, b) with $f(a, b) = 0$ if

$$(f_x(a, b), f_y(a, b)) \neq (0, 0).$$

Otherwise we say that C is *singular* at (a, b) . The curve C is *smooth* or *non-singular* if it is smooth at every point.

If C is nonsingular at $p = (a, b)$, then it has a well-defined tangent line given by

$$T_p C = \{f_x(a, b)(x - a) + f_y(a, b)(y - b) = 0\}.$$

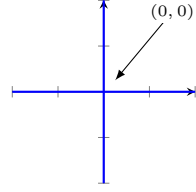
At singular points this is not the case, because the above equation becomes $0 = 0$.

Example 4.3. Any affine line is smooth, because if we define such a line by $f(x, y) = ax + by + c = 0$, then either $a \neq 0$ and so $f_x = a$ is nonzero everywhere, or $b \neq 0$ and so $f_y = b$ is nonzero everywhere.

Example 4.4. Consider the reducible plane curve $C = \{xy = 0\}$. If we set $f(x, y) = xy$ then

$$(f_x, f_y) = (y, x) = (0, 0) \iff (x, y) = (0, 0),$$

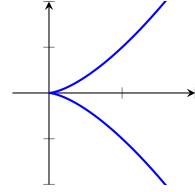
so C has a *nodal* singularity at $(0, 0)$ – where the components $\{x = 0\}$ and $\{y = 0\}$ meet – and is smooth everywhere else.



Example 4.5. The plane curve $C = \{y^2 = x^3\}$ is irreducible, and if $f(x, y) = y^2 - x^3$ then

$$(f_x, f_y) = (-3x^2, 2y) = (0, 0) \iff (x, y) = (0, 0).$$

The curve C has a *cusp* singularity at $(0, 0)$.



Example 4.6. Given $\lambda \neq 0, 1$, we can define the elliptic curve $E \subset \mathbb{K}^2$ by

$$y^2 = x(x - 1)(x - \lambda).$$

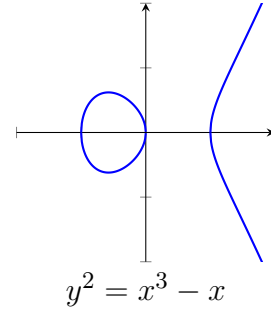
This is the zero set of $f(x, y) = x^3 - (1 + \lambda)x^2 + \lambda x - y^2$, so a point of E is singular if and only if

$$(f_x, f_y) = (-2y, 3x^2 - 2(1 + \lambda)x + \lambda) = (0, 0).$$

From $f_x = 0$ we get $y = 0$, and then $(x, 0) \in E$ if and only if x is 0, 1, or λ . We compute

$$\begin{aligned} f_y(0, 0) &= (0, \lambda), \\ f_y(1, 0) &= (0, 1 - \lambda), \\ f_y(\lambda, 0) &= (0, \lambda(\lambda - 1)), \end{aligned}$$

all of which are nonzero because $\lambda \neq 0, 1$. Thus E is nonsingular.



Example 4.7. Suppose that $f(x, y) = g(x, y)^2 \cdot h(x, y)$. Then the plane curve $\{g = 0\}$ is a subset of the plane curve $\{f = 0\}$, and we have

$$\begin{aligned} f_x &= 2gg_xh + g^2h_x = g(2g_xh + gh_x) \\ f_y &= 2gg_yh + g^2h_y = g(2g_yh + gh_y), \end{aligned}$$

so $\{f = 0\}$ is singular at every point of the curve $\{g = 0\}$. This is a reason to insist that plane curves be defined by polynomials with no repeated factors.

As always, we can extend this notion to projective plane curves. We will restrict

our attention to fields of characteristic zero for simplicity.

Definition 4.8. Suppose that $\text{char}(\mathbb{K}) = 0$, and take a projective plane curve

$$C = \{F(x, y, z) = 0\} \subset \mathbb{P}^2(\mathbb{K}).$$

We say that C is *singular* at a point (a, b, c) if

$$F_x(a, b, c) = F_y(a, b, c) = F_z(a, b, c) = 0.$$

Example 4.9. Suppose that $\text{char}(\mathbb{K}) = 0$ and that $d \geq 1$, and consider

$$C = \{F(x, y, z) = x^d + y^d + z^d = 0\} \subset \mathbb{P}^2.$$

Then we have $F_x = F_y = F_z = 0$ only for $(x, y, z) = (0, 0, 0)$, and this is not a point of $\mathbb{P}^2$, so C is nonsingular.

Note that in the above definition, we didn't actually insist that (a, b, c) belong to C ! This follows automatically from an identity due to Euler:

Proposition 4.10. Let $F(x_0, \dots, x_n) \in \mathbb{K}[x_0, \dots, x_n]$ be homogeneous of degree d . Then

$$\sum_{i=0}^n x_i \cdot F_{x_i}(x_0, \dots, x_n) = d \cdot F(x_0, \dots, x_n).$$

In particular, if $d \neq 0$ in $\mathbb{K}$ (for example, if $\text{char}(\mathbb{K}) = 0$), and if

$$F_{x_i}(a_0, \dots, a_n) = 0 \text{ for } i = 0, 1, \dots, n,$$

then $F(a_0, \dots, a_n) = 0$ as well.

Proof. Since F is homogeneous, we introduce a variable λ and consider the relation

$$F(\lambda x_0, \lambda x_1, \dots, \lambda x_n) = \lambda^d F(x_0, \dots, x_n).$$

We differentiate both sides with respect to λ :

$$\sum_{i=0}^n x_i F_{x_i}(\lambda x_0, \lambda x_1, \dots, \lambda x_n) = d \lambda^{d-1} F(x_0, \dots, x_n).$$

Now setting $\lambda = 1$ proves the desired identity. □

Remark 4.11. If $\text{char}(\mathbb{K})$ is positive then we should also add the condition $F(x, y, z) = 0$ to the definition of a singular point, in order to make sure that singular points of $C \subset \mathbb{P}^2$ really do belong to C .

Remark 4.12. We might be worried about applying the multivariate chain rule in the proof of Proposition 4.10, because over an arbitrary field we don't have any of the usual tools of analysis. This is not a problem, though, because we can define partial derivatives of polynomials as a purely formal operation on $\mathbb{K}[x_0, \dots, x_n]$, and then check that this operation satisfies the usual chain rule when applied to polynomials.

We should check that the definition of smoothness for projective plane curves really does generalize the definition in the affine case.

Proposition 4.13. *Let $\overline{C} \subset \mathbb{P}^2$ be the projectivization of an affine plane curve $C \subset \mathbb{K}^2$. Then $(a, b) \in C$ is singular if and only if $[a, b, 1]$ is a singular point of $\overline{C}$.*

Proof. Write $C = \{f(x, y) = 0\}$, and let $F(x, y, z)$ be the homogenization of f ; then $f(x, y) = F(x, y, 1)$, and $\overline{C} = \{F(x, y, z) = 0\}$. By hypothesis we have $f(a, b) = 0$, so we note the relations $F(a, b, 1) = f(a, b) = 0$ and

$$\begin{aligned} F_x(a, b, 1) &= f_x(a, b), \\ F_y(a, b, 1) &= f_y(a, b). \end{aligned}$$

Certainly if $[a, b, 1] \in \overline{C}$ is singular then $F_x(a, b, 1) = F_y(a, b, 1) = 0$ implies that $f_x(a, b) = f_y(a, b) = 0$, hence $(a, b) \in C$ is singular. Conversely, if $(a, b) \in C$ is singular, then we let $d = \deg(F)$ and set $z = 1$ in Proposition 4.10 to see that

$$d \cdot \underbrace{F(a, b, 1)}_{=f(a,b)=0} = a \underbrace{F_x(a, b, 1)}_{=f_x(a,b)=0} + b \underbrace{F_y(a, b, 1)}_{=f_y(a,b)=0} + F_z(a, b, 1),$$

so $F_z(a, b, 1) = 0$ as well and thus $[a, b, 1]$ is a singular point of $\overline{C}$. □

Similarly, we can say that a smooth point $p = [a, b, c]$ of a projective plane curve $\overline{C}$ defined by the equation $F(x, y, z) = 0$ has tangent line

$$T_p \overline{C} = \{F_x(a, b, c)(x - a) + F_y(a, b, c)(y - b) + F_z(a, b, c)(z - c) = 0\}.$$

(The left side does not look homogeneous, but Euler's identity, Proposition 4.10, guarantees that the constant term equals $-dF(a, b, c) = 0$.) If $\overline{C}$ is the projectivization of an affine plane curve C , and $(a, b) \in C$ is a smooth point, then the tangent line to $\overline{C}$ at $[a, b, 1]$ is the projectivization of the tangent line to C at (a, b) .

The following proposition generalizes Example 4.4.

Proposition 4.14. *Let C be a projective plane curve over an algebraically closed field, and suppose that two different irreducible components $C_1, C_2 \subset C$ intersect at a point $[a, b, c]$. Then C is not smooth at $[a, b, c]$.*

Proof. We write $C = \{F(x, y, z) = 0\}$, and factor F into irreducible polynomials as

$$F = f_1^{\alpha_1} f_2^{\alpha_2} \cdots f_k^{\alpha_k}, \quad \alpha_1, \dots, \alpha_k \geq 1.$$

Proposition 3.13 tells us that $C_1 = \{f_i = 0\}$ and $C_2 = \{f_j = 0\}$ for some indices $i \neq j$, and then by hypothesis we have

$$f_i(a, b, c) = f_j(a, b, c) = 0.$$

We can write $F = f_i f_j g$ where g consists of all the remaining irreducible factors, and then

$$F_x = (f_i)_x f_j g + f_i (f_j)_x g + f_i f_j g_x.$$

Each term on the right has a factor of either f_i or f_j , hence vanishes at (a, b, c) , and so $F_x(a, b, c) = 0$. The same argument says that $F_y(a, b, c) = F_z(a, b, c) = 0$ as well, so C is singular at (a, b, c) . $\square$

We will eventually see (as a consequence of Bézout's theorem) that any two projective plane curves over an algebraically closed field must intersect. Proposition 4.14 will then imply the following.

Theorem 4.15

A smooth projective plane curve over an algebraically closed field is irreducible.

Finally, we note one reason to be especially interested in smooth plane curves over $\mathbb{C}$: they have very nice topology.

Theorem 4.16

Let $C = \{F(x, y, z) = 0\} \subset \mathbb{P}^2(\mathbb{C})$ be a smooth projective plane curve. Then C is a compact *Riemann surface*: it's a 2-dimensional smooth manifold covered by open charts $\phi_i : \Omega_i \xrightarrow{\cong} U_i \subset C$ where $\Omega_i \subset \mathbb{C}$, such that the transition maps $\phi_{ij} : \phi_i^{-1}(U_i \cap U_j) \rightarrow \phi_j^{-1}(U_i \cap U_j)$ between overlapping charts are holomorphic.

Proof sketch. The compactness is Proposition 3.14. For the Riemann surface structure, we know that at every point of C at least one of F_x , F_y , or F_z is nonzero, so we can apply the holomorphic version of the implicit function theorem to express C locally as the graph of a holomorphic function. This provides the desired charts. $\square$

5 Projective transformations

In linear algebra we make a distinction between

- linear maps, which are maps $V \rightarrow W$ between vector spaces satisfying some axioms; and
- matrices, which express these maps in terms of a choice of coordinates on V and on W .

A change of basis of either V or W will change the corresponding matrix in a standard way, but the new matrix still represents the same map. Similarly, until now we've been working with fixed coordinates on $\mathbb{K}^n$ or on $\mathbb{P}^n$, but we should consider two different varieties the same if they're related by a change of coordinates. Since $\mathbb{P}^n$ is the space of lines through the origin in $\mathbb{K}^{n+1}$, and a linear transformation of $\mathbb{K}^{n+1}$ carries lines to lines, any such linear transformation gives rise to an automorphism of $\mathbb{P}^n$, which we will call a *projective transformation*.

We recall that $\mathrm{GL}_n(\mathbb{K})$ denotes the *general linear group* of invertible $n \times n$ matrices with entries in $\mathbb{K}$, or equivalently the $n \times n$ matrices over $\mathbb{K}$ with nonzero determinant.

Lemma 5.1. *An invertible $(n+1) \times (n+1)$ matrix $A = (a_{ij}) \in \mathrm{GL}_{n+1}(\mathbb{K})$ gives rise to a bijection $\Phi_A : \mathbb{P}^n \rightarrow \mathbb{P}^n$ defined by*

$$\Phi_A \left(\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} \right) \mapsto A \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{00}x_0 + a_{01}x_1 + \cdots + a_{0n}x_n \\ a_{10}x_0 + a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{n0}x_0 + a_{n1}x_1 + \cdots + a_{nn}x_n \end{bmatrix}.$$

This defines an action of $\mathrm{GL}_{n+1}(\mathbb{K})$ on $\mathbb{P}^n(\mathbb{K})$, by $A \cdot p = \Phi_A(p)$.

Proof. We should first check that Φ_A is well defined. We fix a point $[x_0, \dots, x_n] \in \mathbb{P}^n$, so that the x_i are not all zero, and we write

$$A \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

Then the y_i cannot all be zero, since A is invertible and thus has kernel $\{0\}$, so $\Phi_A([x_0, \dots, x_n]) = [y_0, \dots, y_n]$ is indeed a point of $\mathbb{P}^n$. Moreover, the value of Φ_A does not depend on the particular coordinates $[x_0, \dots, x_n]$, since if $\lambda \in \mathbb{K}^\times$ then

$$\Phi_A([\lambda x_0, \dots, \lambda x_n]) = [\lambda y_0, \dots, \lambda y_n] = [y_0, \dots, y_n].$$

To see that Φ is a bijection, we let $B \in \mathrm{GL}_{n+1}(\mathbb{K})$ be the inverse of A , and then the relations $AB = I$ and $BA = I$ immediately tell us that $\Phi_A \circ \Phi_B$ and $\Phi_B \circ \Phi_A$ are both the identity map on $\mathbb{P}^n$. $\square$

Question 5.2. For which $A \in \mathrm{GL}_{n+1}(\mathbb{K})$ is Φ_A the identity map? In other words, what is the kernel of the $\mathrm{GL}_{n+1}(\mathbb{K})$ action on $\mathbb{P}^n(\mathbb{K})$?

The projective transformations of $\mathbb{P}^n$ form a group $\mathrm{PGL}_{n+1}(\mathbb{K})$, the *projective general linear group*, which is the quotient of $\mathrm{GL}_{n+1}(\mathbb{K})$ by this kernel.

Example 5.3. A *Möbius transformation* is a rational function $\mathbb{C} \rightarrow \mathbb{C}$ of the form

$$f(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0.$$

If we think of $\mathbb{C}$ as sitting inside $\mathbb{P}^1(\mathbb{C})$ by the map $z \mapsto [z, 1]$, then we can extend f to a map

$$\begin{aligned} \bar{f} : \mathbb{P}^1(\mathbb{C}) &\rightarrow \mathbb{P}^1(\mathbb{C}) \\ [x, y] &\mapsto [ax + by, cx + dy], \end{aligned}$$

since

$$\bar{f}([z, 1]) = [az + b, cz + d] = \left[\frac{az + b}{cz + d}, 1 \right] = [f(z), 1].$$

But then we can write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}),$$

and $\bar{f}$ is simply the projective transformation Φ_A . In other words, Möbius transformations of $\mathbb{C}$ are the same as projective transformations of $\mathbb{P}^1(\mathbb{C})$.

Building on this example, we have an explicit Möbius transformation that sends any three distinct $w_1, w_2, w_3 \in \mathbb{C}$ to $0, 1, \infty$ respectively, namely

$$f(z) = \frac{z - w_1}{z - w_3} \cdot \frac{w_2 - w_3}{w_2 - w_1}.$$

We can turn this into a projective transformation, resulting in the following.

Lemma 5.4. *Given three distinct points $w_1, w_2, w_3 \in \mathbb{P}^1$, there is a unique projective transformation Φ such that $\Phi(w_1) = [0, 1]$, $\Phi(w_2) = [1, 1]$, and $\Phi(w_3) = [1, 0]$.*

Proof. We write $w_i = [a_i, b_i]$ for each i ; for any two indices $i \neq j$, the condition $w_i \neq w_j$ means that $a_i b_j \neq a_j b_i$. With this in mind, we can explicitly check that

$$\Phi([x, y]) = [(a_2 b_3 - a_3 b_2)(b_1 x - a_1 y), (a_2 b_1 - a_1 b_2)(b_3 x - a_3 y)]$$

is a projective transformation of the desired form.

Now suppose that we have two such projective transformations Φ and Φ' . Then the composition $\Phi' \circ \Phi^{-1}$ is a projective transformation that fixes each of $[0, 1]$, $[1, 1]$, and $[1, 0]$. It can be written as Φ_A for some $A \in \text{GL}_2(\mathbb{K})$, and then each of the fixed points gives rise to eigenvectors $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, and $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ of A . This is only possible if all three belong to the same eigenspace, which is then all of $\mathbb{K}^2$, so we have $A = \lambda I$ for some $\lambda \in \mathbb{K}^\times$. But then $\Phi_A = \Phi' \circ \Phi^{-1}$ is the identity, so $\Phi' = \Phi$. $\square$

To prove a similar result one dimension higher, we need the following.

Exercise 5.5. Any three points $w_i = [x_i, y_i, z_i]$ of $\mathbb{P}^2$ are collinear (i.e., lie on a single line) if and only if

$$\det \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix} = 0.$$

Proposition 5.6. Let $w_1, w_2, w_3, w_4 \in \mathbb{P}^2$ be four points, no three of which are collinear. Then there is a unique projective transformation $\Phi : \mathbb{P}^2 \rightarrow \mathbb{P}^2$ satisfying

$$\Phi(w_1) = [1, 0, 0], \quad \Phi(w_2) = [0, 1, 0], \quad \Phi(w_3) = [0, 0, 1], \quad \Phi(w_4) = [1, 1, 1].$$

Proof. Fix coordinates $w_i = [x_i, y_i, z_i]$ for each i , and let

$$A = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix}.$$

Then A is invertible by Exercise 5.5, and Φ_A sends $[1, 0, 0]$ to w_1 , $[0, 1, 0]$ to w_2 , and $[0, 0, 1]$ to w_3 . In other words, if $B = A^{-1}$ then

$$\Phi_B(w_1) = [1, 0, 0], \quad \Phi_B(w_2) = [0, 1, 0], \quad \Phi_B(w_3) = [0, 0, 1].$$

It therefore suffices to assume that $w_1 = [1, 0, 0]$, $w_2 = [0, 1, 0]$, and $w_3 = [0, 0, 1]$, and look for a projective transformation that fixes each of these while sending $w_4 = [x_4, y_4, z_4]$ to $[1, 1, 1]$.

We note that by hypothesis w_4 does not lie on the lines through any of the pairs $\{w_2, w_3\}$, $\{w_1, w_3\}$, or $\{w_1, w_2\}$; these are $\{x = 0\}$, $\{y = 0\}$, and $\{z = 0\}$ respectively, so each of x_4 , y_4 , and z_4 is nonzero. We thus take $\Phi = \Phi_C$ where

$$C = \begin{pmatrix} 1/x_4 & 0 & 0 \\ 0 & 1/y_4 & 0 \\ 0 & 0 & 1/z_4 \end{pmatrix};$$

this fixes w_1 , w_2 , and w_3 while sending $w_4 = [x_4, y_4, z_4]$ to $[1, 1, 1]$, as desired.

The uniqueness follows exactly the same lines as in Lemma 5.4, so we leave the details as an exercise. $\square$

Projective transformations preserve the most important properties of projective plane curves.

Proposition 5.7. *Let $C = \{F(x, y, z) = 0\} \subset \mathbb{P}^2$ be a projective plane curve of degree d , and let $\Phi : \mathbb{P}^2 \rightarrow \mathbb{P}^2$ be a projective transformation. Then:*

1. *The image $\Phi(C)$ is also a projective plane curve of degree d ;*
2. *If C_0 is an irreducible component of C , then $\Phi(C_0)$ is an irreducible component of $\Phi(C)$.*
3. *If $p \in C$ is a smooth point, then $\Phi(p)$ is a smooth point of $\Phi(C)$, and their respective tangent lines satisfy $\Phi(T_p C) = T_{\Phi(p)} \Phi(C)$.*

Proof. (1) We write $\Phi = \Phi_A$ for some $A \in \mathrm{GL}_3(\mathbb{K})$, and let $B = A^{-1}$, with entries $B = (b_{ij})$. Then

$$\begin{aligned} [x_0, \dots, x_n] \in \Phi(C) &\iff \Phi_B([x_0, \dots, x_n]) \in C \\ &\iff \left[\sum_{i=0}^n b_{0i}x_i, \dots, \sum_{i=0}^n b_{ni}x_i \right] \in C \\ &\iff F\left(\sum_{i=0}^n b_{0i}x_i, \dots, \sum_{i=0}^n b_{ni}x_i \right) = 0. \end{aligned}$$

This last expression is a polynomial of degree $\leq d$, so $\Phi(C)$ is a plane curve with

$$\deg(\Phi(C)) \leq \deg(C).$$

Now we apply the same argument to show that if $C' = \Phi(C)$ then $\Phi^{-1}(C') = C$ satisfies $\deg(\Phi^{-1}(C')) \leq \deg(C')$, or equivalently $\deg(C) \leq \deg(\Phi(C))$. We conclude that the two degrees are equal.

(2) Suppose that $\Phi(C_0) = C_1 \cup C_2$ for some plane curves C_1 and C_2 . Then $C_0 = \Phi^{-1}(C_1) \cup \Phi^{-1}(C_2)$, and since C_0 is irreducible it follows that one of the plane curves $\Phi^{-1}(C_j)$ must be equal to C_0 , in which case we apply Φ again to get $C_j = \Phi(C_0)$. Thus $\Phi(C_0)$ is irreducible as well.

(3) We saw above that $\Phi(C)$ is defined by a polynomial

$$G(x_0, \dots, x_n) = F \circ \Phi_B(x_0, \dots, x_n),$$

where here we view Φ_B as acting linearly on $\mathbb{K}^{n+1}$ by $\Phi_B(v) = Bv$. The chain rule tells us that for any $p \in \mathbb{K}^{n+1}$ we have

$$\begin{aligned} \begin{bmatrix} G_{x_0}(p) & G_{x_1}(p) & \cdots & G_{x_n}(p) \end{bmatrix} \\ = \begin{bmatrix} F_{x_0}(\Phi_B(p)) & F_{x_1}(\Phi_B(p)) & \cdots & F_{x_n}(\Phi_B(p)) \end{bmatrix} B, \end{aligned}$$

and since B is invertible, it follows that all of the $G_{x_i}(p)$ are zero if and only if all of the $F_{x_i}(\Phi_B(p))$ are zero. In other words, $q = \Phi_B(p)$ is a singular point of C if and only if $\Phi(q) = p$ is a singular point of $\Phi(C)$.

For the last part, we fix a smooth point $p = (p_0, \dots, p_n)$ of $\Phi(C)$ and let $q = \Phi_B(p)$ be the corresponding smooth point of C ; then we can write

$$q = (q_0, \dots, q_n) = \left(\sum_{i=0}^n b_{0i}p_i, \dots, \sum_{i=1}^n b_{ni}p_i \right).$$

The tangent line $T_q C$ is a projective plane curve defined by

$$\sum_{i=0}^n F_{x_i}(q)x_i = 0,$$

so the same argument as above tells us that $\Phi(T_q C)$ is defined by

$$\sum_{i=0}^n F_{x_i}(q) \left(\sum_{j=0}^n b_{ij}x_j \right) = 0.$$

In particular the coefficient of x_j is

$$\sum_{i=0}^n F_{x_i}(\Phi_B(p))b_{ij} = G_{x_j}(p),$$

and so the defining equation for $\Phi(T_q C)$ also defines $T_p \Phi(C)$, as claimed. $\square$

6 Bézout's theorem, part 1

6.1 Resultants

When are two polynomials $f, g \in \mathbb{K}[x]$ relatively prime? We can use the Euclidean algorithm to compute the greatest common divisor $\gcd(f, g)$, but this doesn't lend

itself to a useful description in terms of varieties. Instead we will develop an algebraic tool, the *resultant* of f and g , that answers this question without even needing to work out the greatest common divisor.

We wish to work in slightly greater generality, allowing our coefficients to be a unique factorization domain (which we will abbreviate as “UFD”) rather than a field. The key fact we will need from commutative algebra is that if R is a UFD, then so is the polynomial ring $R[x]$. (This is a corollary of *Gauss's lemma*, which says that if R is a UFD with field of fractions $\mathbb{F}$, then an irreducible polynomial in $R[x]$ remains irreducible in $\mathbb{F}[x]$.) This tells us that $R = \mathbb{K}[x_1, \dots, x_n]$ is a UFD for any field $\mathbb{K}$.

Lemma 6.1. *Let R be a UFD, and let $f, g \in R[x]$ be polynomials of degrees $d \geq 1$ and $e \geq 1$ respectively. Then f and g have a non-constant common factor if and only if there are nonzero polynomials $a, b \in R[x]$ such that*

$$a(x)f(x) + b(x)g(x) = 0$$

with $\deg(a) \leq e - 1$ and $\deg(b) \leq d - 1$.

Proof. Suppose that f and g have a common divisor h of degree $n \geq 1$, and write

$$f(x) = h(x)q(x), \quad g(x) = h(x)r(x).$$

Then $r(x)f(x) - q(x)g(x) = 0$, and we have

$$\begin{aligned} \deg(r) &= e - n \leq e - 1 \\ \deg(-q) &= d - n \leq d - 1, \end{aligned}$$

so we take $(a, b) = (r, -q)$.

Conversely, suppose that we have a and b satisfying the above hypotheses, and consider the irreducible factorization

$$f = c \cdot f_1^{\alpha_1} f_2^{\alpha_2} \dots f_k^{\alpha_k}$$

of f , with $c \in R$ and all of the f_i having positive degree. Since $af = -bg$, the irreducible factorization of bg must contain terms of the form

$$f_1^{\beta_1} f_2^{\beta_2} \dots f_k^{\beta_k}, \quad \beta_i \geq \alpha_i \text{ for } i = 1, \dots, k.$$

This product cannot possibly divide b , because it has total degree

$$\sum_{i=1}^k \beta_i \deg(f_i) \geq \sum_{i=1}^k \alpha_i \deg(f_i) = \deg(f) > \deg(b).$$

But then at least one of the f_i must divide g , so this f_i divides $\gcd(f, g)$ as well. $\square$

Lemma 6.1 tells us that f and g have a common factor if and only if the polynomials

$$x^{e-1}f, x^{e-2}f, \dots, xf, f \quad \text{and} \quad x^{d-1}g, x^{d-2}g, \dots, xg, x$$

are linearly dependent over R : after all, if we write

$$\begin{aligned} a(x) &= a_{e-1}x^{e-1} + a_{e-2}x^{e-2} + \dots + a_1x + a_0 \\ b(x) &= b_{d-1}x^{d-1} + b_{d-2}x^{d-2} + \dots + b_1x + b_0, \end{aligned}$$

then the relation $af + bg = 0$ becomes

$$\sum_{i=0}^{e-1} a_i(x^i f) + \sum_{j=0}^{d-1} b_j(x^j g) = 0.$$

This is a collection of $d+e$ polynomials, each of which has degree at most $d+e-1$ and is thus specified by a vector of its lowest $d+e$ coefficients (of degrees 0 through $d+e-1$), so we can write down a $(d+e) \times (d+e)$ matrix whose rows are precisely these vectors, and then we need only determine whether the rows are linearly independent.

Definition 6.2. Let R be a UFD and consider a pair of polynomials $f, g \in R[x]$, of the form

$$\begin{aligned} f(x) &= a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0, \\ g(x) &= b_e x^e + b_{e-1} x^{e-1} + \dots + b_1 x + b_0 \end{aligned}$$

where $\deg(f) = d \geq 1$ and $\deg(g) = e \geq 1$. The *resultant* of f and g is the determinant of a $(d+e) \times (d+e)$ matrix, whose rows are the coefficients of $x^i f$ ($i = e-1, e-2, \dots, 0$) and of $x^j g$ ($j = d-1, d-2, \dots, 0$):

$$\mathcal{R}_{f,g} = \det \begin{pmatrix} a_d & a_{d-1} & a_{d-2} & \dots & a_0 & & & & & \\ & a_d & a_{d-1} & \dots & a_1 & a_0 & & & & \\ & & a_d & \dots & a_2 & a_1 & a_0 & & & \\ & & & \ddots & & \ddots & \ddots & \ddots & & \\ & & & & a_d & a_{d-1} & a_{d-2} & \dots & a_0 & \\ b_e & b_{e-1} & b_{e-2} & \dots & b_1 & b_0 & & & & \\ & b_e & b_{e-1} & \dots & b_2 & b_1 & b_0 & & & \\ & & b_e & \dots & b_3 & b_2 & b_1 & b_0 & & \\ & & & \ddots & & \ddots & \ddots & \ddots & \ddots & \\ & & & & b_e & b_{e-1} & b_{e-2} & \dots & b_1 & b_0 \end{pmatrix} \in R.$$

Now Lemma 6.1 and the discussion after it immediately tell us the following.

Theorem 6.3

If R is a UFD and $f, g \in R[x]$ are non-constant polynomials, then f and g share a non-constant common factor if and only if $\mathcal{R}_{f,g} = 0$.

Example 6.4. Let $f(x) = x - a$ and $g(x) = x - b$ for some $a, b \in R$. Then

$$\mathcal{R}_{f,g} = \det \begin{pmatrix} 1 & -a \\ 1 & -b \end{pmatrix} = a - b$$

is zero if and only if $a = b$.

Example 6.5. Let $f(x) = x^4 - 1$ and $g(x) = x^3 - 2x^2 + x - 2$. Then

$$\mathcal{R}_{f,g} = \det \begin{pmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 \\ 1 & -2 & 1 & -2 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & -2 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 & -2 \end{pmatrix} = 0,$$

and sure enough, $f(x) = (x - 1)(x + 1)(x^2 + 1)$ and $g(x) = (x - 2)(x^2 + 1)$ have a common factor of $x^2 + 1$.

This last example seems unnecessary, because we could factor the given polynomials by hand with much less trouble than it would take to compute a 7×7 determinant. But in general factorization is a harder problem than computing determinants, and Theorem 6.3 is useful as an algebraic criterion because the resultant is a polynomial in the coefficients of f and g . For example:

Proposition 6.6. Suppose that the fraction field of R has characteristic zero. Then a polynomial $f \in R[x]$ has no repeated factors if and only if $\mathcal{R}_{f,f'} \neq 0$.

Proof. If f is not square-free, then there are polynomials $g, h \in R[x]$, with g not constant, such that we can write $f = g^2h$. Then we compute that $f' = g(2g'h + gh')$, so g divides $\gcd(f, f')$, hence Theorem 6.3 says that $\mathcal{R}_{f,f'} = 0$.

Conversely, if $\mathcal{R}_{f,f'} = 0$ then there must be some polynomials g and h , with g irreducible and not constant, such that $f = gh$ and g also divides f' . If $f' = gh' + g'h$ is a multiple of g then so is $g'h$, so g divides either g' or h ; the former cannot happen since $\deg(g') < \deg(g)$, so in fact $g \mid h$ and thus $g^2 \mid f$. $\square$

Remark 6.7. We need to assume characteristic zero in the last step of the proof of Proposition 6.6, because otherwise it is possible that $g' = 0$.

We can think of affine space $\mathbb{K}^3$ as the space of polynomials over $\mathbb{K}$ of degree at most 2, via the identification

$$(a, b, c) \mapsto f(t) = at^2 + bt + c.$$

Then $f'(t) = 2at + b$, so by Proposition 6.6, the condition that $f(t)$ is square-free is equivalent to

$$0 \neq \mathcal{R}_{f,f'} = \det \begin{pmatrix} a & b & c \\ 2a & b & 0 \\ 0 & 2a & b \end{pmatrix} = 4a^2c - ab^2 = -a(b^2 - 4ac).$$

In other words, if f is quadratic (i.e., $a \neq 0$) then this is equivalent to the familiar condition that the discriminant $b^2 - 4ac$ must be nonzero.

6.2 Weak Bézout

Suppose we have a pair of projective plane curves

$$\begin{aligned} C &= \{F(x, y, z) = 0\} \subset \mathbb{P}^2 \\ C' &= \{G(x, y, z) = 0\} \subset \mathbb{P}^2, \end{aligned}$$

where F and G are homogeneous of degrees $d \geq 1$ and $e \geq 1$ respectively. We can think of these as polynomials in x over the unique factorization domain $R = \mathbb{K}[y, z]$, writing

$$\begin{aligned} F(x, y, z) &= f_0(y, z)x^d + f_1(y, z)x^{d-1} + \cdots + f_d(y, z) \\ G(x, y, z) &= g_0(y, z)x^e + g_1(y, z)x^{e-1} + \cdots + g_e(y, z) \end{aligned}$$

where each f_i or g_i is homogeneous of degree i . Here the leading coefficients satisfy $f_0(y, z) = F(1, 0, 0)$ and $g_0(y, z) = G(1, 0, 0)$, so if we assume that these are nonzero then F and G have the same degree over $\mathbb{K}$ as they do over $\mathbb{K}[y, z]$. Equivalently, we will assume that

$$[1, 0, 0] \notin C \quad \text{and} \quad [1, 0, 0] \notin C',$$

so that $\deg(F) = d$ and $\deg(G) = e$ in either case.

Proposition 6.8. *Let $F, G \in \mathbb{K}[x, y, z]$ be homogeneous polynomials of degrees $d \geq 1$ and $e \geq 1$ respectively, and suppose that $F(1, 0, 0)$ and $G(1, 0, 0)$ are both nonzero. Viewing these as polynomials in x over $\mathbb{K}[y, z]$, we can then define*

$$\mathcal{R}_{F,G}(y, z) \in \mathbb{K}[y, z]$$

as in Definition 6.2. Then there are polynomials $A, B \in \mathbb{K}[x, y, z]$, with x -degrees $\deg_x(A) \leq e - 1$ and $\deg_x(B) \leq d - 1$, such that

$$\mathcal{R}_{F,G}(y, z) = AF + BG.$$

If $\mathcal{R}_{F,G}(y, z)$ is nonzero, then it is a homogeneous polynomial of degree de .

Proof. We write $\mathcal{R}_{F,G}(y, z) = \det M$, where M is the $(d + e) \times (d + e)$ matrix in Definition 6.2. We build a new matrix N with entries in $\mathbb{K}[x, y, z]$ by taking M and then performing a series of column operations: for each $j = 1, 2, \dots, d + e - 1$, we add x^{d+e-j} times the j th column to the $(d + e)$ th column. The last column of N is then

$$\begin{pmatrix} f_0x^{d+e-1} + f_1x^{d+e-2} + \dots + f_dx^{e-1} \\ f_0x^{d+e-2} + f_1x^{d+e-3} + \dots + f_dx^{e-2} \\ \vdots \\ f_0x^d + f_1x^{d-1} + \dots + f_d \\ g_0x^{d+e-1} + g_1x^{d+e-2} + \dots + g_ex^{d-1} \\ g_0x^{d+e-2} + g_1x^{d+e-3} + \dots + g_ex^{d-2} \\ \vdots \\ g_0x^e + g_1x^{e-1} + \dots + g_e \end{pmatrix} = \begin{pmatrix} x^{e-1}F \\ x^{e-2}F \\ \vdots \\ F \\ x^{d-1}G \\ x^{d-2}G \\ \vdots \\ G \end{pmatrix},$$

while the first $d + e - 1$ columns of N are the same as for M . Since column operations don't change the determinant, we have $\det(M) = \det(N)$. Expanding by minors along the last column of N expresses $\det(N)$ as a sum whose terms are products of one of the entries

$$x^{e-1}F, \dots, xF, F, x^{d-1}G, \dots, xG, G$$

with elements of $\mathbb{K}[y, z]$. Thus $\mathcal{R}_{F,G}(y, z) = \det(M) = \det(N)$ is a $\mathbb{K}[y, z]$ -linear combination of the above entries, meaning we can write it as $AF + BG$ with $\deg_x(A) \leq e - 1$ and $\deg_x(B) \leq d - 1$ as claimed.

Now to show that $\mathcal{R}_{F,G}(y, z)$ is homogeneous, we write $M = (a_{ij})$ and apply the identity

$$\det M = \sum_{\sigma \in S_{d+e}} (-1)^\sigma a_{1,\sigma(1)} a_{2,\sigma(2)} \dots a_{d+e,\sigma(d+e)}.$$

In particular, it suffices to show that each nonzero summand is homogeneous of degree de .

To do so, we write such a term (up to sign) as

$$A_\sigma = \prod_{i=1}^e a_{i,\sigma(i)} \prod_{j=1}^d a_{j+e,\sigma(j+e)}.$$

For $1 \leq i \leq e$, if the entry $a_{i,\sigma(i)}$ is nonzero then it satisfies

$$a_{i,\sigma(i)} = f_{\sigma(i)-i}(y, z),$$

which is homogeneous of degree $\sigma(i) - i$. Similarly, if $a_{j+e,\sigma(j+e)} \neq 0$ where $1 \leq j \leq d$ then

$$a_{j+e,\sigma(j+e)} = g_{\sigma(j+e)-j}$$

which is homogeneous of degree $\sigma(j+e) - j$. Thus if A_σ is nonzero then we have

$$\begin{aligned} \deg(A_\sigma) &= \sum_{i=1}^e (\sigma(i) - i) + \sum_{j=1}^d (\sigma(j+e) - j) \\ &= \sum_{i=1}^{d+e} \sigma(i) - \sum_{i=1}^d i - \sum_{i=1}^e i \\ &= \frac{(d+e)(d+e+1)}{2} - \frac{d(d+1)}{2} - \frac{e(e+1)}{2}, \end{aligned}$$

which is equal to de . □

We should also consider the circumstances under which $\mathcal{R}_{F,G}(y, z)$ is identically zero.

Proposition 6.9. *If $F, G \in \mathbb{K}[x, y, z]$ are homogeneous and $F(1, 0, 0)$ and $G(1, 0, 0)$ are both nonzero, then $\mathcal{R}_{F,G}(y, z)$ is the zero polynomial if and only if F and G share a common non-constant factor in $\mathbb{K}[x, y, z]$.*

Proof. We observe that a divisor $H(x, y, z)$ of $F(x, y, z)$ is non-constant if and only if it has positive x -degree. Indeed, if we could write

$$F(x, y, z) = H(y, z)Q(x, y, z)$$

then H and Q would both be homogeneous, and then

$$H(0, 0)Q(1, 0, 0) = F(1, 0, 0) \neq 0 \implies H(0, 0) \neq 0$$

tells us that $H(y, z)$ must be homogeneous of degree 0, hence constant.

Now Theorem 6.3 says that $\mathcal{R}_{F,G} = 0$ if and only if we can write

$$F(x, y, z) = h(x)q(x), \quad G(x, y, z) = h(x)r(x)$$

for some polynomials h, q, r with coefficients in $\mathbb{K}[y, z]$ such that $\deg(h) > 0$. This is equivalent to the existence of $H, Q, R \in \mathbb{K}[x, y, z]$ satisfying

$$\begin{aligned} F(x, y, z) &= H(x, y, z)Q(x, y, z), \\ G(x, y, z) &= H(x, y, z)R(x, y, z) \end{aligned}$$

and $\deg_x(H) > 0$. By the above discussion we can replace $\deg_x(H) > 0$ with the assertion that H is not constant, since these are equivalent for any divisor H of F , so we conclude that $\mathcal{R}_{F,G} = 0$ if and only if F and G have a non-constant common divisor. $\square$

We have assumed in our discussion of $\mathcal{R}_{F,G}$ that $F(1, 0, 0)$ and $G(1, 0, 0)$ are nonzero, or equivalently that the projective plane curves $\{F = 0\}$ and $\{G = 0\}$ both avoid the point $[1, 0, 0]$. We can always arrange this by finding a suitable projective transformation, because no finite collection of plane curves can cover all of $\mathbb{P}^2$:

Lemma 6.10. *Let $\mathbb{K}$ be algebraically closed, and let $C_1, \dots, C_n \subset \mathbb{P}^2(\mathbb{K})$ be projective plane curves. Then the union $C_1 \cup \dots \cup C_n$ is a proper subset of $\mathbb{P}^2(\mathbb{K})$.*

Proof. If we write $C_i = \{F_i(x, y, z) = 0\}$ for each i , then the union $C_1 \cup \dots \cup C_n$ is the zero set of

$$G(x, y, z) = \prod_{i=1}^n F_i(x, y, z).$$

If the plane curve $\{G = 0\}$ is all of $\mathbb{P}^2$, then for each $c \in \mathbb{K}$ it contains the projective line $\{x - cy = 0\}$, hence by Proposition 3.13 the irreducible polynomial $x - cy$ must divide $G(x, y, z)$. But this is impossible, since the irreducible factorization of G has only finitely many distinct factors. $\square$

We are ready to prove a partial version of Bézout's theorem.

Theorem 6.11: Weak Bézout theorem

Let $\mathbb{K}$ be an algebraically closed field, and let $C, C' \subset \mathbb{P}^2(\mathbb{K})$ be projective plane curves of degrees $d \geq 1$ and $e \geq 1$ respectively. Then

1. The intersection $C \cap C'$ is nonempty.
2. Either $|C \cap C'| \leq de$, or $C \cap C'$ contains a plane curve.

Proof. Write $C = \{F(x, y, z) = 0\}$ and $C' = \{G(x, y, z) = 0\}$, where $\deg(F) = d$ and $\deg(G) = e$. By Lemma 6.10 the union $C \cup C'$ must avoid some point of $\mathbb{P}^2$, so after applying a projective transformation we can assume that neither C nor C' contains the point $[1, 0, 0]$.

Now Proposition 6.8 says that

$$\mathcal{R}_{F,G}(y, z) \in \mathbb{K}[y, z]$$

is either zero or homogeneous of degree de . If it is zero, then we know from Proposition 6.9 that there is some non-constant

$$H(x, y, z) \in \mathbb{K}[x, y, z]$$

such that H divides both F and G , and in particular the plane curve $\{H = 0\} \subset \mathbb{P}^2$ lies in the intersection $C \cap C'$. From now on we will assume that $\mathcal{R}_{F,G}(y, z)$ is not identically zero.

Since the resultant is nonzero, we can use Proposition 3.11 (and the assumption that $\mathbb{K}$ is algebraically closed) to factor it: there are de pairs $(a_i, b_i) \neq (0, 0)$ such that

$$\mathcal{R}_{F,G}(y, z) = \prod_{i=1}^{de} (a_i y + b_i z).$$

At any fixed $(y, z) = (y_0, z_0)$, Theorem 6.3 says that the polynomials

$$F(x, y_0, z_0), G(x, y_0, z_0) \in \mathbb{K}[x]$$

have a common root $x = x_0$ if and only if $\mathcal{R}_{F,G}(y_0, z_0) = 0$, which is equivalent to $a_i y_0 + b_i z_0 = 0$ for some i . This means that there is $x_0 \in \mathbb{K}$ such that

$$[x_0, y_0, z_0] \in C \cap C'$$

if and only if $[y_0, z_0] = [b_i, -a_i]$ for some $i \in \{1, \dots, de\}$. This proves that $C \cap C'$ is nonempty.

It remains to be seen that $|C \cap C'| \leq de$, so let's suppose instead that we have a set $S \subset C \cap C'$ of $de + 1$ distinct points. We apply Lemma 6.10 to see that some point of $\mathbb{P}^2$ does not lie on any of C, C' , or the projective lines through any pair of points of S ; thus after a projective transformation we may assume that none of these passes through $[1, 0, 0]$. As above, we know that $[x, y, z] \in C \cap C'$ implies that $(y, z) \neq (0, 0)$ and that $a_i y + b_i z = 0$ for some i between 1 and de ; in particular, there must be two distinct points

$$[x, y, z], [x', y', z'] \in S$$

corresponding to the same index i . Then the unique line through these two points is $\{a_i y + b_i z = 0\}$, but this contains $[1, 0, 0]$ and we have a contradiction. $\square$

In fact, the proof of Theorem 6.11 tells us slightly more: if C and C' don't have any common components, then $C \cap C'$ contains exactly de points when counted appropriately, in a way that we will make precise later. Specifically, after applying the last projective transformation, we should count each point $[x, y, z] \in C \cap C'$ according to the multiplicity of $[y, z]$ as a root of $\mathcal{R}_{F,G}(y, z)$, and then $\mathcal{R}_{F,G}(y, z)$ has exactly de roots when counted with multiplicity. In particular, if $\mathcal{R}_{F,G}(y, z)$ has no repeated factors then $C \cap C'$ consists of distinct points.

Example 6.12. Let $C = \{xz = y^2\}$ and $C' = \{x = 0\}$. These have degrees 2 and 1 respectively, but

$$C \cap C' = \{[0, 0, 1]\}$$

consists of a single point, and indeed we can check that $\mathcal{R}_{xz-y^2,x}(y, z) = y^2$.

7 Applications to curves of low degree

Theorem 6.11 provides the missing ingredient in the proof of Theorem 4.15, which asserted that if $\mathbb{K}$ is algebraically closed then a smooth projective plane curve over $\mathbb{K}$ must be irreducible. Indeed, if $C \subset \mathbb{P}^2$ is reducible, we can take two different irreducible components $C_1, C_2 \subset C$ and use Bézout's theorem to find some point $p \in C_1 \cap C_2$; then we showed in Proposition 4.14 that p is not a smooth point of C .

The converse is not true: an irreducible curve can have singularities, such as the curve $\{x^2z - y^3 = 0\}$ at $[0, 0, 1]$, but at least it does not have too many singularities.

Proposition 7.1. *Let $\mathbb{K}$ be algebraically closed of characteristic zero, and suppose that $C \subset \mathbb{P}^2(\mathbb{K})$ is irreducible. Then C has finitely many singular points.*

Proof. Let C be defined by an irreducible homogeneous polynomial $F(x, y, z)$ of degree d . At least one of the partial derivatives F_x , F_y , or F_z cannot be identically zero, or else (since $\text{char}(\mathbb{K}) = 0$) F would have been constant; suppose without loss of generality that $F_x \neq 0$. Then F_x is nonzero and homogeneous of degree at most $d - 1$, and moreover F and F_x have no non-constant common factors, since F is irreducible and $\deg(F_x) < \deg(F)$.

Now any singular points of C must satisfy $F = 0$ and $F_x = 0$ (among other conditions), but the intersection

$$\{F(x, y, z) = 0\} \cap \{F_x(x, y, z) = 0\} \subset \mathbb{P}^2$$

does not contain any plane curves – these would correspond to common divisors of F and F_x – so Bézout's theorem says that it contains at most $d(d - 1)$ points. $\square$

We now pursue some more applications of Bézout's theorem, focusing on plane curves of degree at most 3.

7.1 Smooth conics

A *conic* is a projective plane curve of degree 2. Over $\mathbb{R}$ there are several different types of irreducible conics, notably ellipses, parabolas, and hyperbolas (see Exam-

ple 1.9), but over $\mathbb{C}$ they all become projectively equivalent.

Theorem 7.2

Let $\mathbb{K}$ be algebraically closed, and let $C \subset \mathbb{P}^2(\mathbb{K})$ be an irreducible conic. Then C is projectively equivalent to the conic $\{x^2 = yz\}$, and in particular C is nonsingular.

Proof. We know from Proposition 7.1 that all but finitely many points of C are non-singular, so we may take a smooth point $p \in C$ and another point $q \in T_p C$, and apply a projective transformation sending p to $[1, 0, 0]$ and q to $[0, 1, 0]$. Replacing C with its image, we now have $[1, 0, 0] \in C$ and

$$T_{[1,0,0]}C = \{z = 0\}$$

since Proposition 5.7 says that $T_{[1,0,0]}C$ must be the unique line through the images of p and q .

If we write

$$C = \{F(x, y, z) = ax^2 + bxy + cy^2 + dxz + eyz + fz^2 = 0\},$$

then $[1, 0, 0] \in C$ implies that $a = 0$, and description of $T_{[1,0,0]}C$ tells us that

$$F_y(1, 0, 0) = 0 \implies b = 0,$$

so the defining polynomial actually has the form

$$F(x, y, z) = cy^2 + dxz + eyz + fz^2.$$

Moreover, if C is irreducible then F must be irreducible, so $c \neq 0$ (since $z \mid F$ otherwise) and $d \neq 0$ (since F factors into linear terms otherwise).

We now observe that we can write

$$F(x, y, z) = (\sqrt{c}y)^2 + (dx + ey + fz)z,$$

so the map

$$(x, y, z) \mapsto (\sqrt{c}y, dx + ey + fz, -z)$$

takes the conic $\{x^2 - yz = 0\}$ to C . This defines a projective transformation since c and d are nonzero, so C is projectively equivalent to $\{x^2 = yz\}$ as claimed. $\square$

Corollary 7.3. *A conic over an algebraically closed field is smooth if and only if it is irreducible.*

Exercise 7.4. If C is a reducible conic over an algebraically closed field, defined by a degree-2 polynomial with no repeated factors, then it must be a union of two lines. Show that C is projectively equivalent to $\{F(x, y, z) = x^2 + y^2 = 0\}$.

We also note the following fact about conics.

Proposition 7.5. *Any five points in $\mathbb{P}^2$ lie on a conic. If no three of the five points are collinear, then this conic is unique.*

Proof. We write the equation of a general conic C as

$$ax^2 + bxy + cy^2 + dxz + eyz + fz^2 = 0, \quad a, b, c, d, e, f \in \mathbb{K}.$$

Any point $[x_i, y_i, z_i] \in C$ imposes a linear relation on a, b, c, d, e, f by substituting $(x, y, z) = (x_i, y_i, z_i)$ into this equation. Five points of $\mathbb{P}^2$ determine a system of five linear equations in these six variables, which therefore has at least one nonzero solution (a, b, c, d, e, f) , and the corresponding conic contains all five points.

Now suppose that the points $p_1, \dots, p_5 \in \mathbb{P}^2(\mathbb{K})$ lie on two different conics, say C and C' . If $\mathbb{K}$ is not algebraically closed then the defining equations for C and C' also define conics over the algebraic closure $\overline{\mathbb{K}}$ that contain the various p_i , and a given subset of the p_i is collinear in $\mathbb{P}^2(\mathbb{K})$ if and only if it is collinear in $\mathbb{P}^2(\overline{\mathbb{K}})$, so we may as well assume in what follows that $\mathbb{K}$ is algebraically closed.

The intersection $C \cap C'$ has more than $\deg(C) \cdot \deg(C') = 4$ points, so by Bézout's theorem it must contain some irreducible curve C_0 . Then C_0 must be a line, because if it had degree 2 then it would be equal to both C and C' , which are not the same curve. Since C has degree 2 and contains the line C_0 , we must have

$$C = C_0 \cup L$$

where L is also a line. One of these two lines must contain at least three of the p_i , so the non-uniqueness of C implies that three of the p_i are collinear. $\square$

7.2 Intersections of cubics

In this subsection we will repeatedly apply the weak Bézout theorem, Theorem 6.11, to prove the following fact about curves of degree 3, also known as cubics. It asserts that eight points $p_1, \dots, p_8 \in \mathbb{P}^2$ in “general position” uniquely determine a ninth point p_9 such that any plane cubic through $p_1, \dots, p_8$ must also pass through p_9 .

Theorem 7.6: Cayley–Bacharach theorem

Let $\mathbb{K}$ be algebraically closed, and suppose that two projective cubics

$$C = \{F(x, y, z) = 0\}, \quad C' = \{G(x, y, z) = 0\}$$

intersect in exactly nine distinct points $p_1, p_2, \dots, p_9$. If $C'' \subset \mathbb{P}^2$ is a projective cubic through $p_1, \dots, p_8$, then C'' belongs to the *pencil* of cubics

$$\left\{ C_{[a,b]} = \{aF + bG = 0\} \mid [a, b] \in \mathbb{P}^1 \right\}.$$

In particular, C'' also contains p_9 .

We will break the proof of Theorem 7.6 into a series of lemmas, each of which places restrictions on the number of p_i that can lie on a given curve of degree 1 or 2.

Lemma 7.7. *No four of the p_i can be collinear.*

Proof. Supposing otherwise, let L be a line through four of the p_i . Then $|C \cap L| > 3$ and L is irreducible, so Bézout’s theorem says that $L \subset C$; we have $L \subset C'$ by the same argument. But then $C \cap C'$ contains the entire line L , contradiction. $\square$

Lemma 7.8. *No seven of the p_i lie on a single conic.*

Proof. If some conic $D \subset \mathbb{P}^2$ contains seven of the p_i , then Bézout’s theorem tells us once again that both $C \cap D$ and $C' \cap D$ contain a plane curve. If D is irreducible then $C \cap C'$ contains all of D , again a contradiction. Thus D must be a union $L \cup L'$ of two lines, and collectively these contain all seven of the p_i lying on D . But then either L or L' contains at least four of the p_i , contradicting Lemma 7.7. $\square$

Lemma 7.9. *Any five of the p_i determine a unique conic.*

Proof. If no three of the five are collinear, then this is Proposition 7.5. Assuming that three of the five *do* lie on some line L , Lemma 7.7 says that L can’t contain either of the remaining two. Now if L' is the unique line through those remaining two points, then $D = L \cup L'$ is a conic containing all five of the chosen points.

If D' is any other conic through these five points, then Bézout’s theorem says that $D' \cap D = D' \cap (L \cup L')$ must contain either L or L' , so D' is also a union of two lines. But $L \cap L'$ is the only union of two lines containing at least one of L and L' as well as all five of the chosen p_i – by Lemma 7.7, at least two of the five are not on L , and likewise for L' – so in fact $D' = L \cup L' = D$, proving that D is unique. $\square$

Lemma 7.10. *If $C'' = \{H(x, y, z) = 0\}$ is a plane cubic passing through $p_1, \dots, p_8$, and H is not a linear combination of F and G , then no three of $p_1, \dots, p_8$ are collinear.*

Proof. Suppose without loss of generality that p_1, p_2, p_3 all lie on some line L . By Lemma 7.7, none of $p_4, \dots, p_8$ belong to L . On the other hand, Lemma 7.9 says that there is a unique conic D containing $p_4, \dots, p_8$, so $L \cup D$ is a plane cubic containing all of $p_1, \dots, p_8$. Let $q_1 \in L$ be a point different from the p_i , and take another point $q_2 \notin L \cup D$.

Now we take the defining polynomials F, G, H of C, C', C'' and write

$$P(x, y, z) = aF(x, y, z) + bG(x, y, z) + cH(x, y, z)$$

for some $a, b, c \in \mathbb{K}$. The conditions $q_1, q_2 \in \{P(x, y, z) = 0\}$ impose two linear equations on a, b, c , so there must be some nonzero (a, b, c) for which $\{P = 0\}$ contains both q_1 and q_2 . This is a cubic passing through the four points of $p_1, p_2, p_3, q_1 \in L$, so Bézout's theorem says that $L \subset \{P = 0\}$. Then $\{P = 0\}$ is the union of L and a conic through $p_4, \dots, p_8$ (none of which lie on L). Since D is the unique such conic, the zero set of P is precisely $L \cup D$. But $\{P = 0\}$ contains the point q_2 whereas $L \cup D$ does not, and we have a contradiction. $\square$

Lemma 7.11. *If $C'' = \{H(x, y, z) = 0\}$ is a plane cubic passing through $p_1, \dots, p_8$, and H is not a linear combination of F and G , then no six of $p_1, \dots, p_8$ lie on a single conic.*

Proof. Suppose that $p_1, \dots, p_6$ lie on a conic D , and let L be the line through p_7 and p_8 . Then Lemma 7.10 says that D cannot be a union of two lines, because one of them would have to contain at least three of $p_1, \dots, p_6$, so D is irreducible. It also says that none of $p_1, \dots, p_6$ lie on L .

Just as in Lemma 7.10, we take a point $q_1 \in D$ different from $p_1, \dots, p_6$, and another point $q_2 \notin D \cup L$, and the same argument says that we can find $(a, b, c) \neq (0, 0, 0)$ such that the zero locus of the cubic

$$P(x, y, z) = aF(x, y, z) + bG(x, y, z) + cH(x, y, z)$$

contains both q_1 and q_2 . Now $\{P = 0\} \cap D$ has at least the seven points $p_1, \dots, p_6, q_1$, hence by Bézout's theorem the cubic curve $\{P = 0\}$ contains all of the conic D . Its other component is a line through p_7 and p_8 , which must be L , so $\{P = 0\}$ is the union $D \cup L$. The first of these contains q_2 but the second does not, contradiction. $\square$

Proof of Theorem 7.6. Suppose once again that the conic C'' contains each of the points $p_1, \dots, p_8$, but that its defining polynomial $H(x, y, z)$ is not a linear combination of F and G . We start by letting L be the unique line through p_1 and p_2 , which does not contain any of $p_3, \dots, p_8$ by Lemma 7.10. We then let D be the conic through $p_3, \dots, p_7$, which by Lemma 7.9 is also unique; then none of p_1, p_2, p_8 belong to D , by Lemma 7.11, and in particular $L \not\subset D$.

We let $q_1, q_2 \notin \{p_1, p_2\}$ be points of L that do not lie on D , and find once again some triple $(a, b, c) \neq (0, 0, 0)$ such that the zero locus of

$$P(x, y, z) = aF(x, y, z) + bG(x, y, z) + cH(x, y, z)$$

contains both q_1 and q_2 . Then $\{P = 0\}$ is a cubic meeting L at four points, namely p_1, p_2, q_1, q_2 , so by Bézout's theorem the cubic $\{P = 0\}$ must contain L : it is therefore the union of L and a conic D' . But it also contains each of the five points $p_3, \dots, p_7$, which do not lie on L , so then D' contains each of $p_3, \dots, p_7$ and hence $D' = D$. In other words, we have $\{P = 0\} = L \cup D'$, but the left side contains p_8 while the right side does not, contradiction.

We have now shown that $H = sF + tG$ for some pair $(s, t) \neq (0, 0)$. Since each of F and G vanish at p_9 , it follows that H does as well and so $p_9 \in C''$. $\square$

The Cayley–Bacharach theorem allows us to recover the following classical result.

Theorem 7.12: Pascal's theorem

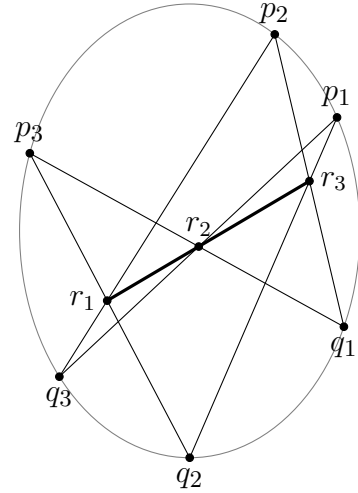
Let $C \subset \mathbb{P}^2$ be a conic, and choose six points $p_1, p_2, p_3, q_1, q_2, q_3 \in C$, no three of which are collinear. Let L_{ij} be the line through p_i and q_j , and let

$$r_1 = L_{23} \cap L_{32},$$

$$r_2 = L_{13} \cap L_{31},$$

$$r_3 = L_{12} \cap L_{21}.$$

Then the points r_1, r_2 , and r_3 are collinear.



A special case is known as Pascal's *hexagrammum mysticum* (mystical hexagon) theorem: we take the hexagon formed by six points on the conic C , no three of them collinear, and let r_1, r_2, r_3 be the points where the lines through pairs of opposite sides intersect. We let $p_1, p_2, p_3, q_1, q_2, q_3$ be the vertices of the hexagon in cyclic order, and then Theorem 7.12 tells us that r_1, r_2, r_3 are collinear.

Proof of Theorem 7.12. We note that the points r_i cannot be equal to any of the p_j

or q_j , by the assumption that no three of the six chosen points are collinear. We may as well also assume that the r_i are all distinct, because otherwise there is nothing to show.

The cubic plane curves

$$L_{12} \cup L_{23} \cup L_{31} \quad \text{and} \quad L_{13} \cup L_{21} \cup L_{32}$$

intersect at exactly the nine points $p_1, p_2, p_3, q_1, q_2, q_3, r_1, r_2, r_3$. If L is the line through r_1 and r_2 then the cubic

$$C \cup L$$

contains eight of these nine points, namely all of the p_i and q_i plus r_1 and r_2 , so by Theorem 7.6 it must contain the ninth point r_3 as well: either $r_3 \in C$ or $r_3 \in L$.

If we suppose that $r_3 \in C$, then by definition r_3 is a point of both the line L_{12} and the conic C , so these intersect in at least p_1, q_2 , and r_3 . But then by Bézout's theorem the line L_{12} must be an irreducible component of C , so C is a union of two lines, and this is impossible because one of these lines must then contain at least three of the points $p_1, p_2, p_3, q_1, q_2, q_3$. We must therefore have $r_3 \in L$, so now the line L contains all three of the points r_i , meaning that they are collinear. $\square$

8 Bézout's theorem, part 2

8.1 More on resultants

We have seen that the resultant of two polynomials $f, g \in R[x]$, where R is a UFD, vanishes precisely when they share a common factor. When f and g split into linear factors, we can make this more precise by expressing the resultant explicitly in terms of the roots of f and g , and this will allow us to prove some useful identities.

Proposition 8.1. *Suppose that the non-constant polynomials $f, g \in R[x]$ factor into linear terms, as*

$$\begin{aligned} f(x) &= a(x - \lambda_1)(x - \lambda_2) \dots (x - \lambda_d) \\ g(x) &= b(x - \mu_1)(x - \mu_2) \dots (x - \mu_e) \end{aligned}$$

where $d, e > 0$. Then the resultant of f and g is given by

$$\mathcal{R}_{f,g} = a^e b^d \prod_{\substack{1 \leq i \leq d \\ 1 \leq j \leq e}} (\lambda_i - \mu_j),$$

so that in particular we have

$$\mathcal{R}_{f,g} = a^e \prod_{i=1}^d (b(\lambda_i - \mu_1) \dots (\lambda_i - \mu_e)) = a^e \prod_{i=1}^d g(\lambda_i).$$

Proof. We may as well assume that f and g are monic, because it follows directly from the definition that $\mathcal{R}_{af,bg} = a^e b^d \cdot \mathcal{R}_{f,g}$ for any $a, b \in R$.

We now start by treating f and g as polynomials not over R , but over the UFD

$$S = R[y_1, \dots, y_d, z_1, \dots, z_e],$$

with $\bar{f} = \prod (x - y_i)$ and $\bar{g} = \prod (x - z_j)$. The homomorphism $S \rightarrow R$ defined by sending each $y_i \mapsto \lambda_i$ and $z_j \mapsto \mu_j$ extends to a homomorphism $S[x] \rightarrow R[x]$ sending $\mathcal{R}_{\bar{f},\bar{g}} \in S[x]$ to $\mathcal{R}_{f,g} \in R[x]$, so we'll compute $\mathcal{R}_{\bar{f},\bar{g}}$ instead. In this setup the polynomials $\bar{f}$ and $\bar{g}$ are homogeneous in the variables x, y_i, z_j , so the proof of Proposition 6.8 tells us that

$$\mathcal{R}_{\bar{f},\bar{g}} \in R[y_1, \dots, y_d, z_1, \dots, z_e]$$

is homogeneous in the y_i and z_j , of degree de .

Now for any $i = 1, \dots, d$ and $j = 1, \dots, e$, we know that if we set $y_i = z_j$ then $\bar{f}$ and $\bar{g}$ share a non-constant (in x) common factor, namely $x - y_i = x - z_j$. This means that if $y_i - z_j = 0$ then $\mathcal{R}_{\bar{f},\bar{g}} = 0$, and so

$$(y_i - z_j) \mid \mathcal{R}_{\bar{f},\bar{g}}.$$

As i and j vary we get de different linear factors, none of which divides any other, and so their product divides $\mathcal{R}_{\bar{f},\bar{g}}$; now by comparing degrees we must have

$$\mathcal{R}_{\bar{f},\bar{g}} = c \prod_{\substack{1 \leq i \leq d \\ 1 \leq j \leq e}} (y_i - z_j)$$

for some $c \in R$.

To determine the constant c , we apply the homomorphism $S \rightarrow R$ that sends each $y_i \mapsto 1$ and each $z_j \mapsto 0$, and we see that

$$c = \mathcal{R}_{(x-1)^d, x^e}.$$

This is by definition the determinant of a block matrix of the form

$$c = \det \left(\begin{array}{c|cccc} * & (-1)^d & 0 & 0 & \dots & 0 \\ * & * & (-1)^d & 0 & \dots & 0 \\ * & * & * & (-1)^d & \dots & 0 \\ * & \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & * & \dots & (-1)^d \\ \hline I_d & 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

where the top right block is $e \times e$, and the ‘*’ entries in the top left and the ‘0’ entries in the bottom right are all $1 \times d$ and $d \times 1$ respectively. If we move the bottom d rows collectively to be above the top e rows, then this matrix becomes lower triangular and we conclude that

$$(-1)^{de} \cdot c = 1^d \cdot ((-1)^d)^e = (-1)^{de},$$

or $c = 1$. □

Proposition 8.1 gives us a straightforward way to deduce some useful new properties of resultants, such as the following.

Proposition 8.2. *Let R be a UFD, and let $g \in R[x]$ be a non-constant polynomial. Then*

$$\mathcal{R}_{x-c,g} = g(c)$$

for any $c \in R$.

Proof. Let $\mathbb{K}$ be an algebraic closure of the fraction field of R . In $\mathbb{K}[x]$ the polynomial g splits into linear factors, so we write $f = x - c$ and apply Proposition 8.1 to get

$$\mathcal{R}_{f,g} = g(c)$$

as an element of $\mathbb{K}$. But by definition this resultant is the same whether we treat f and g as elements of $R[x]$ or of $\mathbb{K}[x]$, so this identity holds equally well in R . □

Proposition 8.3. *Let R be a UFD, and let $f, g, h \in R[x]$ be non-constant polynomials. Then*

$$\mathcal{R}_{f,gh} = \mathcal{R}_{f,g} \cdot \mathcal{R}_{f,h}.$$

Proof. Again we let $\mathbb{K}$ be an algebraic closure of the fraction field of R , and in $\mathbb{K}[x]$ we can write

$$f = a(x - \lambda_1) \dots (x - \lambda_d)$$

and likewise for g and h . Applying Proposition 8.1 several times tells us that in $\mathbb{K}$ we have the identity

$$\begin{aligned} \mathcal{R}_{f,gh} &= a^{\deg(gh)} \prod_{i=1}^d (gh)(\lambda_i) \\ &= \left(a^{\deg(g)} \prod_{i=1}^d g(\lambda_i) \right) \left(a^{\deg(h)} \prod_{i=1}^d h(\lambda_i) \right) \\ &= \mathcal{R}_{f,g} \cdot \mathcal{R}_{f,h}. \end{aligned}$$

These three resultants are the same whether computed in $R[x]$ or in $\mathbb{K}[x]$, so this identity holds equally well in R . $\square$

Proposition 8.4. *Let R be a UFD, and let $f, g, h \in R[x]$ be polynomials so that f , g , and $hf + g$ all have positive degree. If f is monic then $\mathcal{R}_{f,g} = \mathcal{R}_{f,hf+g}$.*

Proof. Again we pass to an algebraic closure $\mathbb{K}$ of the fraction field of R , so that we can write

$$f(x) = (x - \lambda_1) \dots (x - \lambda_d),$$

and then Proposition 8.1 says that

$$\begin{aligned} \mathcal{R}_{f,hf+g} &= \prod_{i=1}^d (hf + g)(\lambda_i) \\ &= \prod_{i=1}^d \left(h(\lambda_i) \underbrace{f(\lambda_i)}_{=0} + g(\lambda_i) \right) \\ &= \prod_{i=1}^d g(\lambda_i) = \mathcal{R}_{f,g}. \end{aligned} \quad \square$$

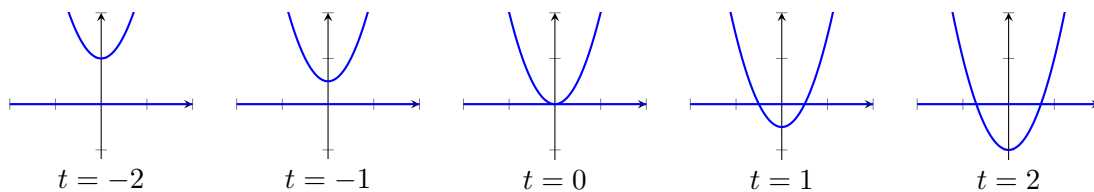
In Proposition 8.4, we don't really need f to be monic; if its leading coefficient is $a \in R$ then we need to adjust the statement by adding factors of $a^{\deg(g)}$ and $a^{\deg(hf+g)}$ where appropriate. We'll mostly be interested in the non-constant factors of the resultants $\mathcal{R}_{f,g}$ and $\mathcal{R}_{f,hf+g}$ anyway.

8.2 Intersection multiplicity and strong Bézout

The weak version of Bézout's theorem, Theorem 6.11, tells us that plane curves of degree d and e with no common component intersect in at most de points. In order to get a more precise count, we need to figure out how points of intersection should be counted with multiplicity. For example, the family of affine plane curves

$$y = x^2 - t, \quad t \in \mathbb{K}$$

intersects the line $y = 0$ in two points $(\pm\sqrt{t}, 0)$ whenever $t \neq 0$, and this suggests that when $t = 0$ the sole point of intersection $(0, 0)$ should be counted twice.



(Note that when $t = -2, -1$ the above pictures over $\mathbb{R}$ do not show the complex points of intersection, but they really do exist.)

The notion of *intersection multiplicity* takes this into account. Here is a list of properties we would like it to have:

Definition 8.5. Let $C, D \subset \mathbb{P}^2$ be projective plane curves, and fix a point $p \in \mathbb{P}^2$. An *intersection multiplicity* is a rule assigning to each $p \in \mathbb{P}^2$ a number

$$I_p(C, D) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$$

satisfying the following list of properties.

1. $I_p(C, D) = I_p(D, C)$.
2. If p is not in the intersection $C \cap D$, then $I_p(C, D) = 0$.
3. If $C \cap D$ contains an irreducible curve C_0 , and $p \in C_0$, then $I_p(C, D) = \infty$.
4. If C and D are distinct lines intersecting at p , then $I_p(C, D) = 1$.
5. If $D = D_1 \cup D_2$ is a union of plane curves, then

$$I_p(C, D) = I_p(C, D_1) + I_p(C, D_2).$$

6. If C and D are defined by $\{F(x, y, z) = 0\}$ and $\{G(x, y, z) = 0\}$, and if $D' \subset \mathbb{P}^2$ is defined by $\{FQ + G = 0\}$ for some homogeneous Q , then

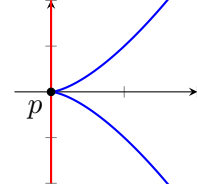
$$I_p(C, D) = I_p(C, D').$$

Most of these properties are not hard to motivate, but the last one is less clear. What it means is that we can “deform” D by adding some multiple of the defining equation for C , and then from the perspective of C the equation defining D hasn't really changed, so the intersection multiplicity shouldn't change either.

Temporarily setting aside the question of whether intersection multiplicity exists, we can see from examples that these rules are enough to determine it. From here on we will identify curves with their defining polynomials, so that if $C = \{F = 0\}$ and $D = \{G = 0\}$ then we write $I_p(F, G) = I_p(C, D)$. We will also not insist that these polynomials be square-free, thinking for example of $\{z^2 = 0\}$ as defining two overlapping copies of the line $\{z = 0\}$.

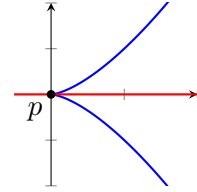
Example 8.6. Let C and D be defined by $y^2z - x^3 = 0$ and $x = 0$, respectively. At $p = [0, 0, 1]$, we compute the intersection multiplicity to be

$$\begin{aligned}
I_p(C, D) &= I_p(y^2z - x^3, x) \\
&= I_p(x, y^2z - x^3) && \text{by (1)} \\
&= I_p(x, y^2z) && \text{by (6)} \\
&= 2I_p(x, y) + I_p(x, z) && \text{by (5)} \\
&= 2 \cdot 1 + 0 = 2 && \text{by (4), (2)}.
\end{aligned}$$



Similarly, if $E = \{y = 0\}$ then we can compute that

$$\begin{aligned}
I_p(C, E) &= I_p(y^2z - x^3, y) \\
&= I_p(y, y^2z - x^3) && \text{by (1)} \\
&= I_p(y, -x^3) && \text{by (6)} \\
&= 3I_p(y, x) = 3 && \text{by (5), (4)}.
\end{aligned}$$



Proposition 8.7. *There is a unique intersection multiplicity $I_p(C, D)$, and it is defined as follows.*

- If p is not in the intersection $C \cap D$, then $I_p(C, D) = 0$.
- If p lies on a common component of C and D , then $I_p(C, D) = \infty$.
- Otherwise, let $C' \subset C$ and $D' \subset D$ be the unions of components that are not common to C and D . Apply a projective transformation so that $[1, 0, 0] \in \mathbb{P}^2$ does not lie on C' , D' , or any of the lines through pairs of points of $C' \cap D'$.

If C' is defined by $\{F = 0\}$ and D' is defined by $\{G = 0\}$, and if $p = [a, b, c]$, then we define $I_p(C, D)$ to be the multiplicity of $[b, c]$ as a root of $\mathcal{R}_{F,G}(y, z)$, or equivalently the largest k such that $(cy - bz)^k$ divides $\mathcal{R}_{F,G}(y, z)$.

Proof. The proof of uniqueness is Theorem 3.18 of Kirwan's book, to which we refer for details: one shows that if $I_p(C, D) = k$ we can use the six axioms to express $I_p(C, D)$ as a combination of intersection multiplicities that are strictly less than k , and so uniqueness follows by induction on k . We note as a consequence that intersection multiplicity must be invariant under projective transformations.

We now check that our definition in terms of resultants satisfies each of the six axioms for $I_p(C, D)$. Property (1) follows from the relation

$$\mathcal{R}_{F,G}(y, z) = (-1)^{\deg(F)\deg(G)} \mathcal{R}_{G,F}(y, z),$$

which is a direct consequence of its definition as a determinant. We have built (2) and (3) into the definition. Properties (5) and (6) follow from the relations

$$\mathcal{R}_{F,GH}(y, z) = \mathcal{R}_{F,G}(y, z) \cdot \mathcal{R}_{F,H}(y, z)$$

and

$$\mathcal{R}_{F,FQ+G}(y, z) = \mathcal{R}_{F,G}(y, z),$$

which are Propositions 8.3 and 8.4, respectively. This leaves only (4).

For (4), we compute directly that two distinct lines $C = \{ax + by + cz = 0\}$ and $D = \{dx + ey + fz = 0\}$ intersect at the point

$$p = [bf - ce, cd - af, ae - bd],$$

whose entries are not all zero because if $C \neq D$ then the matrix $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$ has rank 2 and hence a nonzero 2×2 minor. After possibly applying a projective transformation so that $p \neq [1, 0, 0]$, we have $(af - cd, ae - bd) \neq (0, 0)$, and

$$\begin{aligned} \mathcal{R}_{ax+by+cz, dx+ey+fz}(y, z) &= \det \begin{pmatrix} a & by + cz \\ d & ey + fz \end{pmatrix} \\ &= (ae - bd)y + (af - cd)z. \end{aligned}$$

This resultant is nonzero, and we read off from it the fact that $I_p(C, D) = 1$. $\square$

We can now strengthen Theorem 6.11 to take multiplicities into account.

Theorem 8.8: Bézout's theorem

Let $\mathbb{K}$ be an algebraically closed field, and let $C, D \subset \mathbb{P}^2(\mathbb{K})$ be two projective plane curves of degrees $d \geq 1$ and $e \geq 1$ respectively, with no common component. Then

$$\sum_{p \in C \cap D} I_p(C, D) = de.$$

Proof. We repeat the proof of Theorem 6.11: applying a projective transformation, we may assume that $[1, 0, 0]$ does not lie on C , D , or the lines through any pair of points of $C \cap D$. Letting $C = \{F = 0\}$ and $D = \{G = 0\}$, we saw that points $p = [a, b, c] \in C \cap D$ were in bijection with roots $[b, c]$ of $\mathcal{R}_{F,G}(y, z)$, and now if $[b, c]$ is a root of order k then we know that the corresponding $p = [a, b, c]$ satisfies $I_p(C, D) = k$. Proposition 6.8 says that $\mathcal{R}_{F,G}(y, z)$ has a total of de roots when counted with multiplicity, so we conclude that $\sum_p I_p(C, D) = de$ after all. $\square$

8.3 Intersection multiplicity and tangent lines

Having established Bézout's theorem as a consequence of the definition of intersection multiplicity, we now consider what this means geometrically.

Definition 8.9. We say two curves $C, D \subset \mathbb{P}^2$ are *transverse* at a point $p \in C \cap D$ if p is a smooth point of both curves and $I_p(C, D) = 1$.

The strong version of Bézout's theorem then tells us that if $\mathbb{K}$ is algebraically closed and the curves $C, D \subset \mathbb{P}^2(\mathbb{K})$ have no common components, then the set $C \cap D$ contains exactly $\deg(C) \cdot \deg(D)$ points if and only if C and D meet transversely at every point of intersection.

Proposition 8.10. *Let $p \in \mathbb{P}^2$ be a smooth point on each of the plane curves $C = \{F(x, y, z) = 0\}$ and $D = \{G(x, y, z) = 0\}$. Then $I_p(C, D) = 1$ if and only if the tangent lines $T_p C$ and $T_p D$ are different.*

Since any line $L \subset \mathbb{P}^2$ is smooth, and $T_p L = L$ for all $p \in L$, we conclude:

Corollary 8.11. *Given a smooth point p on a plane curve $C \subset \mathbb{P}^2$ and a line L through p , we have $I_p(C, L) \geq 2$ if and only if $L = T_p C$.*

Proof of Proposition 8.10. We can replace C and D with the irreducible components containing p , and this does not change $I_p(C, D)$, so we might as well do so. If $C = D$ then the tangent lines coincide and $I_p(C, D) = \infty$, so we will further assume that $C \neq D$. Finally, since $I_p(C, D) = I_p(D, C)$ we will assume that $\deg(C) \geq \deg(D) \geq 1$. Now the intersection $C \cap D$ is finite, and we can apply a projective transformation so that $p = [0, 0, 1]$ and $T_p D = \{x = 0\}$, and so that none of C , D , or the lines through pairs of points in $C \cap D$ pass through $[1, 0, 0]$.

Having done this, if we write $C = \{F = 0\}$ and $D = \{G = 0\}$ where

$$\begin{aligned} F(x, y, z) &= f_0(y, z)x^d + f_1(y, z)x^{d-1} + \cdots + f_{d-1}(y, z)x + f_d(y, z) \\ G(x, y, z) &= g_0(y, z)x^e + g_1(y, z)x^{e-1} + \cdots + g_{e-1}(y, z)x + g_e(y, z), \end{aligned}$$

with $d \geq e \geq 1$ and each $f_i(y, z)$ or $g_i(y, z)$ homogeneous of degree i , then $I_p(C, D)$ will be the multiplicity of $[0, 1]$ as a root of $\mathcal{R}_{F,G}(y, z)$, meaning the largest k such that y^k divides $\mathcal{R}_{F,G}(y, z)$. Thus $I_p(C, D) \geq 2$ if and only if y^2 divides $\mathcal{R}_{F,G}(y, z)$.

We now compute the partial derivatives of F and of G at $(0, 0, 1)$: we note from Euler's identity (Proposition 4.10) evaluated at $(x, y, z) = (0, 0, 1)$ that

$$0 = d \cdot F(0, 0, 1) = F_z(0, 0, 1),$$

and then since $p \in C$ is smooth we must have $(F_x(0, 0, 1), F_y(0, 0, 1)) \neq (0, 0)$. We compute that

$$f_d(y, z) = F(0, 0, 1)z^d + F_y(0, 0, 1)yz^{d-1} + \cdots,$$

where the omitted terms are all multiples of y^2 . Then $F(0, 0, 1) = 0$, so y divides $f_d(y, z)$; and in fact y^2 divides $f_d(y, z)$ if and only if $F_y(0, 0, 1) = 0$, which is then equivalent to $T_p C = \{x = 0\}$. Since $T_p D = \{x = 0\}$, the same argument says that $g_e(y, z)$ is a multiple of y^2 , and also that

$$g_{e-1}(0, 1) = G_x(0, 0, 1) \neq 0.$$

We can rescale G to arrange that $g_{e-1}(0, 1) = 1$.

Now if $e = 1$ then $G(x, y, z) = x$ and we compute by Proposition 8.2 that

$$\mathcal{R}_{F,G}(y, z) = \pm \mathcal{R}_{x,F}(y, z) = \pm F(0, y, z) = \pm f_d(y, z).$$

This is a multiple of y^2 if and only if $T_p C = \{x = 0\} = T_p D$, proving the proposition in this case. From now on we will assume that $e \geq 2$.

The resultant of F and G is by definition

$$\mathcal{R}_{F,G}(y, z) = \det \begin{pmatrix} f_0 & f_1 & \dots & f_{d-1} & f_d & & & \\ & f_0 & \dots & f_{d-2} & f_{d-1} & f_d & & \\ & & \ddots & & \ddots & \ddots & \ddots & \\ & & & f_0 & \dots & f_{d-2} & f_{d-1} & f_d \\ & & & & f_0 & \dots & f_{d-2} & f_{d-1} & f_d \\ g_0 & g_1 & \dots & g_{e-1} & g_e & & & \\ & g_0 & \dots & g_{e-2} & g_{e-1} & g_e & & \\ & & \ddots & & \ddots & \ddots & \ddots & \\ & & & g_0 & \dots & g_{e-2} & g_{e-1} & g_e \\ & & & & g_0 & \dots & g_{e-2} & g_{e-1} & g_e \end{pmatrix}.$$

We know that y^2 divides g_e , so we can evaluate $\mathcal{R}_{f,g}(y, z) \pmod{y^2}$, by replacing g_e by 0 and then expanding by minors along the last column:

$$\begin{aligned} \mathcal{R}_{F,G}(y, z) &\equiv \pm f_d \cdot \det \begin{pmatrix} f_0 & f_1 & \dots & f_{d-1} & f_d & & & \\ & f_0 & \dots & f_{d-2} & f_{d-1} & f_d & & \\ & & \ddots & & \ddots & \ddots & \ddots & \\ & & & f_0 & \dots & f_{d-2} & f_{d-1} & f_d \\ g_0 & g_1 & \dots & g_{e-1} & 0 & & & \\ & g_0 & \dots & g_{e-2} & g_{e-1} & 0 & & \\ & & \ddots & & \ddots & \ddots & \ddots & \\ & & & g_0 & \dots & g_{e-2} & g_{e-1} & 0 \\ & & & & g_0 & \dots & g_{e-2} & g_{e-1} \end{pmatrix} \\ &\equiv \pm f_d \cdot \mathcal{R}_{F,H}(y, z) \pmod{y^2} \end{aligned}$$

where we define the new polynomial $H(x, y, z)$ by

$$H(x, y, z) = g_0(y, z)x^{e-1} + g_1(y, z)x^{e-2} + \cdots + g_{e-2}(y, z)x + g_{e-1}(y, z),$$

so that $G = Hx + g_e$. (We note that $\deg(H) = e - 1$ is positive but less than $d = \deg(F)$, and then since F is irreducible the two have no common factors.)

If $T_p C = T_p D = \{x = 0\}$ then f_d is a multiple of y^2 , hence so is $\mathcal{R}_{f,g}(y, z)$ and we have $I_p(C, D) \geq 2$ as desired. Otherwise, if $T_p C \neq T_p D$ then we know that f_d is a multiple of y but not of y^2 , so if we can show that $\mathcal{R}_{F,H}(y, z)$ is not a multiple of y then $y^2 \nmid \mathcal{R}_{F,G}(y, z)$ and hence the desired $I_p(C, D) = 1$ will follow.

We can check that $H(1, 0, 0) = g_0(0, 0) \neq 0$, so $[1, 0, 0]$ does not lie on the curve $\{H(x, y, z) = 0\}$, whose degree is $e - 1 > 0$. Then $\mathcal{R}_{F,H}(y, z)$ is a multiple of y if and only if $[0, 1]$ is a root, or equivalently if and only if

$$F(\lambda, 0, 1) = H(\lambda, 0, 1) = 0$$

for some $\lambda \in \mathbb{K}$. This cannot happen for $\lambda = 0$, since $H(0, 0, 1) = g_{e-1}(0, 1) = 1$. But it cannot happen for $\lambda \neq 0$ either: in this case, we would have

$$G(\lambda, 0, 1) = \lambda \cdot \underbrace{H(\lambda, 0, 1)}_{=0} + \underbrace{g_e(0, 1)}_{=G(0,0,1)=0} = 0,$$

so $[\lambda, 0, 1]$ and $p = [0, 0, 1]$ would be two distinct points of $C \cap D$, and the line $\{y = 0\}$ through them would then pass through $[1, 0, 0]$, a contradiction. So we conclude that if $T_p C \neq T_p D$ then $\mathcal{R}_{F,H}(y, z)$ is not a multiple of y , hence $I_p(C, D) = 1$ after all. $\square$

8.4 Multiple points

We recall that a point p on a projective plane curve $C = \{F(x, y, z) = 0\}$ is smooth if and only if the partial derivatives F_x , F_y , and F_z are not all zero at p . We can refine this by quantifying how far p is from being smooth. Given a triple $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ of nonnegative integers, we will write $|\alpha| = \alpha_1 + \alpha_2 + \alpha_3$ and let

$$\partial^\alpha F = \frac{\partial^{|\alpha|} F}{\partial x^{\alpha_1} \partial y^{\alpha_2} \partial z^{\alpha_3}}$$

for the various partial derivatives of F , which we say have order $|\alpha|$.

Definition 8.12. Let $p = [a, b, c]$ be a point on the projective plane curve

$$C = \{F(x, y, z) = 0\} \subset \mathbb{P}^2.$$

Then we say that p has multiplicity $m_p(C) = k$ if and only if

- all partial derivatives $\partial^\alpha F$ of order $|\alpha| < k$ vanish at (a, b, c) ;

- and some k th order partial derivative of F is nonzero at (a, b, c) .

Note that p is a smooth point of C if and only if $m_p(C) = 1$.

We should check that this does not depend on the coordinates $[a, b, c]$: if F is homogeneous of degree d , then $\partial^\alpha F$ is homogeneous of degree $d - |\alpha|$, and so for $\lambda \in \mathbb{K}^\times$ we have $(\partial^\alpha F)(\lambda a, \lambda b, \lambda c) = 0$ if and only if $(\partial^\alpha F)(a, b, c) = 0$.

Example 8.13. If we let

$$C = \{F(x, y, z) = y^2z - x^3 = 0\}$$

then the point $p = [0, 0, 1]$ has multiplicity 2 – we call it a *double point* – because the first partial derivatives

$$F_x = -3x^2, \quad F_y = 2yz, \quad F_z = y^2$$

all vanish at $(0, 0, 1)$, but $\partial^{(0,2,0)} F = \frac{\partial^2 F}{\partial y^2} = 2z$ does not.

Exercise 8.14. Multiplicity is invariant under projective transformations: in other words, we have $m_p(C) = m_{\Phi(p)}(\Phi(C))$.

Exercise 8.15. Show that if $p \in C$ then $m_p(C) \leq \deg(C)$.

Exercise 8.16. Show that if $p \in C$ and $p \notin D$ then $m_p(C) = m_p(C \cup D)$.

Multiplicity gives us improved estimates of intersection numbers, which we can use to extract more information from Bézout's theorem.

Proposition 8.17. Let $C, D \subset \mathbb{P}^2$ be projective plane curves over a field $\mathbb{K}$ of characteristic zero, and choose a point $p \in C \cap D$. Then

$$I_p(C, D) \geq m_p(C) \cdot m_p(D).$$

Proof. We can assume that C and D have no common components, since if p were to lie on one then we would have $I_p(C, D) = \infty$ anyway. We apply a projective transformation so that $[1, 0, 0]$ is neither in C nor in D , nor on any of the lines through two points of $C \cap D$; and so that $p = [0, 0, 1]$.

We now take $C = \{F = 0\}$ and $D = \{G = 0\}$ for some homogeneous polynomials $F, G \in \mathbb{K}[x, y, z]$, and let $r = m_p(C)$ and $s = m_p(D)$. If we write

$$F(x, y, z) = f_0(y, z)x^d + f_1(y, z)x^{d-1} + \cdots + f_{d-1}(y, z)x + f_d(y, z),$$

with each f_i homogeneous of degree i , then we claim that y^{r-i} divides $f_{d-i}(y, z)$ for each $i = 0, 1, \dots, r$. To see this, we compute for any $j < r - i$ that

$$0 = (\partial^{(i,j,0)} F)(0, 0, 1) = i! \cdot \frac{\partial^j f_{d-i}}{\partial y^j}(0, 1),$$

since $i + j < r = m_p(C)$, and so the $y^j z^{d-i-j}$ -term of $f_{d-i}(y, z)$ is necessarily zero. This means that any nonzero monomial $cy^j z^{d-i-j}$ of $f_{d-i}(y, z)$ must satisfy $j \geq r - i$ and thus be a multiple of y^{r-i} , as claimed.

The above argument says that if $i \leq r$ then we can write $f_{d-i}(y, z) = y^{r-i} \tilde{f}_{d-i}(y, z)$ for some polynomials $\tilde{f}_{d-i}(y, z)$, and similarly if

$$G(x, y, z) = g_0(y, z)x^e + g_1(y, z)x^{e-1} + \dots + g_{e-1}(y, z)x + g_e(y, z)$$

then we can write $g_{e-j}(y, z) = y^{s-j} \tilde{g}_{e-j}(y, z)$ for $j \leq s$. Now the resultant $\mathcal{R}_{F,G}$ is by definition the determinant of the matrix

$$\begin{pmatrix} f_0 & f_1 & f_2 & \dots & y^r \tilde{f}_d & & & & & \\ & f_0 & f_1 & \dots & y^{r-1} \tilde{f}_{d-1} & y^r \tilde{f}_d & & & & \\ & & f_0 & \dots & y^{r-2} \tilde{f}_{d-2} & y^{r-1} \tilde{f}_{d-1} & y^r \tilde{f}_d & & & \\ & & & \ddots & & \ddots & \ddots & \ddots & & \\ & & & & f_0 & f_1 & f_2 & \dots & y^r \tilde{f}_d & \\ & & & & & f_0 & f_1 & \dots & y^{r-1} \tilde{f}_{d-1} & y^r \tilde{f}_d \\ g_0 & g_1 & g_2 & \dots & y^{s-1} \tilde{g}_{e-1} & y^s \tilde{g}_e & & & & \\ & g_0 & g_1 & \dots & y^{s-2} \tilde{g}_{e-2} & y^{s-1} \tilde{g}_{e-1} & y^s \tilde{g}_e & & & \\ & & g_0 & \dots & y^{s-3} \tilde{g}_{e-3} & y^{s-2} \tilde{g}_{e-2} & y^{s-1} \tilde{g}_{e-1} & y^s \tilde{g}_e & & \\ & & & \ddots & & \ddots & \ddots & \ddots & \ddots & \\ & & & & g_0 & \dots & y^{s-3} \tilde{g}_{e-3} & y^{s-2} \tilde{g}_{e-2} & y^{s-1} \tilde{g}_{e-1} & y^s \tilde{g}_e \end{pmatrix}.$$

If we multiply rows $e, e-1, \dots, e-s+1$ of this matrix by $y^s, y^{s-1}, \dots, y$ respectively, and we multiply rows $d+e, d+e-1, \dots, d+e-r+1$ by $y^r, y^{r-1}, \dots, y$ respectively, then we get a new matrix with determinant

$$y^{(s+(s-1)+\dots+1)+(r+(r-1)+\dots+1)} \mathcal{R}_{F,G}(y, z) = y^{r(r+1)/2+s(s+1)/2} \mathcal{R}_{F,G}(y, z).$$

But columns $d+e-1, d+e-2, \dots, d+e-(r+s)$ of this new matrix are multiples of $y^{r+s}, y^{r+s-1}, \dots, y$ respectively, so the new determinant is a multiple of

$$y^{(r+s)+(r+s-1)+\dots+1} = y^{(r+s)(r+s+1)/2}.$$

We conclude that $\mathcal{R}_{F,G}(y, z)$ is a multiple of y^k , where

$$k = \frac{(r+s)(r+s+1)}{2} - \left(\frac{r(r+1)}{2} + \frac{s(s+1)}{2} \right) = rs.$$

Now by definition the intersection multiplicity $I_p(C, D)$ at $p = [0, 0, 1]$ is the largest ℓ such that y^ℓ divides $\mathcal{R}_{F,G}(y, z)$, and we have just seen that $\ell \geq rs$, so the proof is complete. $\square$

Corollary 8.18. *Given plane curves $C, D \subset \mathbb{P}^2$ that meet at a point p , we have $I_p(C, D) = 1$ if and only if p is a smooth point of both curves and $T_p C \neq T_p D$.*

Proof. Proposition 8.17 says that $I_p(C, D) = 1$ only if both C and D are smooth at p , and then Proposition 8.10 says that $I_p(C, D) = 1$ if and only if $T_p C \neq T_p D$. $\square$

Corollary 8.19. *Let $C, D \subset \mathbb{P}^2$ be plane curves with no common components. Then $C \cap D$ consists of exactly $\deg(C) \cdot \deg(D)$ points if and only if for every $p \in C \cap D$, both C and D are smooth at p and $T_p C \neq T_p D$.*

Proof. Bézout's theorem says that $C \cap D$ has exactly $\deg(C) \cdot \deg(D)$ points if and only if each of them has multiplicity 1, so we apply Corollary 8.18. $\square$

As an application of Proposition 8.17, we can refine Proposition 7.1 to bound the number of singular points of a plane curve.

Proposition 8.20. *Let $\mathbb{K}$ be an algebraically closed field of characteristic zero, and let $C \subset \mathbb{P}^2(\mathbb{K})$ be a plane curve of degree d . If C has singular points $p_1, \dots, p_k$ with multiplicities $r_1, \dots, r_k \geq 2$, then*

$$\sum_{i=1}^k r_i(r_i - 1) \leq d(d - 1).$$

Proof. Write $C = \{F(x, y, z) = 0\}$, where F has degree $d \geq 1$ and no repeated factors. Since F is not constant, at least one of F_x , F_y , and F_z is not the zero polynomial; say $F_x \not\equiv 0$ without loss of generality, and let $D = \{F_x = 0\}$. Then D has degree $d - 1$, and C and D have no common components since F is square-free. Each of the p_i belong to $C \cap D$, so Bézout's theorem says that

$$d(d - 1) = \sum_{p \in C \cap D} I_p(C, D) \geq \sum_{i=1}^k I_{p_i}(C, D).$$

Now Proposition 8.17 tells us that $I_{p_i}(C, D) \geq m_{p_i}(C) \cdot m_{p_i}(D)$, and $m_{p_i}(C) = r_i$ while $m_{p_i}(D) \geq r_i - 1$ (exercise: why?), so the result follows. $\square$

Since $r_i(r_i - 1) \geq 2$ for each i , Proposition 8.20 also tells us that there are at most $\frac{1}{2}d(d - 1)$ singular points, and if equality holds then these are all double points. This can happen when C is a union of d lines, no three of which pass through the same point; the singular points are precisely the $\binom{d}{2}$ points where some pair of lines meet.

8.5 Cubic curves

In Theorem 7.2 we classified irreducible conics over an algebraically closed field, and we saw in particular that they are all nonsingular. The situation for irreducible cubic curves is not much worse.

Lemma 8.21. *Let $C \subset \mathbb{P}^2$ be an irreducible cubic curve over an algebraically closed field. If C is not smooth then it has exactly one singular point, of multiplicity 2.*

Proof. Supposing that $p \in C$ is a singular point, we take any other point $q \in C$ and let L be the line through p and q . Since C is irreducible it does not contain L , so Bézout's theorem says that

$$I_p(L, C) + I_q(L, C) \leq \sum_{r \in L \cap C} I_r(L, C) = 3.$$

But Proposition 8.17 says that $I_p(L, C) \geq m_p(C)$ and $I_q(L, C) \geq m_q(C)$, so now

$$2 + m_q(C) \leq m_p(C) + m_q(C) \leq I_p(L, C) + I_q(L, C) \leq 3.$$

This says that $m_q(C) \leq 1$, hence $m_q(C) = 1$ and q is a smooth point of C ; and then $m_p(C) \leq 3 - m_q(C) = 2$, so p is a double point. $\square$

If C is a smooth cubic, then we know by Theorem 4.15 that it is irreducible.

Definition 8.22. Let $C \subset \mathbb{P}^2$ be a smooth curve of degree $d > 0$, defined as $\{F(x, y, z) = 0\}$ for some homogeneous F . The *Hessian matrix* of F is the 3×3 matrix of second partial derivatives, and the *Hessian* of F is its determinant:

$$H_F(x, y, z) = \det \begin{pmatrix} F_{xx} & F_{xy} & F_{xz} \\ F_{yx} & F_{yy} & F_{yz} \\ F_{zx} & F_{zy} & F_{zz} \end{pmatrix}.$$

Then $H_F(x, y, z)$ is a homogeneous polynomial of degree $3(d - 2)$ when $d \geq 2$.

A smooth point $p \in C$ is called an *inflection point* if $H_F(p) = 0$.

The weak form of Bézout's theorem tells us that any smooth curve C of degree at least 3 over an algebraically closed field must have an inflection point, because C intersects the curve $\{H_F = 0\}$ in at least one point.

Example 8.23. If C is a line, then $H_F(x, y, z)$ is the zero polynomial and so every point of C is an inflection point.

Example 8.24. Suppose that $\text{char}(\mathbb{K}) = 0$. If $C = \{x^3 - yz^2 = 0\}$, then its Hessian is

$$H_F(x, y, z) = \det \begin{pmatrix} 6x & 0 & 0 \\ 0 & 0 & -2z \\ 0 & -2z & -2y \end{pmatrix} = -24xz^2.$$

If $-24xz^2 = 0$ then $xyz^2 = 0$, and at a point of C we have $x^4 = xyz^2$, so the points of C where $H_F = 0$ all satisfy $x = 0$. Then $x^3 - yz^2 = 0$ implies that either $y = 0$ or $z = 0$, so

$$C \cap \{H_F = 0\} = \{[0, 1, 0], [0, 0, 1]\}.$$

But $[0, 1, 0]$ is not a smooth point of C , so $[0, 0, 1]$ is the unique inflection point.

We can simplify the Hessian somewhat, and this will lead us to a more intrinsic definition of an inflection point.

Lemma 8.25. *If $F \in \mathbb{K}[x, y, z]$ is homogeneous of degree $d \geq 2$, then its Hessian satisfies*

$$z^2 H_F(x, y, z) = (d-1)^2 \det \begin{pmatrix} F_{xx} & F_{xy} & F_x \\ F_{yx} & F_{yy} & F_y \\ F_x & F_y & \frac{d}{d-1} F \end{pmatrix}.$$

Similar results hold for $x^2 H_F$ and $y^2 H_F$.

Proof. If we multiply the last column of the Hessian matrix by z and apply some column operations, we get

$$zH_F = \det \begin{pmatrix} F_{xx} & F_{xy} & zF_{xz} \\ F_{yx} & F_{yy} & zF_{yz} \\ F_{zx} & F_{zy} & zF_{zz} \end{pmatrix} = \det \begin{pmatrix} F_{xx} & F_{xy} & xF_{xx} + yF_{xy} + zF_{xz} \\ F_{yx} & F_{yy} & xF_{yx} + yF_{yy} + zF_{yz} \\ F_{zx} & F_{zy} & xF_{zx} + yF_{zy} + zF_{zz} \end{pmatrix}.$$

Applying Euler's identity, Proposition 4.10, to each of the degree- $(d-1)$ polynomials F_x , F_y , and F_z gives us

$$zH_F = \det \begin{pmatrix} F_{xx} & F_{xy} & (d-1)F_x \\ F_{yx} & F_{yy} & (d-1)F_y \\ F_{zx} & F_{zy} & (d-1)F_z \end{pmatrix} = (d-1) \det \begin{pmatrix} F_{xx} & F_{xy} & F_x \\ F_{yx} & F_{yy} & F_y \\ F_{zx} & F_{zy} & F_z \end{pmatrix}.$$

Now we multiply the last row of this matrix by z and apply some analogous row operations to get

$$\frac{z^2 H_F}{d-1} = \det \begin{pmatrix} F_{xx} & F_{xy} & F_x \\ F_{yx} & F_{yy} & F_y \\ xF_{xx} + yF_{yx} + zF_{zx} & xF_{xy} + yF_{yy} + zF_{zy} & xF_x + yF_y + zF_z \end{pmatrix},$$

and again by Euler's identity applied to F_x , F_y , and F itself we can simplify this to

$$\frac{z^2 H_F}{d-1} = \det \begin{pmatrix} F_{xx} & F_{xy} & F_x \\ F_{yx} & F_{yy} & F_y \\ (d-1)F_x & (d-1)F_y & dF \end{pmatrix},$$

from which the identity follows. $\square$

Proposition 8.26. *A smooth point $p \in C$ is an inflection point if and only if $I_p(C, T_p C) \geq 3$.*

Proof. Both the Hessian and the intersection multiplicity are preserved under projective transformations, so we will assume that $[1, 0, 0] \notin C$, that $p = [0, 0, 1]$ and that the tangent line $T_p C$ is defined by $\{x = 0\}$. (To do this, we first apply a transformation to arrange that $[1, 0, 0] \notin C \cup T_p C$, then another one fixing $[1, 0, 0]$ and sending p and some other point $q \in T_p C$ to $[0, 0, 1]$ and $[0, 1, 0]$.)

In this case, if C is defined by the equation $F(x, y, z) = 0$ where

$$F(x, y, z) = f_0(y, z)x^d + f_1(y, z)x^{d-1} + \cdots + f_{d-1}(y, z)x + f_d(y, z),$$

then the intersection multiplicity $I_p(C, T_p C)$ is the largest k such that y^k divides

$$\mathcal{R}_{x,F}(y, z) = F(0, y, z) = f_d(y, z),$$

which we have evaluated using Proposition 8.2. We already have $k \geq 2$ by Corollary 8.11, so $I_p(C, T_p C) \geq 3$ is now equivalent to $y^3 \mid f_d(y, z)$, hence to

$$F_{yy}(0, 0, 1) = (f_d)_{yy}(0, 1) = 0.$$

According to Lemma 8.25, we have

$$z^2 H_F(x, y, z) = (d-1)^2 \det \begin{pmatrix} F_{xx} & F_{xy} & F_x \\ F_{yx} & F_{yy} & F_y \\ F_x & F_y & \frac{d}{d-1}F \end{pmatrix}.$$

We wish to evaluate this at $(x, y, z) = (0, 0, 1)$, so we note that $p \in C$ implies $F(0, 0, 1) = 0$ while $T_p C = \{x = 0\}$ says that $F_x(0, 0, 1) \neq 0$ and $F_y(0, 0, 1) = 0$. Then

$$\begin{aligned} H_F(0, 0, 1) &= (d-1)^2 \det \begin{pmatrix} F_{xx}(0, 0, 1) & F_{xy}(0, 0, 1) & F_x(0, 0, 1) \\ F_{yx}(0, 0, 1) & F_{yy}(0, 0, 1) & 0 \\ F_x(0, 0, 1) & 0 & 0 \end{pmatrix} \\ &= -(d-1)^2 (F_x(0, 0, 1))^2 \cdot F_{yy}(0, 0, 1), \end{aligned}$$

and so $H_F(0, 0, 1) = 0$ if and only if $F_{yy}(0, 0, 1) = 0$, which in turn is equivalent to $I_p(C, T_p C) \geq 3$ as claimed. $\square$

Theorem 8.27

Let C be a smooth cubic curve over an algebraically closed field $\mathbb{K}$ of characteristic zero. Then C is projectively equivalent to a curve of the form

$$y^2z = x(x - z)(x - \lambda z),$$

where $\lambda \neq 0, 1$.

Proof. Write $C = \{F(x, y, z) = 0\}$. Then C has an inflection point; after a projective transformation we can assume that $p = [0, 1, 0]$ is an inflection point, with tangent line $T_pC = \{z = 0\}$. If we write

$$F(x, y, z) = ay^3 + (bx + cz)y^2 + (dx^2 + exz + fz^2)y + G(x, z),$$

where $G(x, z)$ is homogeneous of degree three, then $[0, 1, 0] \in C$ and $T_pC = \{z = 0\}$ imply that

$$\begin{aligned} F(0, 1, 0) = 0 &\implies a = 0, \\ F_x(0, 1, 0) = 0 &\implies b = 0, \\ F_z(0, 1, 0) \neq 0 &\implies c \neq 0. \end{aligned}$$

We apply the version of Lemma 8.25 that computes y^2H_F to get

$$y^2H_F(x, y, z) = 4 \det \begin{pmatrix} F_{xx} & F_x & F_{xz} \\ F_x & \frac{3}{2}F & F_z \\ F_{zx} & F_z & F_{zz} \end{pmatrix},$$

and we know that $H_F(0, 1, 0) = 0$ since p is an inflection point, so we substitute $(x, y, z) = (0, 1, 0)$ to get

$$\begin{aligned} 0 &= \det \begin{pmatrix} F_{xx} & 0 & F_{xz} \\ 0 & 0 & F_z \\ F_{zx} & F_z & F_{zz} \end{pmatrix} \Big|_{(x,y,z)=(0,1,0)} \\ &= \det \begin{pmatrix} 2d & 0 & e \\ 0 & 0 & c \\ e & c & 2f \end{pmatrix} = -2c^2d. \end{aligned}$$

Since we know that $c \neq 0$, this means that $d = 0$ as well, and we have

$$F(x, y, z) = cy^2z + (ex + fz)yz + G(x, z).$$

Completing the square since $c \neq 0$ (and since $\text{char}(\mathbb{K}) \neq 2$), we can write

$$F(x, y, z) = cz \left(y + \frac{ex + fz}{2c} \right)^2 + H(x, z)$$

for another homogeneous $H(x, z)$ of degree 3.

If we apply a projective transformation taking $[x, y, z]$ to $[x, y - \frac{1}{2c}(ex + fz), z]$, and then divide by c , we see that in these new coordinates we can write

$$C = \{y^2z = K(x, z)\}.$$

The homogeneous, degree-3 polynomial $K(x, z)$ is not a multiple of z , since otherwise the line $\{z = 0\}$ would be contained in the irreducible C , so we can factor $K(x, z)$ to write the defining equation as

$$y^2z = k(x - az)(x - bz)(x - cz)$$

for some $k \neq 0$. The constants $a, b, c \in \mathbb{K}$ must be distinct, since for example if $a = b$ then $[a : 0 : 1]$ would be a singular point of C . We apply a sequence of projective transformations to change this equation as follows:

$$\begin{aligned} [x, y, z] &\mapsto [x + az, y, z] : & y^2z &= kx(x - b'z)(x - c'z) \\ [x, y, z] &\mapsto [x, y\sqrt{b'k}, z/b'] : & y^2z &= x(x - z)(x - c''z) \end{aligned}$$

and we have $c'' \neq 0, 1$ since C is nonsingular, so now C has the desired form. $\square$

We note that in the proof of Theorem 8.27, we gave an inflection point the coordinates $p = [0, 1, 0]$ with $T_pC = \{z = 0\}$, and both p and the line $\{z = 0\}$ were fixed by subsequent transformations. We can take our equation in its standard form

$$F(x, y, z) = (x^3 + kx^2z + \ell xz^2) - y^2z,$$

and then compute that

$$\begin{aligned} H_F(x, y, z) &= \det \begin{pmatrix} 6x + 2kz & 0 & 2kx + 2\ell z \\ 0 & -2z & -2y \\ 2kx + 2\ell z & -2y & 2\ell x \end{pmatrix} \\ &= 4\ell xz(6x + 2kz) - 4y^2(6x + 2kz) + 2z(2kx + 2\ell z)^2. \end{aligned}$$

From this we can see that $(H_F)_x(0, 1, 0) = -24$ is nonzero, so the Hessian curve $\{H_F = 0\}$ is smooth at $[0, 1, 0]$ and its tangent line there is different from $T_pC = \{z = 0\}$. Proposition 8.10 therefore says that $I_p(C, \{H_F = 0\}) = 1$. But we chose our inflection point $p \in C$ arbitrarily, so this holds for *any* inflection point of C , and since $\deg(F) = \deg(H_F) = 3$ we apply the strong form of Bézout's theorem to conclude the following.

Theorem 8.28

A smooth cubic curve over an algebraically closed field $\mathbb{K}$ of characteristic zero has exactly nine inflection points.

9 Riemann surfaces

We saw in Theorem 4.16 that a smooth plane curve in $\mathbb{P}^2(\mathbb{C})$ has interesting topological structure, beyond what we normally associate to projective plane curves over other fields: it is a *Riemann surface*. Our goal in this section is to investigate what the algebraic structure tells us about the topology, and vice versa.

9.1 Basic definitions

The following is a special case of the definition of a manifold.

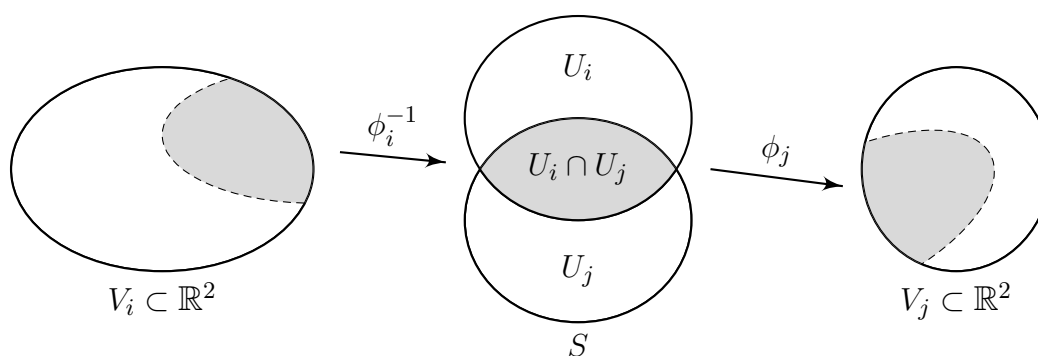
Definition 9.1. A *surface* is a Hausdorff space S together with a countable *atlas* $\{(U_i, \phi_i)\}$, where

- the sets U_i form an open cover of S ;
- each $\phi_i : U_i \rightarrow V_i$ is a homeomorphism onto an open set $V_i \subset \mathbb{R}^2$;
- if $U_i \cap U_j$ is nonempty, then the *transition map*

$$\phi_j \circ \phi_i^{-1} : \underbrace{\phi_i(U_i \cap U_j)}_{\subset V_i \subset \mathbb{R}^2} \xrightarrow{\cong} \underbrace{\phi_j(U_i \cap U_j)}_{\subset V_j \subset \mathbb{R}^2}$$

is continuous. (This is automatically true since ϕ_i and ϕ_j are homeomorphisms, but we'll impose stronger conditions on these maps later.)

The transition maps are defined on the shaded regions in the picture below:



Their continuity allows us to make sense of continuous maps on S . If we ask for the transition maps $\phi_j \circ \phi_i^{-1}$ to be differentiable, we can apply notions from real analysis. Even better, we can identify $\mathbb{R}^2 \cong \mathbb{C}$ and attempt to do complex analysis:

Definition 9.2. A *Riemann surface* is a surface with an atlas $\{(U_i, \phi_i)\}$, where the image of each $\phi : U_i \rightarrow V_i$ is an open subset of $\mathbb{C}$ and the transition maps

$$\phi_j \circ \phi_i^{-1} : \underbrace{\phi_i(U_i \cap U_j)}_{\subset V_i \subset \mathbb{C}} \xrightarrow{\cong} \underbrace{\phi_j(U_i \cap U_j)}_{\subset V_j \subset \mathbb{C}}$$

are holomorphic.

Example 9.3. The projective line $\mathbb{P}^1(\mathbb{C})$ is also known as the *Riemann sphere*. We give it an atlas

$$\{(U_0, \phi_0), (U_1, \phi_1)\},$$

where each U_i and $\phi_i : U_i \rightarrow V_i = \mathbb{C}$ has the form

$$\begin{aligned} U_0 &= \{[x, y] \in \mathbb{P}^1(\mathbb{C}) \mid x \neq 0\}, & \phi_0([x, y]) &= \frac{y}{x} \\ U_1 &= \{[x, y] \in \mathbb{P}^1(\mathbb{C}) \mid y \neq 0\}, & \phi_1([x, y]) &= \frac{x}{y}. \end{aligned}$$

Then $U_0 \cap U_1$ consists of all $[x, y] \in \mathbb{P}^1(\mathbb{C})$ with x and y both nonzero; the transition map $\phi_1 \circ \phi_0^{-1}$ has the form

$$\begin{aligned} \phi_0(U_0 \cap U_1) &\xrightarrow{\phi_0^{-1}} U_0 \cap U_1 \xrightarrow{\phi_1} \phi_1(U_0 \cap U_1) \\ z &\longmapsto [1, z] \longmapsto \frac{1}{z}, \end{aligned}$$

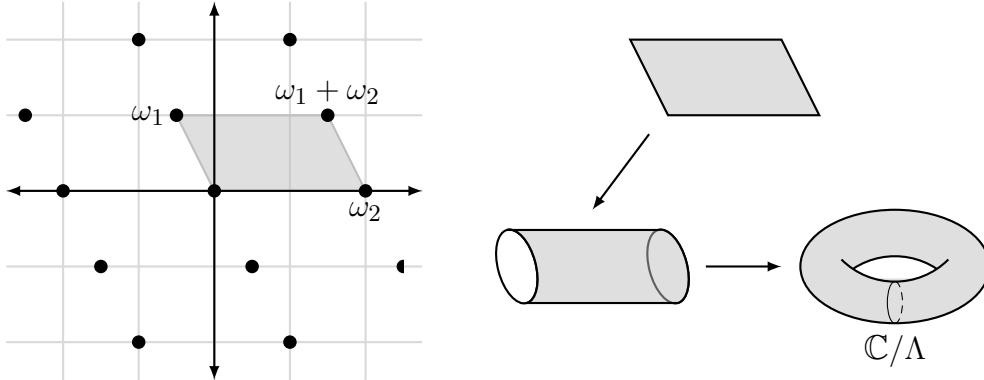
and likewise for $\phi_0 \circ \phi_1^{-1}$. These are holomorphic, so $\mathbb{P}^1(\mathbb{C})$ is a Riemann surface.

Example 9.4. Choose two nonzero complex numbers $\omega_1, \omega_2 \in \mathbb{C}$ whose ratio is not a real number, and define the lattice

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 = \{m\omega_1 + n\omega_2 \mid m, n \in \mathbb{Z}\}.$$

Then the quotient $\mathbb{C}/\Lambda$ is a torus, which we can describe by taking the parallel-

ogram with vertices $0, \omega_1, \omega_2, \omega_1 + \omega_2$ and gluing opposite sides.



We claim that $\mathbb{C}/\Lambda$ is a Riemann surface, which will require us to construct a suitable atlas; in what follows, we will let $\pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda$ be the projection map.

Given any fixed $z \in \mathbb{C}/\Lambda$, we fix a preimage $z_0 \in \pi^{-1}(z) \subset \mathbb{C}$, and let $V \subset \mathbb{C}$ be an open disk centered at z_0 of radius $\epsilon > 0$. If ϵ is small enough then the projection $\pi|_{\bar{V}} : \bar{V} \rightarrow \mathbb{C}/\Lambda$ is a homeomorphism onto its image, so we take $U = \pi(V)$ and $\varphi : U \rightarrow V$ the inverse of $\pi|_V$. Now some finite collection $\{U_i\}$ of these sets suffices to cover the compact $\mathbb{C}/\Lambda$, and we take the corresponding pairs $\{(U_i, \phi_i : U_i \rightarrow V_i)\}$ as an atlas.

With this atlas at hand, the transition maps $\phi_j \circ \phi_i^{-1}$ all have the form $z \mapsto z + c_{ij}$ for some fixed $c_{ij} \in \mathbb{C}$ (exercise: why?), and these are holomorphic. Thus $\mathbb{C}/\Lambda$ is indeed a Riemann surface.

In order to understand when a projective plane curve is a Riemann surface, we will need the following theorem.

Theorem 9.5: Holomorphic implicit function theorem

Let $f(x, y)$ be a holomorphic function in two variables, and suppose for some $(a, b) \in \mathbb{C}^2$ that

$$f(a, b) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(a, b) \neq 0.$$

Then there is an open neighborhood $\Omega \subset \mathbb{C}$ of a , and a holomorphic function $g : \Omega \rightarrow \mathbb{C}$ with $g(a) = b$, such that the graph

$$\{(x, g(x)) \mid x \in \Omega\}$$

of g coincides with the zero set $f^{-1}(0)$ on a neighborhood of (a, b) .

Theorem 9.6

Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a smooth plane curve. Then C is a Riemann surface.

Proof. We need to define an atlas for $C = \{F(x, y, z) = 0\}$. At any point $p = [a, b, c] \in C$, one of the coordinates is nonzero: supposing $c \neq 0$, we can pass to the affine chart $\{z = 1\}$ and think of C near p as the affine plane curve

$$\{f(x, y) = 0\} \subset \mathbb{C}^2, \quad f(x, y) = F(x, y, 1).$$

Since C is smooth, we know by Proposition 4.13 that $p = (a, b)$ is a smooth point of this affine curve. Supposing without loss of generality that $f_y(a, b) \neq 0$, Theorem 9.5 says that we have an open neighborhood $V \subset \mathbb{C}$ of a and a holomorphic function $g : V \rightarrow \mathbb{C}$ such that on some neighborhood W of p , we have

$$C \cap W = \{(x, g(x)) \mid x \in V\}.$$

We let $U = C \cap W$, and then we define $\phi : U \rightarrow V$ by $\phi(x, y) = x$. Finitely many of these suffice to cover C , since Proposition 3.14 guarantees that C is compact, so we take the corresponding finite collection $\{(U_i, \phi_i)\}$ to be our atlas.

This definition does not appear to use the holomorphic g , but in fact we need it to check that the transition maps are holomorphic. For example, if $f_x(a, b)$ and $f_y(a, b)$ are both nonzero, then we could have constructed two different charts: either we use $f_y(a, b) \neq 0$ to write C locally as

$$U = \{(x, g(x)) \mid x \in V\}, \quad \phi(x, y) = x,$$

or we use $f_x(a, b) \neq 0$ to write C locally as

$$U' = \{(h(y), y) \mid y \in V'\}, \quad \phi'(x, y) = y.$$

These charts overlap near (a, b) , with $h(g(x)) = x$ and $g(h(y)) = y$; we have

$$\begin{array}{ccccc} \phi(U \cap U') & \xrightarrow{\phi^{-1}} & U \cap U' & \xrightarrow{\phi'} & \phi'(U \cap U') \\ x & \longmapsto & (x, g(x)) & \longmapsto & g(x), \end{array}$$

so this transition map is holomorphic since g is, and likewise for $\phi \circ (\phi')^{-1} = h$. Any other transition map between overlapping charts either has this form or is the identity, so C is a Riemann surface after all. $\square$

Remark 9.7. The proof of Theorem 9.6 only uses local properties of C near smooth points. In particular, if C is a *singular*, irreducible projective plane curve, then its set $\text{Sing}(C)$ of singular points is finite by Proposition 7.1, and the same argument tells us that $C \setminus \text{Sing}(C)$ is also a Riemann surface, even though it is not compact.

9.2 Holomorphic maps

Now that we have many examples of Riemann surfaces, we can define appropriate notions functions and maps between them.

Definition 9.8. Let S be a Riemann surface, with an atlas $\{(U_i, \phi_i : U_i \rightarrow V_i)\}$. A continuous function

$$f : S \rightarrow \mathbb{C}$$

is *holomorphic* if each composition

$$V_i \xrightarrow{\phi_i^{-1}} U_i \xrightarrow{f} \mathbb{C}$$

is a holomorphic function.

Proposition 9.9. If S is a compact, connected Riemann surface, then any holomorphic function $f : S \rightarrow \mathbb{C}$ is constant.

Proof. Supposing that f is not constant, the modulus $|f|$ is a continuous function on S , which is maximized at some $p \in S$ since S is compact. We take any chart

$$\phi : U \rightarrow V \subset \mathbb{C}$$

where $p \in U$, and then $f \circ \phi^{-1} : V \rightarrow \mathbb{C}$ is a holomorphic function with $|f \circ \phi^{-1}|$ maximized at the interior point $\phi(p)$ of V . The maximum modulus principle now says that $f \circ \phi^{-1}$ is constant on V , hence f is constant on the neighborhood U of p .

Letting $c = f(p)$, it follows that $f^{-1}(c)$ is an open subset of C , but it is also closed. Thus $f^{-1}(c)$ contains the connected component containing p , which is S itself. $\square$

Proposition 9.9 says that there are no interesting holomorphic functions on a smooth plane curve, but there *are* interesting maps to other Riemann surfaces:

Definition 9.10. Let S and S' be Riemann surfaces, with respective atlases

$$\{(U_i, \phi_i : U_i \rightarrow V_i)\} \quad \text{and} \quad \{(W_i, \psi_i : W_i \rightarrow X_i)\}.$$

A continuous map $f : S \rightarrow S'$ is *holomorphic* at $p \in S$ if each composition

$$V_i \xrightarrow{\phi_i^{-1}} U_i \xrightarrow{f} W_j \xrightarrow{\psi_i} X_j, \quad p \in U_i, f(p) \in W_j$$

is holomorphic. If this holds at every $p \in S$ then f is *holomorphic*. If f is also a bijection with holomorphic inverse, then it is *biholomorphic*.

Example 9.11. Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a smooth plane curve with $[0, 0, 1] \notin C$. Then the projection $f : C \rightarrow \mathbb{P}^1(\mathbb{C})$ defined by

$$f([x, y, z]) = [x, y]$$

is a holomorphic map.

For example, if $C = \{x^2 + y^2 + z^2 = 0\}$ then this is a 2-to-1 map almost everywhere: the preimage

$$f^{-1}([x, y]) = \left\{ [x, y, \pm \sqrt{-(x^2 + y^2)}] \right\}$$

consists of two points if and only if $x^2 + y^2 \neq 0$. The remaining points of $\mathbb{P}^1(\mathbb{C})$ are $[x, y] = [1, \pm i]$, where $f^{-1}([1, \pm i]) = [1, \pm i, 0]$ is a single point.

Exercise 9.12. Given holomorphic maps $f : S \rightarrow S'$ and $g : S' \rightarrow S''$ between Riemann surfaces, prove that $g \circ f : S \rightarrow S''$ is also holomorphic.

9.3 Meromorphic functions

We recall some definitions from complex analysis. The key fact is that holomorphic functions $f : U \rightarrow \mathbb{C}$ defined on an open set $U \subset \mathbb{C}$ are *analytic*: at every point $a \in U$, we can write f as a power series

$$f(z) = \sum_{j=0}^{\infty} c_j(z-a)^j = c_0 + c_1(z-a)^1 + c_2(z-a)^2 + \dots$$

that converges on an open neighborhood of a .

Definition 9.13. Suppose that f is holomorphic at $a \in U$, and that for some $k \geq 1$ we have a power series

$$f(z) = \sum_{j=k}^{\infty} c_j(z-a)^j = c_k(z-a)^k + c_{k+1}(z-a)^{k+1} + \dots, \quad c_k \neq 0.$$

Then we say that f has a *zero of order k* at a .

Similarly, a holomorphic function $f : U \setminus \{a\} \rightarrow \mathbb{C}$ has a Laurent series expansion

$$f(z) = \sum_{j=-\infty}^{\infty} c_j(z-a)^j = \dots + c_{-1}(z-a)^{-1} + c_0 + c_1(z-a)^1 + \dots$$

that converges on a (punctured) neighborhood of a .

Definition 9.14. We say that f has a *pole of order k* at a , where $k \geq 1$, if $c_{-k} \neq 0$ and $c_j = 0$ for all $j < -k$; that is, if

$$f(z) = \sum_{j=-k}^{\infty} c_j(z-a)^j, \quad c_{-k} \neq 0.$$

If instead $c_j = 0$ for infinitely many $j < 0$, then we say that f has an *essential singularity* at a .

We can deduce from these power series and Laurent series expansions that both zeroes and poles of f are necessarily isolated, unless f is identically zero. We will adopt the notation that if f has a pole at a , then $f(a) = \infty$; then we can think of a map with some collection of poles (but no essential singularities) as a well-defined map to $\mathbb{C} \cup \{\infty\}$, which we will interpret as $\mathbb{P}^1(\mathbb{C})$.

Definition 9.15. A function $f : U \rightarrow \mathbb{C} \cup \{\infty\}$ is *meromorphic* if there are finitely many points $z_1, \dots, z_k \in U$ such that f is holomorphic on $U \setminus \{z_1, \dots, z_k\}$ and has poles at each of the z_i .

This leads naturally to the following analogue of Definition 9.8.

Definition 9.16. Let S be a Riemann surface with atlas $\{(U_i, \phi_i : U_i \rightarrow V_i)\}$. A function

$$f : S \rightarrow \mathbb{C} \cup \{\infty\}$$

is *meromorphic* if each composition

$$V_i \xrightarrow{\phi_i^{-1}} U_i \xrightarrow{f} \mathbb{C} \cup \{\infty\}$$

is a meromorphic function.

We have seen that compact Riemann surfaces do not admit holomorphic functions, other than constant ones; a much harder theorem asserts that every compact Riemann surface *does* have non-constant meromorphic functions. We will give a proof in the case of algebraic curves.

Proposition 9.17. *Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a smooth projective plane curve. Then there exist non-constant, meromorphic functions $f : C \rightarrow \mathbb{C}$.*

Proof. Write $C = \{F(x, y, z) = 0\}$, where $F \in \mathbb{C}[x, y, z]$ is an irreducible homogeneous polynomial. Pick any two irreducible, relatively prime polynomials $G, H \in \mathbb{C}[x, y, z]$ of the same degree, such that F does not divide H . Then we

define

$$f([x, y, z]) = \frac{G(x, y, z)}{H(x, y, z)}.$$

This is well-defined, meaning that it does not depend on our choice of coordinates $[x, y, z]$, precisely because G and H are homogeneous and $\deg(G) = \deg(H)$.

The denominator $H(x, y, z)$ only vanishes at finitely many points of C by Bézout's theorem; away from these points the function f is holomorphic, because C is covered by charts on which f can be expressed as a rational function. At the zeroes of H it has only poles rather than essential singularities, because in these same charts we can multiply f by a polynomial (namely H) to make it holomorphic. Thus f is meromorphic. $\square$

Proposition 9.18. *Let S be a connected Riemann surface. Then there is a bijection*

$$\left\{ \begin{array}{l} \text{meromorphic functions} \\ f : S \rightarrow \mathbb{C} \cup \{\infty\} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{holomorphic maps } \tilde{f} : S \rightarrow \mathbb{P}^1(\mathbb{C}) \\ \text{other than } g(x) = [1, 0] \end{array} \right\},$$

defined by sending a meromorphic $f : S \rightarrow \mathbb{C}$ to the map

$$\tilde{f}(p) = \begin{cases} [f(p), 1], & f(p) \in \mathbb{C} \\ [1, 0], & f(p) = \infty. \end{cases}$$

Proof. We will equip $\mathbb{P}^1(\mathbb{C})$ with the atlas $\{(U_0, \phi_0), (U_1, \phi_1)\}$ from Example 9.3, and use this to show that $\tilde{f}$ is holomorphic. Then it is clear that the map $f \mapsto \tilde{f}$ is a bijection, with inverse taking a holomorphic map $\tilde{f} = [x, y] \not\equiv [1, 0]$ (for suitable $x, y : S \rightarrow \mathbb{C}$ that are never simultaneously zero) to the meromorphic $f = \frac{x}{y}$.

Fix a point $a \in S$, and suppose for now that $f(a) \neq \infty$. Then there is a chart $\psi : U \rightarrow V$, with $a \in U \subset S$ and $V \subset \mathbb{C}$, which we may shrink if needed to assume that $f(U) \subset \mathbb{C}$. The image $\tilde{f}(U)$ lands in the chart $U_1 = \{[x, y] \mid y \neq 0\}$ on $\mathbb{P}^1(\mathbb{C})$, and the composition

$$V \xrightarrow{\psi^{-1}} U \xrightarrow{f|_U} \mathbb{C} \xrightarrow{\phi_1^{-1}} \mathbb{P}^1(\mathbb{C})$$

is equal to $\tilde{f} \circ \psi^{-1}$ on V . The composition $f|_U \circ \psi^{-1} : V \rightarrow \mathbb{C}$ is holomorphic, since f was assumed to be a holomorphic function on U , so it follows that $\tilde{f}|_U$ is holomorphic as well.

Now suppose instead that $f(a) = \infty$, and take a chart $\psi : U \rightarrow V$ where $a \in U$. Then the map $(f \circ \psi^{-1}) : V \rightarrow \mathbb{C} \cup \{\infty\}$ is meromorphic, with a pole at b of some order $k \geq 1$, and near b it has the form

$$(f \circ \psi^{-1})(z) = \sum_{j=-k}^{\infty} c_j(z-b)^j = \frac{1}{(z-b)^k} \sum_{j=0}^{\infty} c_{j+k}(z-b)^j$$

with $c_{-k} \neq 0$. Shrinking U and V so that $f|_U$ is never zero, the composition

$$V \xrightarrow{\psi^{-1}} U \xrightarrow{\frac{1}{f}} \mathbb{C}$$

is now holomorphic, with a zero of order k at b . It sends $x \neq b$ to $\frac{1}{f(\psi^{-1}(x))}$, and b to 0. But then the composition

$$V \xrightarrow{\psi^{-1}} U \xrightarrow{\frac{1}{f}} \mathbb{C} \xrightarrow{\phi_0^{-1}} \mathbb{P}^1(\mathbb{C})$$

sends $x \neq b$ to $[1, \frac{1}{f(\psi^{-1}(x))}] = [f(\psi^{-1}(x)), 1]$, and it sends b to $\phi_0^{-1}(0) = [1, 0]$, so it is equal to $\tilde{f} \circ \psi^{-1}$ on V . Thus $\tilde{f}$ is holomorphic on U in this case as well, and so it is holomorphic everywhere. $\square$

Thus we have shown that any smooth projective plane curve $C \subset \mathbb{P}^2(\mathbb{C})$ admits non-constant holomorphic maps to $\mathbb{P}^1(\mathbb{C})$.

9.4 Degrees of holomorphic maps

Holomorphic maps between compact Riemann surfaces are much better behaved than arbitrary smooth maps. For example:

Proposition 9.19. *Let $f : S \rightarrow S'$ be a non-constant holomorphic map between compact, connected Riemann surfaces. Then $f^{-1}(c)$ is finite for every $c \in S'$.*

Proof. Suppose that some $f^{-1}(c)$ is infinite. Since S is compact, the set $f^{-1}(c) \subset S$ has an accumulation point a , which must also lie in the closed set $f^{-1}(c)$. Passing to charts $\phi : U \rightarrow V$ around $a \in S$ and $\phi' : U' \rightarrow V'$ around $f(a) = c \in S'$, and shrinking U and V if needed so that $f(U) \subset U'$, we get a holomorphic map

$$g : V \xrightarrow{\phi^{-1}} U \xrightarrow{f|_U} U' \xrightarrow{\phi'} V'.$$

The holomorphic function $h : V \rightarrow \mathbb{C}$ defined by $h(z) = g(z) - \phi'(c)$ is zero on all of $\phi(f^{-1}(c) \cap U)$, which includes the accumulation point $\phi(a)$, so $\phi(a)$ is a non-isolated zero of h and this implies that h must be identically zero. But then g is constant on V , so f is constant on the whole neighborhood U of a .

This argument shows that the set A of accumulation points of $f^{-1}(c)$ is an open subset of S . At the same time, by continuity we have $A \subset f^{-1}(c)$ and so any accumulation point of A is again an element of A : in other words, A is also closed. Thus A must be all of S , contradicting the assumption that f is non-constant. $\square$

To understand the cardinality of $f^{-1}(c)$, we must find suitable local models for f near each point. We write $\mathbb{D} \subset \mathbb{C}$ for the open unit disk centered at the origin.

Proposition 9.20. *Let $f : S \rightarrow S'$ be a holomorphic map between Riemann surfaces, with $f(a) = b$. Suppose that f is not constant on a neighborhood of a . Then there exist*

- an integer $n \geq 1$,
- a chart $\phi : U \rightarrow \mathbb{D}$ around a , with $\phi(a) = 0$;
- and a chart $\phi' : U' \rightarrow \mathbb{D}$ around b , with $\phi'(b) = 0$

such that $f(U) \subset U'$ and the diagram

$$\begin{array}{ccc} U & \xrightarrow{f} & U' \\ \phi \downarrow & & \downarrow \phi' \\ \mathbb{D} & \xrightarrow{z \mapsto z^n} & \mathbb{D} \end{array}$$

commutes. Moreover, the integer n is uniquely determined by f near a .

Proof. We start by choosing our charts ϕ and ϕ' arbitrarily, and then translating their codomains to ensure that $\phi(a) = 0$ and $\phi'(b) = 0$. Then since the composition $\phi' \circ f \circ \phi^{-1}$ is holomorphic and non-constant and it sends 0 to 0, it admits a power series of the form

$$(\phi' \circ f \circ \phi^{-1})(z) = c_n z^n + c_{n+1} z^{n+1} + \cdots = c_n z^n g(z),$$

where $n \geq 1$ and $c_n \neq 0$, and

$$g(z) = 1 + \frac{c_{n+1}}{c_n} z + \frac{c_{n+2}}{c_n} z^2 + \cdots.$$

For $|z|$ sufficiently small, we know that $g(z)$ stays arbitrarily close to 1, so we can take an n th root $h(z) = g(z)^{1/n}$, which is also holomorphic, and which satisfies $h(0) \neq 0$ since $g(0) = 1$. Letting d be an n th root of c_n , it follows that in these coordinates we have

$$(\phi' \circ f \circ \phi^{-1})(z) = (d \cdot z \cdot h(z))^n$$

for all z in some small neighborhood of 0.

The function $k(z) = dz \cdot h(z)$ satisfies $k(0) = 0$ and $k'(0) = dh(0) \neq 0$, so we can apply the holomorphic implicit function theorem (Theorem 9.5) to $F(x, y) = x - k(y)$ to get a holomorphic function $\tilde{k}$, defined on a neighborhood of 0, such that $x - k(y) = 0$ precisely when $y = \tilde{k}(x)$: in other words, $\tilde{k}$ is locally an inverse to k .

We use $\tilde{k}$ to reparametrize the chart ϕ , and this replaces $(\phi' \circ f \circ \phi^{-1})(z) = k(z)^n$ with $(\phi' \circ f \circ \phi^{-1})(z) = z^n$ on a small enough neighborhood of 0. Now we rescale each chart to get the desired map $\mathbb{D} \rightarrow \mathbb{D}$.

The only remaining claim is the uniqueness of n . In fact, we can describe n topologically: the above model for f near a , of the form $z \mapsto z^n$, implies that

- given *any* neighborhoods U of a and U' of b , there are points $y \in U'$ with at least n different preimages in U , meaning that $|f^{-1}(y) \cap U| \geq n$;
- and there are neighborhoods U and U' of a and b such that no point of U' has more than n preimages in U .

Thus the integer n is uniquely determined. $\square$

Definition 9.21. Let $f : S \rightarrow S'$ be a holomorphic map between Riemann surfaces, and suppose that f is not constant on a neighborhood of the point $s \in S$. We define the *ramification index* of f at s by

$$v_f(s) = n,$$

where $n \geq 1$ is the unique integer provided by Proposition 9.20 such that f is modeled by $z \mapsto z^n$ at s .

If $v_f(s) > 1$, then we say that s is a *ramification point* of f .

Lemma 9.22. If $f : S \rightarrow S'$ is holomorphic, then its ramification points are isolated.

Proof. At any ramification point $s \in S$, the map f can be written in local coordinates as $z \mapsto z^n$, and the only ramification point of this model is $z = 0$. Thus f has no other ramification points in the corresponding neighborhood of s . $\square$

Example 9.23. Let $C = \{x^2 + y^2 + z^2 = 0\} \subset \mathbb{P}^2(\mathbb{C})$ and consider the map $f : C \rightarrow \mathbb{P}^1$ given by

$$f([x, y, z]) = [x, y],$$

as in Example 9.11. We saw that each $[x, y] \in \mathbb{P}^1$ has preimages

$$f^{-1}([x, y]) = \left\{ [x, y, \pm \sqrt{-(x^2 + y^2)}] \right\},$$

consisting of exactly two points unless $[x, y] = [1, \pm i]$.

We first consider $p = [1, i, 0]$. We can define a chart $\phi : U \rightarrow \mathbb{D}$ as the inverse of

$$\phi^{-1}(z) = [1, i\sqrt{1+z^2}, z],$$

where on $\mathbb{D}$ we have chosen the branch of $\sqrt{1+z^2}$ with nonnegative real part so that $\phi^{-1}(0) = [1, i, 0]$. (Here we are defining U as the image $\phi^{-1}(\mathbb{D})$.) In the local charts U on C and U_0 on $\mathbb{P}^1$, we have

$$\begin{array}{ccccccc} \mathbb{D} & \xrightarrow{\phi^{-1}} & U & \xrightarrow{f} & \mathbb{P}^1 & \xrightarrow{\phi_0} & \mathbb{C} \\ z & \longmapsto & [1, i\sqrt{1+z^2}, z] & \longmapsto & [1, i\sqrt{1+z^2}] & \longmapsto & i\sqrt{1+z^2}. \end{array}$$

After subtracting i from the latter and applying the binomial theorem, it follows that f can be modeled near $[1, i, 0]$ by

$$z \mapsto i \left(\sqrt{1+z^2} - 1 \right) = \frac{i}{2}z^2 + O(z^3).$$

Thus $v_f([1, i, 0]) = 2$, and similarly $v_f([1, -i, 0]) = 2$.

Any other point $p \in C$ lies in the affine chart $\{z \neq 0\}$, say $p = [x, y, 1]$, and then the other point of $f^{-1}([x, y])$ is $[x, y, -1] = [-x, -y, 1]$. With this in mind we can take a very small neighborhood $U \subset C$ around p such that $f|_U$ is injective, and it follows that $v_f(p) = 1$.

Proposition 9.24. *Given a holomorphic map $f : S \rightarrow S'$ between compact Riemann surfaces, define a function $d_f : S' \rightarrow \mathbb{Z}_{\geq 0}$ by*

$$d_f(s') = \sum_{s \in f^{-1}(s')} v_f(s).$$

Then d_f is locally constant.

Proof. The sum is well-defined, because Proposition 9.19 says that $f^{-1}(s')$ is always finite. Now if we fix $s' \in S'$ and write $f^{-1}(s') = \{s_1, \dots, s_k\}$, we can simultaneously choose disjoint neighborhoods $U_i \subset S$ of each s_i and a neighborhood $U' \subset S'$ of s' , so that each $f|_{U_i} : U_i \rightarrow U'$ has a local model of the form $z \mapsto z^{n_i}$ where $n_i = v_f(s_i)$.

If $s'' \in S'$ is sufficiently close to s' , so that $f^{-1}(s'') \subset \bigcup_i U_i$, then by the above models it will have exactly n_i preimages inside each U_i , all with $v_f = 1$. Thus

$$d_f(s'') = \sum_{i=1}^k \sum_{s \in f^{-1}(s'') \cap U_i} \underbrace{v_f(s)}_{=1} = \sum_{i=1}^k n_i = d_f(s'). \quad \square$$

Since the function d_f is locally constant, it is in fact constant whenever S' is connected, we can turn its unique value into an invariant of the map f .

Definition 9.25. Let $f : S \rightarrow S'$ be a holomorphic map between compact, connected Riemann surfaces. Then the *degree* of f , denoted $\deg(f)$, is defined as the value of $d_f(s')$ for any $s' \in S'$.

Lemma 9.26. Let $f : S \rightarrow S'$ be a holomorphic map between compact Riemann surfaces. Then for all but finitely many $s' \in S'$ we have

$$\deg(f) = |f^{-1}(s')|.$$

Proof. Lemma 9.22 and the compactness of S tell us that f has only finitely many ramification points $s_1, \dots, s_k$. Thus for any point

$$s' \in S' \setminus \{f(s_1), f(s_2), \dots, f(s_k)\},$$

the preimage $f^{-1}(s')$ has no ramification points, and so we compute that

$$d_f(s') = \sum_{s \in f^{-1}(s')} v_f(s) = \sum_{s \in f^{-1}(s')} 1 = |f^{-1}(s')|. \quad \square$$

Proposition 9.27. Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a smooth plane curve of degree $d \geq 2$ that does not pass through the point $[0, 0, 1]$. Then the holomorphic map $f : C \rightarrow \mathbb{P}^1$ defined by $f([x, y, z]) = [x, y]$ has degree d .

Proof. A point of C lies in the preimage $f^{-1}([a, b])$ if and only if it has the form $[a, b, c]$ for some $c \in \mathbb{C}$, or equivalently if and only if it lies on the line $L_{[a, b]} = \{bx - ay = 0\}$. Lemma 9.26 thus tells us that

$$\deg(f) = |L_{[a, b]} \cap C|$$

for all but finitely many $[a, b]$, so the proposition will follow if we can also prove that the right side also equals $d = \deg(C)$ for all but finitely many $[a, b]$.

According to Bézout's theorem, we have

$$\deg(C) = \sum_{p \in L_{[a, b]} \cap C} I_p(L_{[a, b]}, C).$$

Since each such $p = [a, b, c]$ is a smooth point of the curve $C = \{F(x, y, z) = 0\}$, Corollary 8.11 tells us that $I_p(L_{[a, b]}, C) > 1$ if and only if $L_{[a, b]}$ is the tangent line $T_p C$, which means that $T_p C = \{bx - ay = 0\}$ and so $F_z(a, b, c) = 0$. In this case

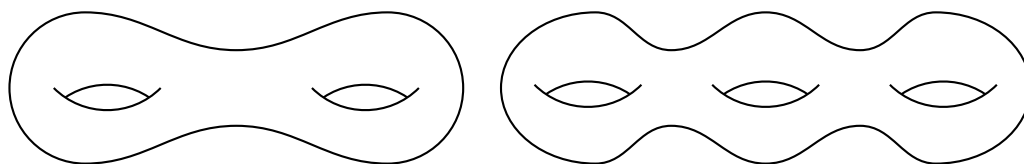
p lies at the intersection of the irreducible curve C and $\{F_z = 0\}$, so by Bézout's theorem there are only finitely many such p . (Exercise: why does $d \geq 2$ imply that F_z is not identically zero?) Thus for all but finitely many $[a, b]$ the line $L_{[a,b]}$ always meets C with intersection multiplicity 1. In other words, we have

$$\deg(C) = |L_{[a,b]} \cap C|$$

for all but finitely many $[a, b]$, and so $\deg(C) = \deg(f)$. $\square$

9.5 The Riemann–Hurwitz formula

Let S be a compact, connected Riemann surface. Then S is in particular a smooth orientable surface, and these have been classified up to homeomorphism since the mid-19th century. Every such S is homeomorphic to a 2-sphere with some number $g \geq 0$ of handles attached, and this number g uniquely determines S up to homeomorphism; it is called the *genus* of S . The surfaces shown below have genus 2 and 3, respectively.



We can compute the genus of a given surface in terms of any triangulation, which is a convenient combinatorial decomposition of the surface into triangles.

Definition 9.28. A *triangulation* of a surface S is a triple (V, E, F) , where

- the set V consists of some points of S , called *vertices*;
- the set E of *edges* consists of the images of embeddings $[0, 1] \hookrightarrow S$, such that
 - each edge intersects V exactly at its endpoints;
 - the interiors of any two edges are disjoint;
 - only finitely many edges meet at each vertex;
- and each element of F , called a *face*, is the closure of a component of $S \setminus (V \cup E)$ and contains exactly three edges, such that
 - each face contains exactly three edges;
 - each edge belongs to exactly two faces; and

- the intersection of any two faces is either empty, a single vertex, or a single edge.

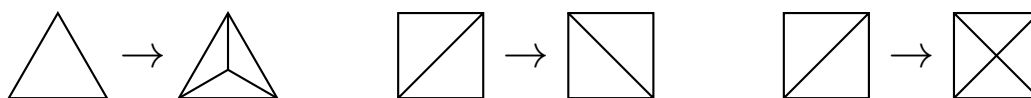
Theorem 9.29: Radó (1925)

Every compact Riemann surface admits a triangulation.

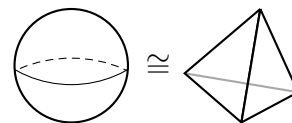
If S has a triangulation (V, E, F) with finitely many faces, then it also has finitely many edges and vertices, and we can define the *Euler characteristic*

$$\chi(S) = |V| - |E| + |F|.$$

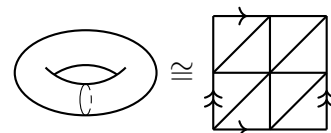
We can check that this does not change when we make simple changes to the triangulation, such as a barycentric subdivision (also known as a “1-3 Pachner move”), a “2-2 Pachner move”, or an edge subdivision, all of which are pictured below.



Example 9.30. The sphere S^2 , which has genus 0, is homeomorphic to a tetrahedron. This gives rise to a triangulation with 4 vertices, 6 edges, and 4 faces, so $\chi(S^2) = 4 - 6 + 4 = 2$.



Example 9.31. The torus T^2 , which has genus 1, can be built from a square by gluing pairs of opposite sides. Dividing this square into four smaller squares and dividing each of these into two triangles gives a triangulation with 4 vertices, 12 edges, and 8 faces, so $\chi(T^2) = 4 - 12 + 8 = 0$.

**Theorem 9.32**

The Euler characteristic of a compact Riemann surface does not depend on the choice of triangulation. If S is homeomorphic to a surface of genus g , then $\chi(S) = 2 - 2g$.

Proof (sketch). One way to prove that $\chi(S)$ does not depend on (V, E, F) is to borrow from the algebraic topology module: we can define the simplicial homology

$H_*^\Delta(S)$ and singular homology H_* of S and prove that they are isomorphic. The former is by construction finitely generated and satisfies

$$\chi(S) = |V| - |E| + |F| = \text{rank}(H_0^\Delta(S)) - \text{rank}(H_1^\Delta(S)) + \text{rank}(H_2^\Delta(S)),$$

while the latter is an invariant of S , and since $H_*^\Delta(S) \cong H_*(S)$ it follows that $\chi(S)$ is an invariant as well.

We have already proved in Examples 9.31 and 9.30 that $\chi(S) = 2 - 2g$ if S has genus $g \leq 1$, so we proceed by induction. Given a surface S_g of genus $g \geq 0$, one way to produce a surface S_{g+1} of genus $g+1$ is to take triangulations (V, E, F) of S_g and (V', E', F') of T^2 , remove a face from each, and glue S_g and T^2 together along the boundaries of these faces. This naturally produces S_{g+1} with a triangulation (V'', E'', F'') , where

$$\begin{aligned} |V''| &= |V| + |V'| - 3, \\ |E''| &= |E| + |E'| - 3, \\ |F''| &= |F| + |F'| - 2 \end{aligned}$$

and so the Euler characteristic of S_{g+1} is given by

$$\begin{aligned} \chi(S_{g+1}) &= (|V| + |V'| - 3) - (|E| + |E'| - 3) + (|F| + |F'| - 2) \\ &= (|V| - |E| + |F|) + (|V'| - |E'| + |F'|) - 2 \\ &= (2 - 2g) + (0) - 2 \\ &= 2 - 2(g + 1), \end{aligned}$$

completing the induction. □

The Riemann–Hurwitz formula relates the Euler characteristics of two Riemann surfaces in terms of a holomorphic map between them.

Theorem 9.33: Riemann–Hurwitz

Let $f : S \rightarrow S'$ be a non-constant holomorphic map between compact, connected Riemann surfaces. Then

$$\chi(S) = \deg(f)\chi(S') - \sum_{p \in S} (v_f(p) - 1).$$

We remark that the sum on the right side of the Riemann–Hurwitz formula makes sense, because f has only finitely many ramification points by Lemma 9.22 and so only finitely many summands are nonzero.

Proof of Theorem 9.33. We cover S and S' by charts so that every point of S belongs to a chart where f has the form $z \mapsto z^n$ for some $n \geq 1$. We then pick a triangulation (V', E', F') of S' and refine it by subdividing faces and edges, so that

- the image $f(q)$ of every ramification point $q \in S$ belongs to V' ,
- and every face of F' is contained in some chart on S' .

We use this to build a triangulation (V, E, F) of S , where $V = f^{-1}(V')$ consists of all the preimages of the vertices of S' , and likewise $E = f^{-1}(E')$. Since each face of F' is a triangle in one of these model charts, we can check directly that its preimage in S is also a union of triangles, and so V and E determine the faces F of a triangulation of S .

Let $d = \deg(f)$. Then $d = \sum_{p \in f^{-1}(q)} v_f(p)$ for any $q \in S$, so we can compute that

$$\begin{aligned} |V| &= |f^{-1}(V')| = \sum_{q \in V'} \sum_{p \in f^{-1}(q)} 1 \\ &= \sum_{q \in V'} \left(d - \sum_{p \in f^{-1}(q)} (v_f(p) - 1) \right) \\ &= d|V'| - \sum_{p \in S} (v_f(p) - 1). \end{aligned}$$

Here we have extended the last sum from $p \in f^{-1}(V')$ to all $p \in S$, but this does not change anything because all of the ramification points already lie in $f^{-1}(V')$. Similarly, we have

$$|E| = d|E'| \quad \text{and} \quad |F| = d|F'|,$$

because the interior of every edge of E' or face of F' is disjoint from V , hence from the image of the ramification points of f , and thus has exactly d disjoint preimages. We put all of this together to get

$$\begin{aligned} \chi(S) &= |V| - |E| + |F| \\ &= \left(d|V'| - \sum_{p \in S} (v_f(p) - 1) \right) - d|E'| + d|F'| \\ &= d\chi(S') - \sum_{p \in S} (v_f(p) - 1), \end{aligned}$$

exactly as promised. □

Example 9.34. Consider the smooth plane curve $C = \{x^3 + y^3 + z^3 = 0\} \subset \mathbb{P}^2(\mathbb{C})$, which does not contain $[0, 0, 1]$, and define the map $f : C \rightarrow \mathbb{P}^1$ by $f([x, y, z]) = [x, y]$. Then $\deg(f) = \deg(C) = 3$ by Proposition 9.27.

For any point $[a, b] \in \mathbb{P}^1$, the preimage $f^{-1}([a, b])$ consists of all $[a, b, c]$ where $c^3 = -(a^3 + b^3)$, so there is one point for each third root of $a^3 + b^3$: we have

$$|f^{-1}([a, b])| = \begin{cases} 3, & a^3 + b^3 \neq 0 \\ 1, & a^3 + b^3 = 0. \end{cases}$$

The points of $\mathbb{P}^1$ with exactly one preimage are therefore $[-1, \omega]$ where $\omega^3 = 1$, and the corresponding ramification points $[-1, \omega, 0]$ must have index 3 since $\sum_{p \in f^{-1}(q)} v_f(p)$ is supposed to equal $\deg(f) = 3$. Thus

$$\chi(C) = 3\chi(\mathbb{P}^1) - \sum_{\omega^3=1} (v_f([-1, \omega, 0]) - 1) = 3 \cdot 2 - \sum_{\omega^3=1} 2 = 0.$$

But this is equal to $2 - 2g$ where g is the genus of C , so C has genus 1.

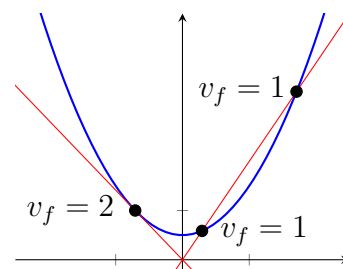
9.6 The degree-genus formula

In this section, we will study a smooth plane curve $C \subset \mathbb{P}^2(\mathbb{C})$ that does not pass through $[0, 0, 1]$, and we will apply the Riemann–Hurwitz formula to the holomorphic map

$$\begin{aligned} f : C &\rightarrow \mathbb{P}^2 \\ [x, y, z] &\mapsto [x, y] \end{aligned}$$

to determine its genus. In order to do so, we need to determine the ramification points of f together with their indices.

We can interpret $f^{-1}([a, b])$ as the intersection of C with the line $\{bx - ay = 0\}$, and Bézout’s theorem says that usually there should be exactly $\deg(C)$ points in this preimage; the exceptions occur when that line is tangent to C . Throughout the next several lemmas we will show that the points where this happens are the ramification points of f , and compare the ramification index to the intersection multiplicity at these points.



For the sake of notation we fix the smooth curve $C = \{G(x, y, z) = 0\}$ and the map f as above.

Lemma 9.35. Suppose that a point $p = [a, b, 1] \in C$ lies in a chart of the form $\phi : U \rightarrow V$, where

$$\phi^{-1}(x) = [x, h(x), 1]$$

for some holomorphic function $h : V \rightarrow \mathbb{C}$ with $h(a) = b$. Then

- p is a ramification point of f if and only if $ah'(a) = b$,
- and if p is a ramification point, then $v_f(p) = 2$ if and only if $h''(a) \neq 0$.

Proof. In these local coordinates, we can write f as a composition

$$\begin{array}{ccccccc} V & \xrightarrow{\phi^{-1}} & U & \xrightarrow{f} & \mathbb{P}^1 & \xrightarrow{\phi_0} & \mathbb{C} \\ x & \longmapsto & [x, h(x), 1] & \longmapsto & [x, h(x)] & \longmapsto & \frac{h(x)}{x} \end{array}$$

if $x \neq 0$, and as $(\phi_1 \circ f \circ \phi^{-1})(x) = \frac{x}{h(x)}$ otherwise. We will assume that $f(x) = \frac{h(x)}{x}$ in these coordinates, since the other case is essentially the same. We have

$$f'(a) = \frac{ah'(a) - h(a)}{a^2} = \frac{ah'(a) - b}{a^2},$$

which is zero if and only if $ah'(a) = b$, so from the power series

$$f(x) = b + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \dots$$

we see that $p = [a, b, 1]$ is a ramification point of f if and only if $ah'(a) = b$. In this case we similarly compute with some additional effort that

$$f''(a) = \frac{a^2 h''(a) - 2ah'(a) + 2h(a)}{a^3} = \frac{a^2 h''(a) - 2(ah'(a) - b)}{a^3} = \frac{h''(a)}{a},$$

so then $v_f(p) > 2$ if and only if $h''(a) = 0$. □

Lemma 9.36. Suppose that $p = [a, b, 1] \in C$ lies in a chart of the form $\phi^{-1}(x) = [x, h(x), 1]$, with $h : V \rightarrow \mathbb{C}$ holomorphic and $h(a) = b$. Then p is an inflection point of C if and only if $h''(a) = 0$.

Proof. If $d = \deg(G)$ is equal to 1, then every point of C is an inflection point, and $h(x)$ must be linear, so $h''(x) = 0$ for all $x \in V$. Supposing now that $d \geq 2$, we differentiate both sides of

$$g(x, h(x)) = G(x, h(x), 1) = 0$$

at $x = a$ and substitute $y = b$ to get

$$g_x(a, b) + g_y(a, b)h'(a) = 0.$$

We note that $g_y(a, b) \neq 0$, since otherwise we have $g_x(a, b) = 0$ as well and then p would not be a smooth point of C .

Now Lemma 8.25 says that the Hessian of G satisfies

$$\begin{aligned} \frac{H_G(a, b, 1)}{(d-1)^2} &= \det \begin{pmatrix} G_{xx} & G_{xy} & G_x \\ G_{yx} & G_{yy} & G_y \\ G_x & G_y & 0 \end{pmatrix} \\ &= 2g_x g_y g_{xy} - g_y^2 g_{xx} - g_x^2 g_{yy} \end{aligned}$$

and the relation $g_x(a, b) = -g_y(a, b)h'(a)$ from above turns this into

$$\begin{aligned} \frac{H_G(a, b, 1)}{(d-1)^2} &= -2g_y^2 g_{xy} h'(a) - g_y^2 g_{xx} - g_y^2 g_{yy} h'(a)^2 \\ &= -g_y(a, b)^2 (g_{xx}(a, b) + 2g_{xy}(a, b)h'(a) + g_{yy}(a, b)h'(a)^2) \\ &= -g_y(a, b)^2 \cdot \left[\frac{d^2}{dx^2} \underbrace{(g(x, h(x)))}_{=0 \text{ near } a} \right]_{x=a} - g_y(a, b)h''(a) \\ &= g_y(a, b)^3 h''(a). \end{aligned}$$

Since $g_y(a, b) \neq 0$ we conclude that $H_G(a, b, 1) = 0$ vanishes, meaning that $p = [a, b, 1]$ is an inflection point of C , if and only if $h''(a) = 0$. $\square$

Lemma 9.36 leads to a generalization of Theorem 8.28, which was about plane cubics.

Theorem 9.37

Suppose that the smooth plane curve $C \subset \mathbb{P}^2(\mathbb{C})$ has degree $d \geq 2$. Then C has at most $3d(d-2)$ inflection points.

Proof. At a point of the form $p = [a, b, 1] \in C$, we suppose without loss of generality that $g_y(a, b) \neq 0$, so that C has a chart $\phi : U \rightarrow V$ of the form $\phi^{-1}(x) = [x, h(x), 1]$ with h holomorphic.

We saw in Lemma 9.36 that $p = [a, b, 1]$ is an inflection point if and only if $h''(a) = 0$. Then if every point of U is an inflection point, we must have $h''(x) = 0$ for all $x \in V$, so $h(x) = cx + d$ for some $c, d \in \mathbb{C}$. In this case the open neighborhood $U \subset C$ belongs to the line $\{cx - y + dz = 0\}$, and this line intersects the irreducible C in infinitely many points, so Bézout's theorem says that they coincide. But then $\deg(C) = 1$, a contradiction.

It now follows that some point of U is not an inflection point. The intersection of C with $\{H_G = 0\}$ is therefore a proper subset of the irreducible C , so by Bézout's theorem it must contain at most $\deg(G) \cdot \deg(H_G) = d \cdot 3(d-2)$ points. $\square$

Proposition 9.38. *A point $p \in C$ is a ramification point of f if and only if $[0, 0, 1] \in T_p C$. If $\deg(C) \geq 2$ and $p \in C$ is a ramification point, then $v_f(p) = 2$ if and only if p is not an inflection point of C .*

Proof. We will assume throughout that $p = [a, b, 1]$. The case where $p = [a, b, 0]$ follows exactly the same outline, but requires us to redo Lemmas 9.35 and 9.36 with slightly different coordinate charts; we leave it as an exercise. Having fixed $p = [a, b, 1]$ now, we observe that $[0, 0, 1] \in T_p C$ if and only if $G_z(a, b, 1) = 0$, so we will prove that the latter is equivalent to p being a ramification point.

We can pass to the affine chart $\{z \neq 0\} \subset \mathbb{P}^2$, where C is defined by $g(x, y) := G(x, y, 1) = 0$. One of the partial derivatives $g_x(a, b)$, $g_y(a, b)$ is nonzero since p is a smooth point; assuming without loss of generality that $g_y(a, b) \neq 0$, we take a chart $\phi : U \rightarrow V$ near $[a, b, 1]$ of the form

$$\phi^{-1}(x) = [x, h(x), 1],$$

where $a \in V \subset \mathbb{C}$ and $h : V \rightarrow \mathbb{C}$ is holomorphic, with $h(a) = b$.

Lemma 9.35 now says that p is a ramification point if and only if $ah'(a) = b$. Recalling the relation

$$G_x(a, b, 1) + G_y(a, b, 1)h'(a) = 0$$

from the proof of Lemma 9.36, we can multiply by a and rearrange to get

$$aG_x(a, b, 1) + bG_y(a, b, 1) + G_z(a, b, 1) = (b - ah'(a))G_y(a, b, 1) + G_z(a, b, 1).$$

Now the left side is zero since $p \in T_p C$, so from the right side we have

$$G_z(a, b, 1) = (ah'(a) - b)G_y(a, b, 1),$$

and since $G_y(a, b, 1) = g_y(a, b)$ is nonzero it follows that $G_z(a, b, 1) = 0$ if and only if $ah'(a) = b$, which is equivalent to p being a ramification point.

Supposing now that $p = [a, b, 1]$ is a ramification point, Lemmas 9.35 and 9.36 say respectively that $v_f(p) = 2$ if and only if $h''(a) \neq 0$, and that $h''(a) \neq 0$ if and only if p is not an inflection point of C . $\square$

Theorem 9.39: Degree-genus formula

Let $C \subset \mathbb{P}^2(\mathbb{C})$ be a smooth plane curve of degree $d \geq 1$. Then C is a Riemann surface of genus

$$g = \frac{(d-1)(d-2)}{2}.$$

Exercise 9.40. Prove that the Fermat curve $C = \{x^d + y^d + z^d = 0\}$ is smooth, and therefore has genus $\frac{1}{2}(d-1)(d-2)$.

Proof of Theorem 9.39. First, when $d = 1$ we know that C is projectively equivalent to the line $\{z = 0\}$, which the map f sends biholomorphically onto $\mathbb{P}^1$, so we may assume that $d \geq 2$ from now on. We apply a projective transformation to ensure that $[0, 0, 1]$ does not lie on either C or any of the tangent lines through inflection points of C , of which there are finitely many by Theorem 9.37.

We apply the Riemann–Hurwitz formula, Theorem 9.33, to $f : C \rightarrow \mathbb{P}^1$ to get

$$\chi(C) = d\chi(\mathbb{P}^1) - \sum_{p \in C} (v_f(p) - 1).$$

Proposition 9.38 says that a point $p = [a, b, c] \in C$ is a ramification point if and only if $[0, 0, 1] \in T_p C$, or equivalently if and only if $G_z(a, b, c) = 0$; and that in this case we have $v_f(p) = 2$ since we have arranged that p is not an inflection point of C . Thus C has genus g , where

$$2 - 2g = 2d - n,$$

where n is the number of ramification points of C , and it remains to determine n .

As above, a ramification point p lies at the intersection of C and the degree- $(d-1)$ curve $D = \{G_z = 0\}$, and Bézout's theorem says that

$$d(d-1) = \sum_{p \in C \cap D} I_p(C, D).$$

We will show that $n = d(d-1)$ by proving that $I_p(C, D) = 1$ for all such p . We already know that p is a smooth point of C , and that $[0, 0, 1] \in T_p C$, so by Proposition 8.10 it will suffice to show that p is also a smooth point of D but that $[0, 0, 1] \notin T_p D$.

To see that D is smooth at $p = [a, b, c]$, we note that the Hessian

$$H_G(a, b, c) = \det \begin{pmatrix} G_{xx} & G_{xy} & G_{xz} \\ G_{yx} & G_{yy} & G_{yz} \\ G_{zx} & G_{zy} & G_{zz} \end{pmatrix}$$

is nonzero since p is not an inflection point. Thus the bottom row of the Hessian matrix is not identically zero at p , and the entries of this row are the partial derivatives of G_z at (a, b, c) , so p must be a smooth point of $D = \{G_z = 0\}$. These entries are the coefficients of the equation for $T_p D$, so $[0, 0, 1] \in T_p D$ if and only if $G_{zz}(a, b, c) = 0$. Supposing without loss of generality that $a \neq 0$ (since if $a = b = 0$ then $p \notin C$), Lemma 8.25 says that

$$\frac{a^2 H_G(a, b, c)}{(d-1)^2} = \det \begin{pmatrix} \frac{d}{d-1} G & G_y & G_z \\ G_y & G_{yy} & G_{yz} \\ G_z & G_{zy} & G_{zz} \end{pmatrix},$$

and since $p \in C \cap D$ we have $G(a, b, c) = G_z(a, b, c) = 0$, hence

$$\frac{a^2 H_G(a, b, c)}{(d-1)^2} = \det \begin{pmatrix} 0 & G_y & 0 \\ G_y & G_{yy} & G_{yz} \\ 0 & G_{zy} & G_{zz} \end{pmatrix} = -G_y(a, b, c)^2 G_{zz}(a, b, c).$$

Again the left side is nonzero, so $G_{zz}(a, b, c) \neq 0$, hence $[1, 0, 0] \notin T_p D$ as claimed.

To summarize, we have shown that $I_p(C, D) = 1$ at every point of $C \cap D$, so the map f has exactly $n = d(d-1)$ ramification points. We now have

$$2 - 2g = 2d - n = 2d - d(d-1),$$

which we rearrange to get the desired $g = \frac{1}{2}(d-1)(d-2)$. □